AI Exposure and Registered Unemployment in Spain: Early Evidence from Occupation-Level Administrative Data*

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Abstract

This paper provides early occupation-level evidence on whether labor-market outcomes in Spain have begun to diverge according to exposure to generative AI. We combine administrative records for 502 four-digit occupations through March 2026 with an exposure measure based on observed AI use. A continuous-treatment difference-in-differences (DiD) design compares occupations within broad occupational families in each month following ChatGPT's November 2022 release. A 10 percentage-point increase in exposure is associated with 1.2 percent more registered unemployment during the first 25 months and 1.7 percent thereafter, although the later estimate is imprecise. The unemployment gradient emerges gradually rather than as an immediate post-release break, while new contracts show no persistent response. These patterns provide early evidence of differential labor-market adjustment across occupations with different exposure to generative AI.

Keywords: generative AI, unemployment, hiring, occupations, Spain, continuous-treatment difference-in-differences.

JEL Classification: J23, J24, J63, J64, O33.

1 Introduction

The rapid diffusion of generative AI has renewed a classic labor-market question: do new automation technologies displace workers, or do they reorganize tasks and increase demand for complementary labor? The question is especially difficult in the first years after a technology shock. Adoption

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is uneven, exposed occupations differ from unexposed occupations before the shock, and early labor-market effects may appear as churn, hiring changes, or occupational reallocation rather than immediate net job destruction. These concerns are particularly salient for younger workers. In its 2026 report on global youth employment, the International Labour Organization (ILO) estimates that 6.1 percent of jobs held by young people aged 15–29 fall within the occupational categories most exposed to AI. Many of these jobs overlap with middle-skilled clerical and administrative occupations in which youth employment has declined since 2023 (ILO, 2026). Whether this exposure has begun to translate into adverse labor-market outcomes remains an open empirical question.

This paper provides early occupation-level evidence from Spain on whether labor-market outcomes have begun to diverge according to exposure to generative AI following the public release of ChatGPT in November 2022 and the subsequent diffusion of generative AI tools. Rather than asking whether generative AI has raised aggregate unemployment, the analysis focuses on whether registered unemployment and hiring have evolved differently across occupations with different exposure to the technology. We combine monthly administrative records on registered unemployment and newly registered contracts by four-digit occupation (CNO4) from the Spanish Public Employment Service (*Servicio Público de Empleo Estatal*, SEPE) with an occupation-level AI exposure measure adapted to the Spanish National Classification of Occupations. The resulting panel covers 502 CNO4 occupations from January 2021 through March 2026. Registered unemployment is a stock measured at the end of each month, whereas newly registered contracts are a monthly flow. Examining both outcomes provides evidence on the accumulation of unemployment and the evolution of hiring activity.

The empirical design combines a common change in technological availability—the public release of ChatGPT—with cross-occupational variation in potential exposure to generative AI. Our time-invariant exposure measure builds on the measure developed by Anthropic for O*NET occupations (Massenkoff and McCrory, 2026) and is mapped to Spanish CNO4 occupations. We implement a continuous-treatment DiD design that compares changes in labor-market outcomes before and after November 2022 across occupations with different exposure intensities. Our preferred specifications interact AI exposure with indicators defined relative to ChatGPT's release and absorb CNO4 fixed effects (FE) and CNO1d-by-month FE.[1] Identification therefore comes from comparisons among differently exposed four-digit occupations within the same broad occupational family and month. The family-by-month effects absorb shocks common to related occupations, including family-specific demand conditions and differences in their post-pandemic recovery. The choice of CNO1d balances comparability against identifying variation: finer CNO2d-by-month comparisons are potentially more homogeneous but provide substantially less within-family variation in AI exposure.

The dynamic estimates for registered unemployment remain close to zero around ChatGPT's release

[1]The dynamic specification interacts AI exposure with separate event-time indicators, with October 2022 omitted as the reference month. The static specification distinguishes an adjustment period (event times 0–24, November 2022 through November 2024) from a later period (event times 25–40, December 2024 through March 2026). Both are estimated using two-way FE (TWFE) regressions, with standard errors clustered at the four-digit occupation level.

and become positive only gradually. Confidence intervals are wide and generally include zero. This delayed response is more consistent with gradual organizational adjustment than with an immediate displacement shock. Its timing also coincides with the continued diffusion of generative AI and the introduction of more capable agentic tools, including Claude Code and Codex in 2025, although the design cannot isolate the effects of individual products. Because registered unemployment is observed as a monthly stock, the data cannot determine whether its relative accumulation reflects greater inflows into unemployment, slower exits from the register, or both.

Motivated by this gradual dynamic path, we summarize the post-release estimates over an adjustment period and a later period. A 10 percentage-point increase in AI exposure is associated with a 1.2 percent larger stock of registered unemployment from November 2022 through November 2024 and a 1.7 percent larger stock from December 2024 through March 2026. Only the adjustment-period estimate is statistically distinguishable from zero, and equality between the two period effects cannot be rejected. The corresponding estimates for new contracts are 1.4 and 1.3 percent, neither of which is statistically significant.

The age-specific estimates suggest heterogeneity in labor-market responses. Among individuals aged 30–39, a 10 percentage-point increase in AI exposure is associated with 1.7 percent more registered unemployment during the adjustment period and 3.4 percent more thereafter; the later estimate is statistically significant at the 5 percent level. The corresponding estimates are 1.5 and 2.2 percent for individuals under 30 and 0.9 and 1.2 percent for those aged 40 or older. The under-30 estimates are statistically indistinguishable from those for ages 30–39 in both periods ($p = 0.897$ and $p = 0.320$, respectively). In the later period, however, the estimate for ages 30–39 differs from that for ages 40 or older ($p = 0.035$). Equality across all three groups cannot be rejected at conventional levels ($p = 0.105$). The point estimates are largest for individuals aged 30–39, but the formal cross-group tests do not establish a sharp age threshold. This pattern does not appear to be explained by the concentration of this group in more exposed occupations.[2]

The contract estimates also vary by age. A 10 percentage-point increase in AI exposure is associated with 2.2 percent more new contracts for individuals under 30 during the adjustment period and 2.6 percent more thereafter; both estimates are statistically significant. The corresponding effects are 1.3 and 1.7 percent for individuals aged 30–39 and 0.3 and −0.1 percent for those aged 40 or older, none of which are individually significant. In a specification pooling the two younger groups, equality with individuals aged 40 or older is rejected during both the adjustment ($p = 0.008$) and later ($p = 0.002$) periods. The contract results therefore provide clearer evidence of a younger-versus-older distinction than the unemployment estimates. Because contracts are measured as monthly flows, however, these positive responses may reflect greater labor-market turnover rather than more stable employment.

The preferred estimates show no clear heterogeneity by individual gender. For registered unem-

[2]Weighting CNO4d exposure by each occupation's average pre-treatment registered unemployment yields mean scores of 0.113, 0.119, and 0.118 for individuals under 30, aged 30–39, and aged 40 or older, respectively.

ployment, a 10 percentage-point increase in exposure is associated with increases of 0.8 and 1.6 percent for men and 1.0 percent in both periods for women. The corresponding contract estimates are 1.4 and 1.6 percent for men and 1.1 and 0.9 percent for women. None of these estimates is individually statistically significant, and their similar magnitudes provide no clear indication of differential adjustment by gender.

Additional heterogeneity analyses suggest a stronger unemployment gradient in less-feminized occupations and in Madrid and Barcelona, although these patterns are accompanied by less favorable pre-treatment diagnostics and are therefore treated as suggestive.

Finally, we assess whether alternative estimators yield the same qualitative result. Both alternatives produce larger positive unemployment estimates. For a 10 percentage-point increase in exposure, the CNO1d-stratified continuous difference-in-differences (CDiD) estimator implies increases of 3.4 percent during the adjustment period and 7.1 percent thereafter. Synthetic difference-in-differences (SDiD) yields upper-tail effects of 2.7 and 4.1 percent, respectively. The estimates for new contracts are more method-dependent and generally less precise. Because CDiD aggregates marginal responses differently from TWFE, while SDiD estimates a binary high-versus-lower-exposure contrast, agreement in direction is more informative than differences in magnitude.

This paper is most directly related to the emerging empirical literature on the labor-market effects of generative AI, which has focused primarily on the United States and the Nordic countries. Using U.S. payroll data, Brynjolfsson et al. (2025) document relative employment declines among workers aged 22–25 in highly exposed occupations, whereas Massenkoff and McCrory (2026) find no systematic unemployment response and only tentative evidence of slower hiring among younger workers. Nordic evidence is similarly mixed. Humlum and Vestergaard (2025) document rapid chatbot adoption and extensive task reorganization in Denmark but find precise null effects on earnings and hours. Facius and Iacono (2026) find no robust displacement in Norwegian administrative data and show that backdated synthetic comparisons can generate apparent effects before ChatGPT. Lodefalk et al. (2026) document an age gradient in Swedish hiring, while online labor markets exhibit sharper short-run reductions in demand for directly exposed freelance services (Hui et al., 2024). Using task-level exposure measures that vary across firms, occupations, and time, Hampole et al. (2025) find that more AI-exposed tasks experience lower subsequent labor demand, although aggregate employment effects are more modest because substitution is partly offset by productivity-driven increases in labor demand at adopting firms. Taken together, this evidence suggests that the early labor-market effects of generative AI depend on adoption, task content, workers' career stage, and the particular adjustment margin being measured.

Second, the paper builds on the task-based literature on technological change and occupational exposure. Autor et al. (2003) and Acemoglu and Restrepo (2019) emphasize that automation reallocates tasks between labor and capital and can generate displacement and reinstatement simultaneously. Exposure measures developed by Felten et al. (2021), Felten et al. (2023), and Eloundou et al. (2024) translate technological capabilities into occupation-level measures. Massenkoff and

McCrory (2026) extend this approach by combining theoretical capabilities with observed work-related AI use. Our analysis also relates to the task-autonomy framework in Garicano et al. (2026a), which classifies jobs into three tiers, with those in the first tier most susceptible to replacement by AI.

Exposure nevertheless remains distinct from adoption. Using nationally representative worker survey data linked to detailed occupations and tasks, Bick et al. (2026) show that standard occupational exposure measures explain only part of the variation in realized GenAI adoption and that usage measures based on platform chat logs differ conceptually from worker-level adoption measures. Using firm websites across four European countries, Garbers and Gregory (2026) show that realized adoption depends strongly on firms' human capital and digital capabilities. Linked Danish survey and administrative data in Pulito et al. (2026) similarly show that common occupational exposure measures differ in their ability to predict actual use.

Third, the paper relates to the literature on productivity, task reorganization, and worker complementarity. Experimental evidence shows that generative AI can improve performance in writing and problem-solving tasks (Noy and Zhang, 2023). Cruces et al. (2026) find larger gains for workers with less formal education in a randomized business problem-solving exercise. Freund and Mann (2026) show theoretically and quantitatively that automation can generate heterogeneous wage effects within the same occupation by changing the relative importance of the tasks that remain. Evidence for Germany based on patent-based measures of AI advances that predate generative AI shows that AI-related task changes occur mostly within detailed occupations, and that workers respond mainly by changing employers rather than occupations, with modest employment losses and earnings losses concentrated among low-skilled workers (Gathmann et al., 2024). These findings imply that occupational exposure need not translate one-for-one into job loss: the same technology may displace workers specializing in automatable tasks while increasing demand for complementary skills. At the aggregate level, Acemoglu (2025) similarly emphasizes that the macroeconomic consequences of AI depend on the share of tasks affected, task-level productivity gains, and the balance between automation and complementary uses, implying that potentially large occupational exposure need not translate into equally large aggregate employment effects.

This paper contributes to these literatures by providing early, high-frequency evidence on how labor-market outcomes have evolved across occupations with different exposure to generative AI. First, it uses administrative data for Spain at the four-digit occupation level, a level of detail unavailable in standard labor-force survey estimates, to track registered unemployment and hiring at monthly frequency. Second, it adapts an observed-use exposure measure to Spanish occupations and exploits comparisons among differently exposed occupations within broad occupational families and months. Third, it documents whether the emerging occupational divergence differs across worker and occupation characteristics and relates its timing to independent evidence on the diffusion of generative AI in Spain. The contribution is deliberately reduced-form and occupation-level: the paper documents differential labor-market adjustment by AI exposure rather than estimating the

aggregate employment effect of generative AI, firm-level adoption effects, or individual worker transitions.

The remainder of the paper proceeds as follows. Section 2 describes the administrative data, exposure measure, and evidence on AI adoption in Spain. Section 3 presents the empirical strategy. Section 4 reports the baseline estimates, heterogeneity analyses, and robustness checks, and Section 5 concludes. Appendix Sections A and B report the principal identification diagnostics and supporting age and gender results, while additional validation exercises, heterogeneity analyses, robustness checks, estimator comparisons, and the aggregate calibration appear in Online Appendix Sections O.A–O.I.

2 Data

This section describes the data used in the empirical analysis. We first introduce the administrative labor-market data from the Spanish Public Employment Service (*Servicio Público de Empleo Estatal*, SEPE). We then describe the construction of the occupation-level AI exposure measure, which maps Anthropic/O*NET occupations to four-digit occupations in the Spanish National Classification of Occupations 2011 (CNO-11, hereafter CNO4d). Finally, we explain how these data sources are combined to construct the estimation panel, define the outcome variables, and present summary statistics.

2.1 Administrative labor-market data

The main labor-market data come from SEPE, which publishes monthly statistics on registered unemployment and registered contracts through the Registered Labor Market Activity (Movimiento Laboral Registrado, MLR) system (Servicio Público de Empleo Estatal, 2026).[3] Monthly statistics by four-digit occupations (CNO4d) are available from January 2021 through March 2026, and occupation codes are defined according to the Spanish National Classification of Occupations 2011 (CNO-11).[4]

The data span the period from January 2021 through March 2026, allowing us to construct a monthly occupation-level panel that tracks changes in registered unemployment and hiring flows before and after the public launch of ChatGPT in November 2022. The analysis relies exclusively on SEPE administrative statistics and does not use Social Security records, which contain information on unemployment insurance and unemployment assistance receipt. Consequently, the registered

[3] Information is available through the SEPE website (https://www.sepe.es/HomeSepe/que-es-observatorio/informacion-mt-por-ocupacion.html), which provides a search interface for occupation-specific statistics. To construct a machine-readable dataset, we scraped the information for all occupations at the four-digit level.

[4] Historical data are available at https://www.sepe.es/HomeSepe/que-es-observatorio/informacion-mt-por-ocupacion/historico-informacion-mt-por-ocupacion.html. However, these statistics are reported only at the two-digit occupation level and follow the earlier CNO-94 classification.

unemployment counts include both individuals who receive unemployment benefits and those who do not.

Registered unemployment (*parados*) is defined as the stock of unemployed workers registered with SEPE at the end of each month.[5] It should therefore be interpreted as an administrative measure of formal unemployment registration rather than as a measure of total unemployment. The broader population measure of unemployment is provided by the Labor Force Survey (Encuesta de Población Activa, EPA). As a validation exercise, Online Appendix Figure O.A.1 compares aggregate unemployment in the SEPE data with the corresponding EPA series. The two series closely track each other and exhibit very similar trends throughout the sample period. Registered contracts (*contratos*) correspond to new employment contracts registered during the month and therefore measure the monthly flow of new hires.

The main panel contains 502 CNO4d occupations observed over 63 months, yielding a perfectly balanced panel with 31,626 occupation-month observations. We also construct separate panels disaggregated by province, gender, and age. These dimensions are published by SEPE as separate tabulations rather than as a single joint cross-tabulation, so cells combining multiple dimensions (e.g., province × age or gender × age) cannot be recovered. For the age analysis, we aggregate the published age categories into three groups: workers younger than 30, workers aged 30–39, and workers aged 40 and older.

2.2 AI exposure measure

2.2.1 Anthropic measure

Our occupation-level measure of AI exposure is based on the observed exposure index developed by Massenkoff and McCrory (2026). The index combines three sources of information: (i) occupation task descriptions from O*NET, (ii) task-level measures of the theoretical capabilities of large language models (LLMs) developed by Eloundou et al. (2024), and (iii) observed usage patterns from the Anthropic Economic Index, which records how Claude is used in real-world settings.[6] The resulting measure captures the extent to which occupations perform tasks that are both technically feasible for generative AI and already observed in professional AI use. The analysis also documents a substantial gap between the theoretical capabilities of LLMs and their observed use in professional

[5] According to the official SEPE methodology, registered unemployment corresponds to active employment demands registered with the Public Employment Services on the last working day of the month that satisfy the statistical criteria for registered unemployment. These criteria exclude employed jobseekers, workers without immediate availability, and other categories established by the Ministerial Order of 11 March 1985. See Ministerio de Trabajo y Economia Social (2023).

[6] The Anthropic methodology first identifies tasks that are theoretically feasible for large language models, then measures whether these tasks are actually observed in work-related Claude interactions. Tasks involving automated use receive greater weight than augmentative uses, and occupation-level exposure is obtained as a time-weighted average across tasks. The index is estimated using Claude.ai usage data collected during 2025 and released as part of the 2026 Anthropic Economic Index. See the online appendix of Massenkoff and McCrory (2026) for a detailed description of the methodology.

settings, suggesting that AI adoption remains well below its technological frontier.

Unlike earlier exposure measures based solely on task descriptions or expert assessments, the Anthropic AI index incorporates observed AI usage in addition to technological capability. Because the index is based on observed Claude interactions, it captures realized platform use, although such chat-log-based measures should not be interpreted as direct measures of worker- or firm-level adoption (Bick et al., 2026). The measure is published as a single cross-sectional occupation-level index, which we treat as time-invariant over our sample period. By construction, the exposure score is bounded between zero and one,[7] with higher values indicating greater observed exposure to generative AI. The original Anthropic index is reported for 756 occupations defined using the O*NET classification.[8] We therefore map these occupations to four-digit occupations in the Spanish National Classification of Occupations 2011 (CNO4d), as described in the next subsection.

2.2.2 Mapping to Spanish occupations

The Anthropic AI exposure index is reported for 756 occupations defined under the O*NET classification, whereas the Spanish labor-market data are organized using 502 four-digit occupations in the Spanish National Classification of Occupations 2011 (CNO4d). We therefore construct a semantic mapping between the two occupational classifications using multilingual text embeddings, allowing occupations from both systems to be represented in a common semantic space.

Let $i = 1, \ldots, N$ index Anthropic/O*NET occupations and $j = 1, \ldots, J$ index Spanish CNO4d occupations. For each Anthropic/O*NET occupation i, we embed the occupation title and O*NET description, yielding an embedding vector $x_i \in \mathbb{R}^d$, where d denotes the embedding dimension. The corresponding observed AI exposure score provided by Anthropic is denoted by y_i. For each Spanish CNO4d occupation j, we embed a structured semantic record containing the occupation code, title, definition, typical tasks, included examples, and related or excluded occupations, yielding an embedding vector $x_j \in \mathbb{R}^d$. Both sets of embeddings are generated using Qwen3-Embedding-4B.[9]

Similarity between Anthropic/O*NET and CNO4d occupations is measured using cosine similarity:

$$c_{ji} = \frac{x_j^\top x_i}{\|x_j\| \, \|x_i\|},$$

where c_{ji} denotes the cosine similarity between Spanish occupation j and Anthropic/O*NET occupation i.[10]

[7] The index can be interpreted as the share of occupation-specific work time associated with tasks exhibiting observed AI usage, after applying the weighting procedure described by Massenkoff and McCrory (2026).

[8] The data are available at https://huggingface.co/datasets/Anthropic/EconomicIndex/tree/main/labor_market_impacts.

[9] Qwen3-Embedding-4B is a multilingual embedding model with approximately two thousand dimensions. Its multilingual design is particularly important in this application because the source texts are written in different languages: Spanish for CNO4d occupations and English for Anthropic/O*NET occupations.

[10] As a robustness check, we consider one alternative mapping. A cosine-weighted measure assigns Anthropic/O*NET occupations to their nearest CNO4d occupation and computes, for each CNO4d occupation, a

For each Spanish occupation, we identify the Anthropic/O*NET occupation with the highest cosine similarity:

$$i^*(j) = \arg\max_i c_{ji}.$$

The baseline AI exposure assigned to Spanish occupation j is then given by

$$\widehat{y}_j = y_{i^*(j)},$$

that is, the observed Anthropic exposure score of its nearest Anthropic/O*NET occupation. The resulting $\widehat{y}_j$ constitutes our baseline occupation-level AI exposure measure.

2.2.3 Validation

The construction of the Spanish AI exposure measure relies on the assumption that occupational AI adoption patterns are sufficiently similar across countries for the Anthropic exposure measure to provide an informative ranking of AI exposure in Spain. The observed exposure measure uses global Claude usage classified under the U.S. O*NET taxonomy, rather than usage restricted to the U.S. labor market (Massenkoff and McCrory, 2026). We assess the validity of the resulting exposure measure through three complementary exercises: external validation, distributional validation, and face validity.

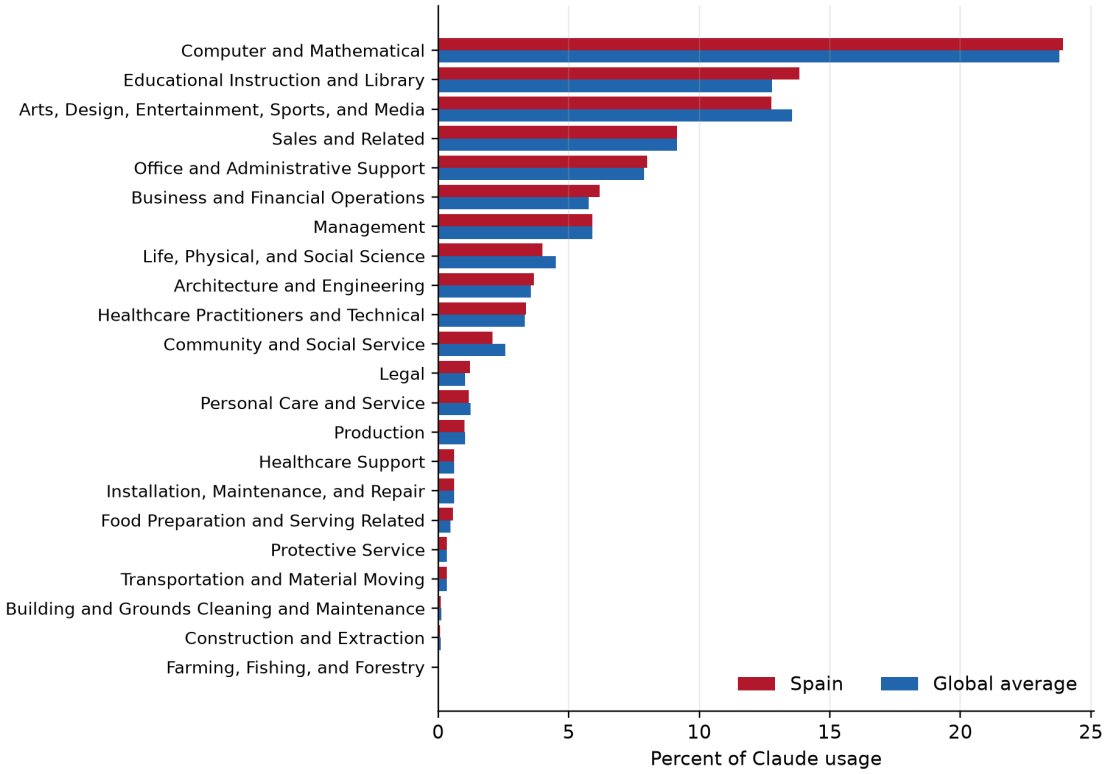
Our first validation exercise examines whether occupational AI usage patterns are similar in Spain and globally. Figure 1 compares the distribution of Claude usage across broad occupational groups using the latest Anthropic Economic Index. The two distributions are remarkably similar. Computer and mathematical occupations account for the largest share of Claude usage in both distributions, with education, arts and media, sales, business occupations, and office support also accounting for substantial shares. Differences between Spain and the global average are generally small, suggesting that occupational AI usage follows similar patterns in Spain and globally. This evidence supports the external validity of applying the Anthropic exposure ranking to the Spanish labor market. A similar occupation-level imputation strategy is also adopted by recent European evidence on generative AI and labor markets, including Facius and Iacono (2026).

similarity-weighted average of the assigned Anthropic exposure scores,

$$\widehat{y}_j^{CW} = \frac{\sum_{i \in A_j} c_{ji} y_i}{\sum_{i \in A_j} c_{ji}},$$

where A_j is the set of Anthropic/O*NET occupations assigned to CNO4d occupation j. If A_j is empty, the nearest-neighbor exposure is used instead.

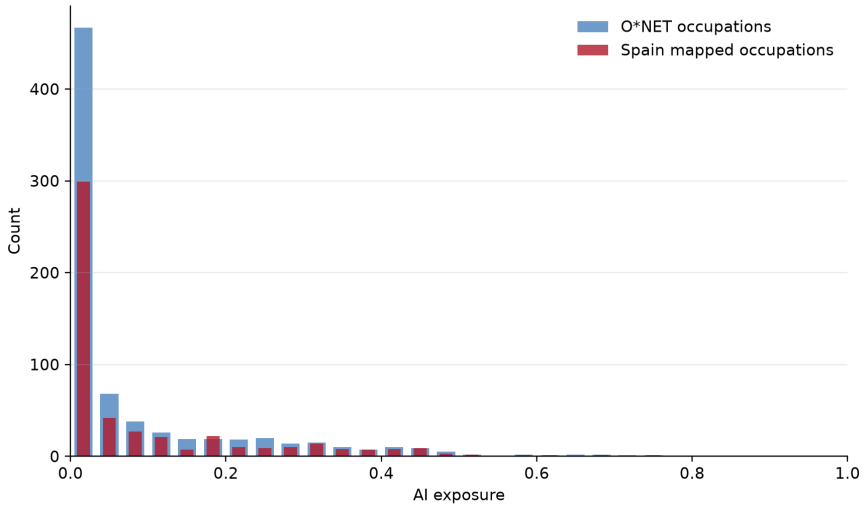
Figure 1: Distribution of Claude usage by major occupational group: Spain and the global average



Notes: The figure reports the percentage of Claude usage accounted for by each major occupational group according to the Anthropic Economic Index. Spain refers to Claude interactions originating in Spain; the global average refers to pooled worldwide Claude usage, rather than an unweighted average across countries.
Source: Anthropic Economic Index (May 2026).

Our second validation exercise evaluates whether the semantic mapping preserves the overall distribution of AI exposure when transferring the measure from the O*NET occupational classification to Spanish CNO4d occupations. Figure 2 compares the distribution of exposure scores for the original Anthropic/O*NET occupations and the mapped Spanish occupations. The mapped distribution closely reproduces both the concentration of occupations with very low AI exposure and the upper tail of highly exposed occupations observed in the original O*NET data. This close correspondence indicates that the semantic mapping preserves the overall distribution of AI exposure despite the differences between the two occupational classifications.

Figure 2: Distribution of AI exposure before and after mapping



Notes: The figure compares the distribution of the original Anthropic AI exposure scores for O*NET occupations with the distribution obtained after mapping the exposure measure to Spanish CNO4d occupations using the nearest-neighbor semantic matching procedure.

Finally, we assess the face validity of the mapped exposure measure by examining the occupations with the highest and lowest exposure levels. Online Appendix Table O.B.1 reports the ten occupations with the highest and lowest AI exposure after mapping the Anthropic measure to Spanish CNO4d occupations. The ranking is economically intuitive. Computer programmers, software specialists, financial analysts, and data-entry clerks appear among the most exposed occupations, whereas occupations involving predominantly manual or physical tasks are concentrated among the least exposed. The close correspondence between the ranking and prior expectations provides additional reassurance that the semantic mapping preserves economically meaningful relationships across occupational classifications.

2.2.4 Timing of adoption

The release of ChatGPT marks a sharp change in access to generative AI, but not a common date of adoption by Spanish firms. Survey evidence instead points to a gradual and uneven diffusion. In the harmonized survey of enterprises with at least ten employees, the share of Spanish firms using at least one covered AI technology was 7.7 percent in 2021, 11.8 percent in 2022, 9.6 percent in 2023, and 11.3 percent in 2024 (Observatorio Nacional de Tecnología y Sociedad, 2023, 2024; Izquierdo et al., 2025). These annual observations do not form a smooth monotonic series, partly because the survey captures a broad set of AI technologies and their measurement evolved as the technology diffused.

A separate survey carried out by the Bank of Spain and conducted in November 2024 found that 19.9 percent of responding firms used predictive or generative AI and that 18.1 percent used generative

AI specifically (Izquierdo et al., 2025). Most adoption was still shallow: about 60 percent of AI-using firms described their use as experimental or pilot use. At the individual level, the share of Spanish residents aged 18–75 reporting frequent ChatGPT use rose from 4 percent in 2023 to 14 percent in 2024 and 28 percent in 2025 (Funcas, 2026). The firm and individual series measure different populations and concepts, so their levels should not be compared directly. Together, however, they show that effective exposure expanded over several years rather than appearing fully in November 2022.

Figure 3 summarizes this evidence and relates it to the empirical windows. We interpret event times 0–24 as an adjustment period in which firms experimented with the technology and reorganized tasks. Event times 25–40 constitute a later period in which adoption had more time to affect hiring and unemployment. These windows do not assume that adoption stopped after 24 months, nor are the survey indicators used as treatment variables. They provide institutional support for allowing the exposure gradient to evolve gradually after the common availability shock.

2.3 Panel construction and summary statistics

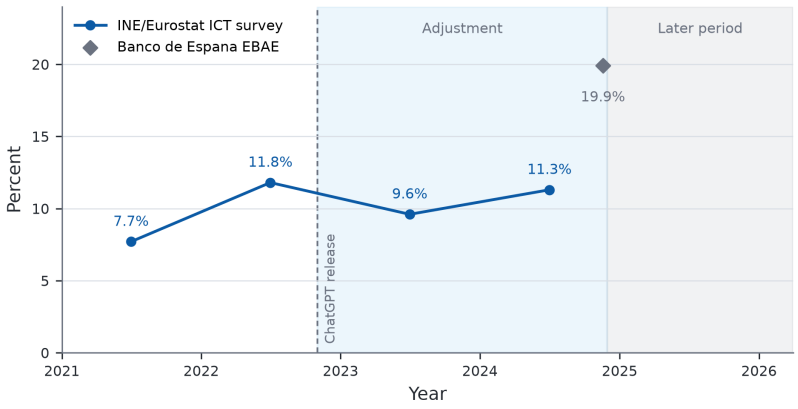
We construct the estimation panel by merging the monthly SEPE occupation panel with the occupation-level AI exposure measure described in the previous subsection, using the four-digit occupation code (CNO4d) as the matching key. Since the Anthropic AI exposure index is cross-sectional, the resulting exposure measure is time-invariant, while the labor-market outcomes are observed monthly. The final estimation sample is a balanced panel of 502 occupations observed over 63 consecutive months, yielding 31,626 occupation-month observations.

The main outcome variables are the logarithm of the number of registered unemployed workers (denoted by U) and the logarithm of the number of newly registered contracts (denoted by C).[11] These are the outcome variables considered in the empirical analysis.

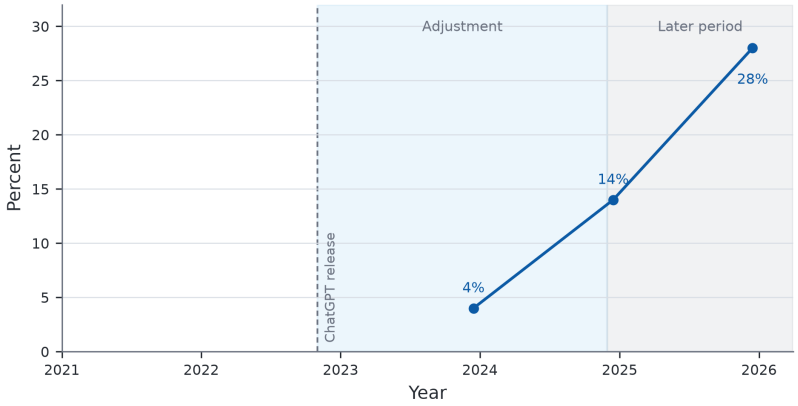
Table 1 reports summary statistics for the estimation sample. The table shows substantial variation in labor-market outcomes across occupation-month observations. It also indicates that the two AI exposure measures have very similar distribution.

[11] The baseline specification uses $\log(U)$ and $\log(C)$, excluding occupation-month observations with zero counts. There are two occupation-month observations (0.006% of the sample) with zero registered unemployed workers and 342 observations (1.08% of the sample) with zero newly registered contracts. As a robustness check, we re-estimate all specifications using the transformations $\log(U+1)$ and $\log(C+1)$, thereby retaining the full estimation sample. The results are quantitatively very similar to the baseline estimates.

Figure 3: AI adoption and generative-AI use in Spain



(a) Firms using AI



(b) Frequent ChatGPT use among adults

Notes: Panel (a) reports the share of Spanish enterprises with at least 10 employees using at least one AI technology covered by the harmonized ICT survey. The gray diamond reports the broader Banco de España (BdeE) EBAE estimate for Nov. 2024, which includes experimental use and is not directly comparable to the annual ICT series. Panel (b) reports the share of Spanish residents aged 18–75 who use ChatGPT frequently. The dashed vertical line marks the public release of ChatGPT in Nov. 2022. Shading identifies the adjustment period (Nov. 2022–Nov. 2024) and later period (Dec. 2024–March 2026) used in the regressions. Sources: ONTSI, BdeE, and Funcas (Observatorio Nacional de Tecnología y Sociedad, 2023, 2024; Izquierdo et al., 2025; Funcas, 2026).

Table 1: Summary statistics for SEPE occupation-level variables and AI exposure indices

	Mean	SD	Min	Max	Obs.
Outcomes					
Registered unemployed (in levels)	5,693	25,338	0	467,292	31,626
New registered contracts (in levels)	2,776	11,081	0	216,727	31,626
Registered unemployed (in logs)	6.718	1.944	0.000	13.055	31,624
New registered contracts (in logs)	5.871	2.088	0.000	12.286	31,284
AI exposure index					
AI index - cosine similarity	0.084	0.136	0.000	0.745	31,626
AI index - cosine-weighted	0.084	0.128	0.000	0.628	31,626

Notes: Labor-market data come from SEPE. The occupation-level AI exposure measures map Anthropic/O*NET occupations to four-digit occupations in the Spanish National Classification of Occupations 2011 (CNO-11).

3 Empirical strategy

The empirical strategy exploits the public release of ChatGPT in November 2022 and the subsequent diffusion of generative AI to estimate whether labor-market outcomes evolved differently across occupations with different exposure. Because AI became broadly available at essentially the same time for all occupations, identification comes from cross-occupation differences in AI exposure rather than from staggered treatment timing. Furthermore, the evidence on the timing of adoption summarized above motivates a design that allows the response to emerge gradually rather than imposing an immediate constant effect.

The analysis uses monthly panel data for four-digit occupations (CNO4d), comparing changes within occupations over time. Throughout the paper, the baseline measure of AI exposure is the cosine-nearest exposure measure developed in Section 2.2.2. This measure is fixed over time and assigns each occupation a predicted AI exposure score based on the mapping between Anthropic/O*NET and Spanish CNO4d occupations.

3.1 Empirical model

The baseline specification relates occupation-level labor-market outcomes to the interaction between AI exposure—a time-invariant measure of treatment intensity—and a full set of monthly event-time indicators pre- and post-Chat GPT release to capture the dynamic effects:

$$Y_{jt} = \alpha_j + \lambda_{g(j),t} + \sum_{k \in \mathcal{K},\, k \neq -1} \beta_k D_j \mathbf{1}\{e_t = k\} + \varepsilon_{jt}, \tag{1}$$

where Y_{jt} is alternatively the logarithm of registered unemployed workers—i.e., $\log(U)$—or the

logarithm of newly registered contracts—i.e., $\log(C)$—for CNO4d occupation j in year-month t.[12] Occupation FE α_j absorb permanent differences across CNO4d occupations, including the level effect of the time-invariant dose. The FE $\lambda_{g(j),t}$ subsume common year-month effects and absorb any shock shared by occupations in CNO1d family g in month t. $A_j \in [0,1]$ is the baseline AI exposure score and define $D_j = A_j/0.10$, so one unit of D_j corresponds to a 10 percentage-point increase in exposure.

$\mathbf{1}\{e_t = k\}$ is the set of event-time dummies, with $\mathcal{K} = \{-21, \ldots, 40\}$. Event time $e_t = 0$ is November 2022 and October 2022 ($e_t = -1$) is the omitted reference month. Each β_k, our coefficients of interest, measures the change in the within-CNO1d exposure gradient at event time k, relative to October 2022, the omitted reference month. The within-family interpretation follows from the inclusion of CNO1d-by-year-month FE.[13] The event study serves two purposes. First, it shows whether the response appears immediately or accumulates as firms adopt the technology, which is better than imposing a constant post-treatment effect. In addition, it provides diagnostics for differential trends before ChatGPT's release—i.e., testing whether the pre-treatment coefficients are jointly equal to zero and statistically indistinguishable from one another—.[14]

Baseline standard errors allow arbitrary serial correlation and heteroskedasticity within CNO4 occupations. Because exposure is assigned at the CNO4 level, this is the natural primary clustering level. We also cluster by CNO3 to allow correlated shocks across more closely related four-digit occupations. Point estimates are unchanged under this alternative, while the number of independent clusters falls from 502 to 170.

For reporting purposes, we define $P_t^A = \mathbf{1}\{0 \leq e_t \leq 24\}$, which covers the adjustment period during which firms were beginning to adopt AI, and $P_t^L = \mathbf{1}\{25 \leq e_t \leq 40\}$, which covers the later period, when AI tools were more widely used by firms and workers. We then estimate the following equation:

$$Y_{jt} = \alpha_j + \lambda_{g(j),t} + \beta_A D_j P_t^A + \beta_L D_j P_t^L + \varepsilon_{jt} \tag{2}$$

Again the coefficients, β_A and β_L are identified from differences in exposure among CNO4d occupations inside the same broad family and year-month. In this log-outcome specification, $100\beta_p$ is the approximate percentage change associated with a 10 percentage-point increase in exposure

[12] The baseline specification excludes occupation-month observations with zero counts when taking logarithms. There are two occupation-month observations (0.006% of the sample) with zero registered unemployed workers and 342 observations (1.08% of the sample) with zero newly registered contracts. As a robustness check, we re-estimate all specifications using the logarithm of each labor market outcome plus one, thereby retaining the full estimation sample. The results are quantitatively very similar to the baseline estimates.

[13] This comparison is more demanding than not including family group times monthly dummies but less restrictive than CNO2-by-month effects, for which within-family exposure support is limited.

[14] We report two Wald diagnostics over the full, early, and recent pre-treatment windows. The joint-nullity test evaluates whether the selected β_k coefficients are jointly zero. The flatness test evaluates whether they are equal to one another while allowing their common level to differ from zero. The latter is informative when normalization to October 2022 shifts the entire pre-period path without producing a differential slope.

during phase $p \in \{A, L\}$; the exact transformation is $100[\exp(\beta_p) - 1]$. The coefficients are marginal exposure gradients, not average treatment effects from a binary design.

Because publicly available administrative data do not provide monthly labor-force denominators by occupation, the baseline analysis uses the logarithm of occupation-level unemployment counts rather than occupation-specific unemployment rates. Consequently, changes in U_{ot} should be interpreted as approximately percentage changes in the number of registered unemployed workers within an occupation rather than changes in the occupation-specific unemployment rate.

3.2 Identification and interpretation

The preferred specification identifies the exposure gradient from differences across CNO4 occupations within the same CNO1 family. Then, identification relies on conditional parallel trends: absent the diffusion of generative AI after November 2022, differently exposed occupations within a family group would have experienced similar changes in labor-market outcomes. By allowing each broad occupational family to follow an unrestricted monthly path, CNO1d-by-month FE make these comparisons more local and absorb shocks common to the family. As mentioned above, we use event-study diagnostics to assess whether differential trends were already present before treatment. We use CNO1d rather than CNO2d to balance comparability and identifying variation: CNO2d-by-month effects yield narrower comparisons but leave too little within-family exposure variation for precise estimation.

Because AI exposure is continuous, the causal parameters of interest differ from the average treatment effect on the treated (ATT) in a binary design, as discussed by Callaway et al. (2024). A dose-specific ATT measures the effect of exposure level d, relative to zero exposure, among occupations observed at that dose. The average causal response on the treated, or ACRT, instead measures the effect of a marginal increase in exposure around d for those occupations.[15] In our linear specification, β_A and β_L are period-specific within-family exposure gradients. They capture whether labor-market outcomes changed more in occupations with greater AI exposure during the adjustment and later periods, respectively. Interpreting these gradients as ACRTs requires the stronger assumptions discussed below.

The standard conditional parallel-trends assumption restricts untreated potential outcomes and identifies dose-specific effects relative to zero exposure. Interpreting differences across exposure levels as ACRTs requires strong parallel trends (Callaway et al., 2024), which additionally assumes that, within each CNO1d family, occupations receiving different exposure levels would experience the same average outcome path if assigned the same dose.[16] This assumption rules out systematic selection into exposure intensity based on dose-specific effects. Importantly, standard and strong

[15] Following Callaway et al. (2024), define $ATT_t(\ell \mid d, g) \equiv \mathbb{E}[Y_{jt}(\ell) - Y_{jt}(0) \mid D_j = d, g(j) = g]$ and $ACRT_t(d \mid d, g) \equiv \partial ATT_t(\ell \mid d, g)/\partial \ell|_{\ell=d}$. The ATT is a level effect relative to zero exposure, whereas the ACRT is the slope of the dose-response function at d.

[16] Let $\Delta Y_{jt}(d) \equiv Y_{jt}(d) - Y_{j,-1}(0)$. Formally, strong parallel trends requires $\mathbb{E}[\Delta Y_{jt}(d) \mid D_j > 0, g(j) = g] = \mathbb{E}[\Delta Y_{jt}(d) \mid D_j = d, g(j) = g]$ for every relevant dose d, CNO1d family g, and post-treatment month t.

parallel trends produce the same observable estimands and have identical implications for pre-treatment coefficients. Event-study tests therefore cannot distinguish between them: favorable pre-treatment evidence is consistent with standard parallel trends but may or may not provide supportive evidence on the stronger assumption required to interpret the exposure gradients as ACRTs.

4 Results

4.1 Baseline results

Figure 4 places the analysis in the context of Spain's post-pandemic labor-market recovery. Registered unemployment declines markedly across occupations over most of the sample period, with the steepest decline occurring in 2021 as labor-market conditions normalized following the COVID-19 shock. Prior to the public release of ChatGPT, occupations with different levels of AI exposure follow broadly similar downward trends, although some differential movement is visible during this early recovery period.[17] The similar pre-release paths are reassuring, but they are not sufficient for identification because exposure groups may differ in their recovery from the pandemic and in other occupation-specific shocks. After late 2022, however, the decline becomes noticeably slower among highly exposed occupations, leading to a gradual divergence in unemployment trajectories. Middle-exposure occupations also begin to separate from the zero-exposure group, although the pattern is less pronounced than for highly exposed occupations. In contrast, newly registered contracts—a monthly flow measure—exhibit substantially greater month-to-month variation than registered unemployment and show no comparable post-2022 divergence across exposure groups. Overall, the descriptive evidence suggests that labor-market outcomes began to diverge across occupations with different levels of AI exposure after the public release of ChatGPT, particularly for registered unemployment.

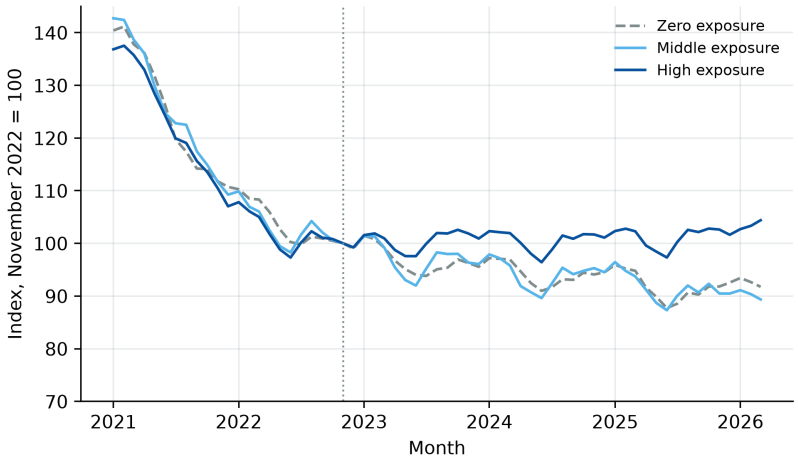
Figure 5 reports the preferred dynamic estimates.[18] The preferred specification asks whether outcomes diverged after ChatGPT's release across occupations with different exposure, while comparing occupations within the same broad CNO1d family in every month. This restriction is economically important: it removes shocks shared by related occupations, such as changes in sectoral demand or the pace of the post-pandemic recovery, without requiring those shocks to follow a parametric trend.[19] For ease of interpretation, coefficients are reported for a 10 percentage-point

[17] The year 2021 represents an unusually transitional period for the Spanish labor market. The post-pandemic recovery proceeded unevenly across activities and occupations, while extraordinary employment-retention measures, including the COVID-related ERTE schemes, remained in place. Because the incidence of these shocks and policies may have differed systematically across occupations, we report a complementary specification that excludes 2021.

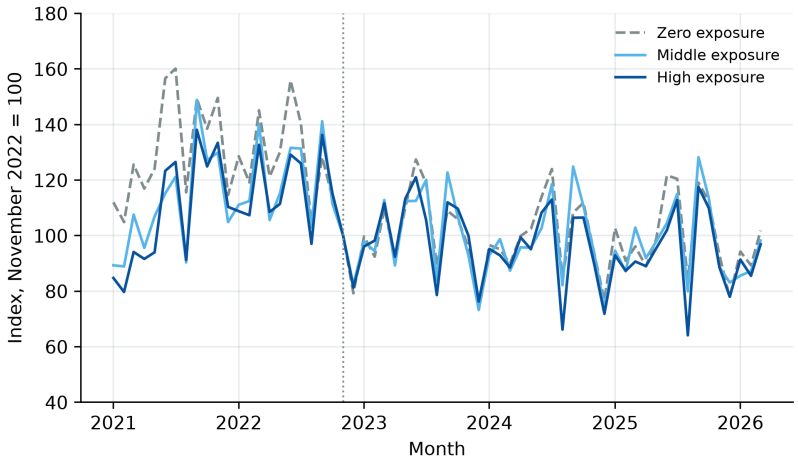
[18] Online Appendix O.C reports results from using a binary-treatment that classifies occupations with AI exposure above the 75th-percentile cutoff of 0.1169 as treated and all remaining occupations as controls.

[19] For CNO4 occupation j, CNO1 family $g(j)$, and event time k, the event-study regression is $Y_{jt} = \alpha_j + \lambda_{g(j),t} + \sum_{k \neq -1} \beta_k (A_j/0.10) \mathbf{1}\{t - T_0 = k\} + \varepsilon_{jt}$. The event window runs from -21 to 40, November 2022 is T_0, and October 2022 is omitted. Occupation FE absorb permanent differences across CNO4 occupations, CNO1-by-month FE absorb

Figure 4: Descriptive patterns by AI exposure group



(a) Number of registered unemployed



(b) Number of new contracts

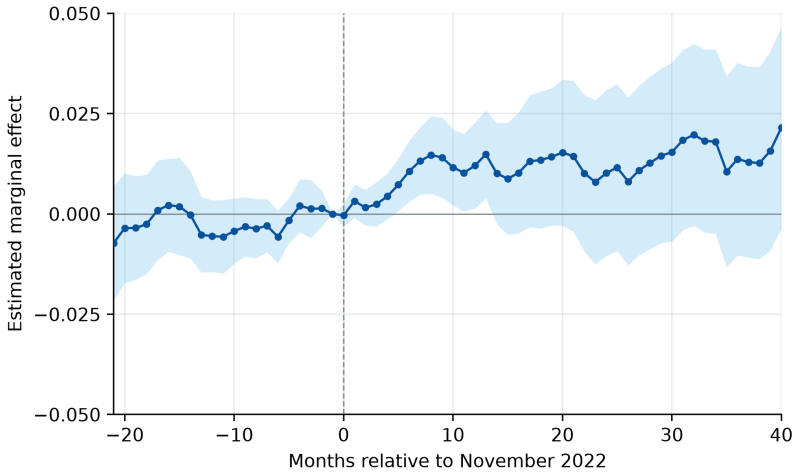
Notes: The figure plots monthly averages of the outcomes across occupations within each AI exposure group. Panel A reports the evolution of the average number of registered unemployed workers, while Panel B reports the average number of newly registered contracts. Each series is indexed to November 2022 = 100. Occupations are grouped into three categories based on their AI exposure: zero-exposure occupations, middle-exposure occupations, and high-exposure occupations. The middle-exposure group includes occupations with AI exposure greater than zero but below the seventy-fifth percentile, while the high-exposure group includes occupations with AI exposure above the seventy-fifth percentile (0.1169). The vertical dashed line marks November 2022, the month of the public release of ChatGPT.

increase in AI exposure relative to October 2022, the omitted reference month.

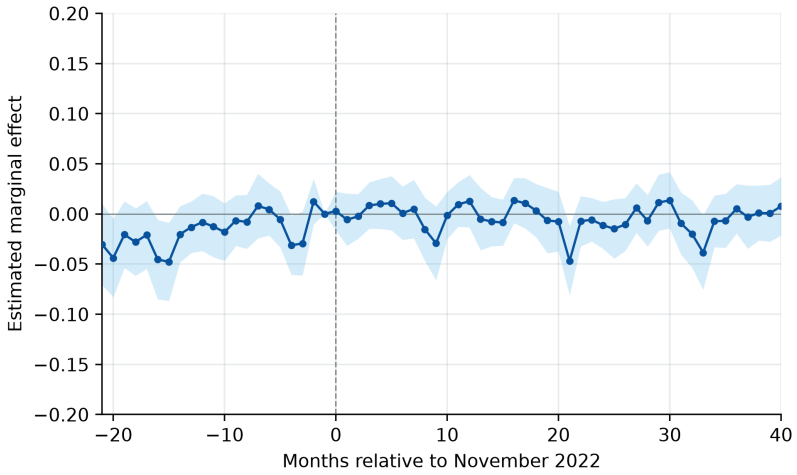
In panel (a), the unemployment gradient is close to zero around the release date and becomes positive only after several months. A 10 percentage-point increase in exposure is associated with roughly 1–2 percent more registered unemployment over much of the post-treatment period. The estimates remain positive near the end of the sample, but their confidence intervals widen and

shocks common to occupations in the same broad family, and standard errors are clustered by CNO4.

Figure 5: AI exposure and labor market outcomes. Event-study estimates.



(a) Number of registered unemployed



(b) Number of new contracts

Notes: The figure reports estimates of the preferred event-study specification for the total CNO4d occupation panel. The dependent variables are the logarithm of registered unemployed workers in panel (a) and the logarithm of new contracts in panel (b). Each marker is the marginal effect of a 10 percentage-point increase in AI exposure at the indicated event month. October 2022 (event time −1) is the omitted reference month; the vertical dashed line marks November 2022, the public release of ChatGPT. All regressions include CNO4d, and CNO1d-by-year-month FE. Shaded areas report 95 percent confidence intervals based on standard errors clustered by CNO4d.

frequently include zero. One possible explanation is heterogeneity in the timing and intensity of AI adoption across occupations. A gradual diffusion of the technology would imply that average effects accumulate over time while remaining relatively imprecisely estimated. As shown below, the estimated responses also vary across some demographic and geographic subgroups, which may contribute to the observed imprecision. Panel (b) shows no comparable persistence in new contracts: the coefficients fluctuate around zero and are imprecisely estimated. Overall, the panels are consistent with a gradual increase in the stock of registered unemployment that is more pronounced

in occupations with higher AI exposure, rather than with a discrete contraction in the monthly flow of new hiring. These aggregate patterns alone cannot determine whether the increase in registered unemployment is driven by more workers losing their jobs or by unemployed workers taking longer to find employment.

The estimated pre-treatment coefficients fluctuate around zero and display no clear monotonic pattern or systematic drift before the public release of ChatGPT. Despite the absence of a clear visual pattern, formal tests of the joint significance and flatness of the pre-treatment coefficients reject over some event windows. For registered unemployment, the joint-nullity and flatness tests reject over event times -21 through -2 (both $p = 0.003$), but not over the more recent window -10 through -2 ($p = 0.367$ and 0.280). The full-period rejection is therefore driven by the earlier part of the pre-treatment period, which largely overlaps with the exceptional post-pandemic adjustment in 2021 discussed above. This motivates the complementary specification in Table 2 that excludes 2021. Importantly, excluding 2021 does not materially change the estimated post-release unemployment gradient, indicating that the magnitude of the post-release estimates is not particularly sensitive to the inclusion of the unusual labor-market adjustment observed during the first year of the sample. For new contracts, the corresponding tests reject over the full window ($p = 0.032$ and 0.045) and remain borderline over the recent window ($p = 0.049$ and 0.035).[20]

Overall, the event-study evidence is consistent with a relative increase in registered unemployment among more AI-exposed occupations following the public release of ChatGPT. At the same time, the rejection of the joint pre-treatment tests over the full pre-treatment period suggests that this interpretation should be viewed with some caution, even though the estimated pre-treatment coefficients display no clear systematic pattern and the more recent pre-treatment window does not reject for registered unemployment. As shown below, several of the age- and gender-specific estimates exhibit more stable pre-treatment dynamics, suggesting that part of the irregularity in the aggregate estimates may reflect the pooling of groups with heterogeneous labor-market adjustment.

Table 2 summarizes the post-treatment path in two periods. The adjustment period covers November 2022 through November 2024, when firms were learning about and beginning to deploy generative AI. The later period covers December 2024 through March 2026, when adoption had more time to affect staffing and job search.[21]

Estimates in column 1, which only includes CNO4d FE and year-month FE, imply 1.9 percent more registered unemployment during the adjustment period and 3.8 percent more in the later period for a 10 percentage-point increase in exposure. Restricting identification to within-CNO1d comparisons attenuates the estimates to 1.2 and 1.7 percent in column 2. The adjustment-period coefficient is statistically significant at the 5 percent level; the later coefficient is positive but less precisely estimated. The hypothesis of equal effects across the two periods cannot be rejected

[20] Appendix Table A.2 reports pre-treatment diagnostics for alternative event windows corresponding to the specifications reported in Table 2.

[21] The table estimates $Y_{jt} = \alpha_j + \lambda_t + \theta_{g(j),t} + \beta_A(A_j/0.10)\mathbf{1}\{0 \le e_t \le 24\} + \beta_L(A_j/0.10)\mathbf{1}\{25 \le e_t \le 40\} + \varepsilon_{jt}$, with and without $\theta_{g(j),t}$ CNO1d-by-month effects.

Table 2: Impact of AI exposure on labor market outcomes

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
AI exposure × adjustment period	0.019***	0.012**	0.012**	0.033***	0.014	0.004
	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.038***	0.017	0.016	0.029***	0.013	0.003
	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)	(0.010)
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No
p-value: equal phase effects	0.001	0.486	0.486	0.488	0.807	0.806
Pre-treatment joint-null p-value	< 0.001	0.003	0.367	< 0.001	0.032	0.049
Observations	31,122	31,122	25,600	30,787	30,786	25,317

Notes: Entries are marginal effects from occupation-month regressions. The adjustment period covers November 2022 through November 2024 (event times 0–24); the later period covers December 2024 through March 2026 (event times 25–40). All available pre-treatment months form the omitted category. The dependent variables are logarithms. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021, when pandemic recovery, employment-retention policies, and unusually high telework intensity may have generated differential occupational trends. Exposure is divided by 0.10, so each coefficient corresponds to a 10 percentage-point increase and can be interpreted approximately as a proportional change in the outcome. Standard errors are clustered by CNO4. The phase-equality row tests whether the adjustment- and later-period coefficients are equal. The pre-treatment joint-null row tests whether all available pre-treatment event-study coefficients equal zero: event times -21 through -2 in columns that include 2021 and event times -10 through -2 in columns 3 and 6. October 2022 is omitted. These are marginal effects, not ATTs. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

($p = 0.486$). Column 3 excludes 2021, when the uneven recovery from the pandemic, employment-retention policies, and unusually high telework intensity may have generated occupational trends unrelated to generative AI. The unemployment estimates remain similar, at 1.2 and 1.6 percent. The attenuation after adding CNO1d-by-month effects is economically informative: part of the unconditional relationship reflects shocks shared by broad occupational families rather than variation between comparable CNO4d occupations.

The contract estimates display a similar attenuation. Coefficients in column 4 are 3.3 and 2.9 percent, while the preferred estimates in column 5 are 1.4 and 1.3 percent. Neither preferred estimate is statistically distinguishable from zero, and the two periods are nearly identical ($p = 0.807$). Excluding 2021 in column 6 reduces the estimates to 0.4 and 0.3 percent, providing no evidence of a persistent contract response.

AI exposure appears to be associated with an accumulation of registered unemployment without a corresponding broad contraction in hiring flows. Taken together, these results point toward adjustment through unemployment inflows, exits, or duration, although the available data do not allow us to distinguish among these margins. In particular, the absence of a persistent contract response does not imply unchanged exits from unemployment, since contract registrations are not linked to individual unemployment spells and are not a one-to-one measure of exits from registered unemployment.

4.2 Heterogeneity by age and gender

Table 3 examines heterogeneity by age. In the preferred specification, a 10 percentage-point increase in AI exposure is associated with 1.7 percent more registered unemployment among workers aged 30–39 during the adjustment period and 3.4 percent more thereafter. Only the later-period estimate is statistically significant. The corresponding estimates are 1.5 and 2.2 percent for workers under age 30 and 0.9 and 1.2 percent for workers aged 40 or older; none is individually distinguishable from zero. Although the point estimates are largest for workers aged 30–39, the formal cross-group tests provide only suggestive evidence of age heterogeneity. Appendix Table B.2 does not reject equality across all three age groups in the later period ($p = 0.105$). Pairwise tests reject equality between workers aged 30–39 and those aged 40 or older ($p = 0.035$), but not between workers aged 30–39 and workers under age 30 ($p = 0.320$). Thus, the estimates support a larger response among workers aged 30–39 relative to older workers, but not the stronger conclusion that the response is uniquely concentrated in this age group.

The results for new contracts exhibit a different age profile. For workers under age 30, the preferred estimates imply increases of 2.2 percent during the adjustment period and 2.6 percent thereafter, both statistically significant. The corresponding estimates are 1.3 and 1.7 percent for workers aged 30–39 and close to zero for workers aged 40 or older. When the two younger groups are pooled, the estimated AI-exposure gradient differs significantly between workers under 40 and

Table 3: Heterogeneity in the effects of AI exposure

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Age: under 30						
AI exposure × adjustment period	0.008	0.015	0.015*	0.033***	0.022**	0.011
	(0.009)	(0.010)	(0.009)	(0.009)	(0.010)	(0.008)
AI exposure × later period	0.028**	0.022	0.021	0.027**	0.026**	0.015
	(0.014)	(0.016)	(0.015)	(0.011)	(0.013)	(0.013)
p-value: equal phase effects	0.005	0.461	0.466	0.290	0.519	0.513
Observations	30,493	30,493	25,035	30,160	30,139	24,788
Pre-trend joint-null p-value	0.036	0.307	0.668	< 0.001	0.061	0.045
Panel B. Age: 30–39						
AI exposure × adjustment period	0.037***	0.017*	0.018**	0.030***	0.013	0.005
	(0.008)	(0.009)	(0.009)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.075***	0.034**	0.036**	0.031***	0.017	0.010
	(0.014)	(0.015)	(0.015)	(0.010)	(0.011)	(0.010)
p-value: equal phase effects	0.000	0.023	0.024	0.809	0.439	0.432
Observations	30,776	30,776	25,316	30,270	30,255	24,861
Pre-trend joint-null p-value	0.006	0.594	0.486	< 0.001	0.128	0.019
Panel C. Age: 40 or older						
AI exposure × adjustment period	0.017***	0.009	0.007	0.028***	0.003	-0.007
	(0.004)	(0.005)	(0.005)	(0.009)	(0.009)	(0.007)
AI exposure × later period	0.032***	0.012	0.011	0.026**	-0.001	-0.010
	(0.009)	(0.011)	(0.011)	(0.010)	(0.011)	(0.010)
p-value: equal phase effects	0.006	0.604	0.604	0.615	0.538	0.547
Observations	31,099	31,099	25,583	30,442	30,435	25,014
Pre-trend joint-null p-value	< 0.001	0.034	0.746	< 0.001	0.077	0.019
Panel D. Gender: men						
AI exposure × adjustment period	0.017***	0.008	0.008	0.032***	0.014	0.003
	(0.005)	(0.007)	(0.006)	(0.009)	(0.009)	(0.007)
AI exposure × later period	0.035***	0.016	0.015	0.032***	0.016	0.005
	(0.010)	(0.012)	(0.012)	(0.011)	(0.012)	(0.011)
p-value: equal phase effects	0.003	0.312	0.304	0.977	0.819	0.817
Observations	31,058	31,058	25,550	30,535	30,530	25,118
Pre-trend joint-null p-value	< 0.001	0.135	0.433	< 0.001	0.016	0.038
Panel E. Gender: women						
AI exposure × adjustment period	0.014**	0.010	0.008	0.022**	0.011	0.003
	(0.006)	(0.007)	(0.006)	(0.009)	(0.009)	(0.007)
AI exposure × later period	0.030***	0.010	0.008	0.010	0.009	0.001
	(0.010)	(0.012)	(0.012)	(0.011)	(0.012)	(0.010)
p-value: equal phase effects	0.010	0.995	0.988	0.024	0.786	0.789
Observations	30,813	30,813	25,339	30,207	30,188	24,807
Pre-trend joint-null p-value	0.014	0.035	0.856	0.003	0.115	0.077
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

Notes: Each panel reports subgroup-specific marginal effects for the adjustment period (event times 0–24) and later period (event times 25–40), relative to all pre-treatment months. Age categories are under 30, 30–39, and 40 or older. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Exposure is measured in 10 percentage-point units, and standard errors are clustered by CNO4. Equality rows test whether the two phase effects are equal. Pre-treatment rows test the joint null over all available event-study coefficients: event times -21 through -2 when 2021 is included and -10 through -2 otherwise. October 2022 is omitted. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

those aged 40 or older in both periods. These estimates indicate that exposure is associated with greater contracting activity among younger workers, especially those under age 30. They should not, however, be interpreted directly as evidence of net job creation or employment stability. The SEPE variable counts contracts rather than workers or employment spells, so an increase may reflect additional hiring, repeated short contracts, or greater labor-market churn. The churn interpretation is consistent with German evidence that AI exposure raises employer-to-employer mobility without increasing occupational switching (Gathmann et al., 2024), although that evidence refers to pre-generative AI and to worker-level data. Similarly, the joint increase in contracts and registered unemployment cannot distinguish increased unemployment inflows from slower exits out of unemployment.

A pre-treatment composition exercise does not support the hypothesis that the 30–39 result arises because these workers are disproportionately concentrated in more exposed occupations. Each age panel contains the same 502 CNO4 occupations, making the unweighted exposure distribution identical across age groups. Weighting occupations by pre-treatment registered unemployment produces mean exposure scores of 0.113 for workers under age 30, 0.119 for workers aged 30–39, and 0.118 for workers aged 40 or older. When occupations are weighted by new contracts, the corresponding means are 0.075, 0.066, and 0.051. The effective regression samples also have nearly identical mean exposure across age groups. Taken together, the estimates are suggestive of different labor-market responses across age groups within similarly exposed occupations, rather than simply reflecting differential sorting into exposed occupations. Spain's comparatively protracted school-to-work transition offers one possible interpretation: workers in their thirties may remain relatively early in their career progression and possess less tenure or employment protection than older workers. This interpretation remains provisional because the data do not observe tenure, contract type, AI adoption, unemployment inflows, or unemployment duration. The subgroup pre-treatment diagnostics and dynamic estimates are reported in Appendix Table B.1 and Figures B.1–B.2.[22]

The gender-specific estimates show substantially less heterogeneity. In the preferred specification, the registered-unemployment gradients are 0.8 and 1.6 percent for men and 1.0 percent in both periods for women. The corresponding contract estimates are also similar across genders, and none is individually statistically significant. We therefore find no clear evidence of differential adjustment by gender.

We also examine heterogeneity by occupational feminization and geography. The unemployment gradient is stronger among less-feminized occupations and in Madrid and Barcelona, but these patterns are accompanied by less favorable pre-treatment diagnostics. We therefore treat these results as suggestive and report the full analyses in Sections O.E and O.F of the Online Appendix.

[22]SEPE omits its CNO4-by-age breakdown for May 2024. We recover uniquely identified integer cells by inverting each June level and the corresponding June-over-May percentage change, with the observed May CNO4 total used as an adding-up validation. Cells that are not uniquely identified remain missing.

4.3 Robustness and additional analyses

Online Appendix Table O.D.1 examines whether the baseline result is an artifact of outcome coding or exposure construction. Replacing $\log(Y)$ with $\log(Y+1)$ leaves the preferred unemployment estimates almost unchanged: 1.2 percent in the adjustment period and 1.7 percent later. Cosine-weighted exposure produces somewhat larger estimates of 1.8 and 2.9 percent. Online Appendix Table O.D.2 and Figures O.D.1–O.D.2 show that the identifying diagnostics remain the principal limitation across measures.

No single CNO1 family drives the result: the leave-one-family-out estimates in Online Appendix Figure O.D.3 remain positive and close to the full-sample estimate. As a final outcome check, Online Appendix Table O.D.3 replaces monthly contracts with registrations accumulated over the current and preceding 11 months. The preferred estimates imply increases of 1.5 percent during the adjustment period and 0.9 percent later, but neither is statistically distinguishable from zero. Online Appendix Figure O.D.4 similarly shows no persistent post-treatment divergence in the cumulative flow. The trailing measure reduces high-frequency volatility but mechanically smooths and delays changes because adjacent observations share 11 component months. It remains a cumulative hiring flow, rather than a stock of active contracts, and is therefore best interpreted as a robustness check on the contract results.

Additional exposure checks assess whether the baseline result depends on Anthropic's observed-use score. The first uses the BLS classification, which assigns occupations to low, moderate, high, or very high AI-exposure categories.[23] Online Appendix Table O.D.4 shows that this classification is related to, but not identical to, the baseline ranking. Online Appendix Table O.D.5 provides mixed evidence: relative to low-exposure occupations, the unemployment response is concentrated in the very-high category rather than increasing monotonically across all four groups. The associated pre-treatment diagnostics also remain less favorable than in the preferred baseline (see Online Appendix Figure O.D.5).

The second check uses the language-model AIOE score in Felten et al. (2021) and Felten et al. (2023), a capability-based measure whose occupational coverage is narrower than that of the Anthropic score.[24] Online Appendix Table O.D.6 replaces Anthropic exposure with LM AIOE. The capability-based estimates retain a positive unemployment gradient and small contract responses, broadly matching the direction of the preferred baseline, although their magnitudes are not mechanically comparable because the scores use different scales. Online Appendix Figure O.D.6 reports the corresponding dynamics.

A third check uses the local TEV decision model to assign the highest-probability U.S. occupation, construct a probability-weighted exposure conditional on a fixed 24-candidate cosine shortlist, and

[23] The BLS measure combines evidence on the potential use of AI with observed use and is mapped from detailed U.S. occupations to Spanish CNO4 occupations through the fixed crosswalk.

[24] LM AIOE measures the overlap between occupational tasks and language-model capabilities. We map it through the same SOC-to-CNO4 crosswalk and express it as a percentile rank among matched occupations.

predict exposure directly from Spanish occupational information. Online Appendix Table O.D.7 reports the results. In the preferred within-CNO1 specification, the highest-probability and shortlist-weighted measures imply unemployment increases of approximately 1.6 and 1.7 percent during the adjustment period and 2.2 and 2.6 percent later, per 10 percentage points of the respective exposure measure. The direct score gives larger coefficients, approximately 6.0 and 14.2 percent, on a different model-predicted scale. Contract estimates are small and imprecise for all three measures. The full-sample pre-treatment joint tests reject for both outcomes under each measure, and excluding 2021 does not resolve every diagnostic. These checks therefore show a positive unemployment association under alternative constructions, while retaining measurement and identification concerns. Online Appendix Figures O.D.7 and O.D.8 report the dynamics.

Finally, we examine whether the baseline findings depend on potentially misleading occupational matches. Online Appendix O.D uses TEV to screen the baseline crosswalk, reuses its fixed-shortlist matching output for flagged cases, and applies a documented assistant-led source review to all 502 occupations. The resulting crosswalk changes 123 baseline codes, including 56 explicit review corrections. Online Appendix Table O.D.8 implies unemployment increases of approximately 1.6 percent during the adjustment period and 2.4 percent later per 10 percentage points of reviewed-match observed exposure; both coefficients are statistically significant at the 5 percent level. Contract estimates remain close to zero and imprecise. Full-sample pre-treatment tests reject for both outcomes, so this exercise does not resolve the identifying concerns. Retained cross-national proxies and the absence of independent human adjudication also limit its interpretation as a matching-quality check.

Alternative estimators, task bundling, and aggregate calibration. To conserve space, we also defer three additional exercises to the Online Appendix. First, Online Appendix O.G reports CDiD and SDiD estimates that construct the counterfactual differently from the preferred within-CNO1 TWFE specification. Both alternatives point to a positive unemployment response, although the estimated magnitudes are larger and the estimands and identifying assumptions differ from the baseline; the contract results are more method-dependent and generally less precise. We therefore treat these estimators as complementary robustness exercises rather than replacements for the preferred design.

Second, Online Appendix O.H examines tier 1 jobs with separable outputs relative to pooled strongly bundled and relational jobs (tiers 2 and 3), using the local TEV classification. The preferred within-CNO1 estimates show higher registered unemployment in tier 1 and fewer new contracts, although the later contract difference is imprecise. The full-sample pre-treatment joint tests reject for both unemployment and contracts ($p = 0.010$ and $p = 0.007$). Excluding 2021 leaves a rejection for unemployment ($p = 0.005$) while the contract test does not reject ($p = 0.226$). These diagnostics warrant interpreting the estimates as differences in occupational adjustment rather than evidence that stronger task bundling protects jobs.

Third, Online Appendix O.I presents an illustrative partial-equilibrium calibration that translates the occupation-level estimates into an aggregate unemployment magnitude. The calculation is intended only to provide an order of magnitude: it applies the estimated exposure gradient across occupations and abstracts from spillovers, general-equilibrium adjustment, and any common effect of AI that does not vary with occupational exposure. For these reasons, we do not interpret the exercise as identifying the aggregate causal effect of generative AI on unemployment in Spain.

5 Conclusion

This paper provides early evidence on whether labor-market outcomes in Spain have begun to diverge across occupations according to their exposure to generative AI. The preferred design compares differently exposed CNO4 occupations within the same CNO1 family and month following ChatGPT's public release. A 10 percentage-point increase in exposure is associated with 1.2 percent more registered unemployment during the adjustment period and 1.7 percent more thereafter. The later estimate is imprecise, the two period effects are not statistically different, and new contracts show no precise persistent response. AI exposure appears to be associated with an accumulation of registered unemployment without a corresponding negative exposure gradient in new contract registrations. Taken together, these outcomes point toward adjustment through unemployment inflows, exits, or duration, although the available data do not allow us to distinguish among these margins.

The age-specific estimates provide suggestive evidence of heterogeneity. Individuals aged 30–39 exhibit the largest later-period unemployment gradient, while those under 30 show a more positive contract response, although the cross-group tests do not support a sharp distinction across all age categories. Spain's relatively late labor-market entry and prolonged early-career adjustment offer one possible interpretation. The under-30 category combines established workers with individuals still entering employment, whereas the 30–39 group may better capture workers who are already attached to an occupation but remain relatively mobile and have accumulated less tenure than older workers.[25] This interpretation remains suggestive because the data do not observe labor-market entry, tenure, or worker transitions, and the cross-group tests do not establish a sharp age threshold.

The results also illustrate why occupational exposure and realized adoption should be distinguished. November 2022 created broad access to generative AI, but Spanish evidence indicates that firm adoption remained limited and often experimental well into 2024, while frequent individual use continued to increase through 2025. The gradual post-release divergence in registered unemployment is consistent with this diffusion path, but it does not establish that firm adoption caused the

[25] Quintini and Martin (2014), using primarily 2011 labor-force data, estimate that the age at which 50 percent of young people are employed and no longer enrolled in education was 26.7 in Spain, higher than in the other seven advanced economies in their sample. More recent OECD evidence continues to document comparatively low youth employment and participation rates and difficult school-to-work transitions in Spain (OECD, 2023).

occupational differences. The exposure score is time-invariant and external to the Spanish labor-market outcomes, but it is derived partly from observed AI use in the Anthropic data rather than from adoption by individual Spanish firms.

The preferred estimates point to a gradual relative accumulation of registered unemployment in more AI-exposed occupations following the public release of ChatGPT. This pattern is robust to excluding the unusual labor-market adjustment of 2021 and is qualitatively consistent with alternative estimators. At the same time, the rejection of the pre-treatment diagnostics over the full pre-period remains an important limitation of the identifying design. The results therefore speak most directly to differential labor-market adjustment across occupations with different AI exposure, rather than to the aggregate effect of generative AI on Spain's unemployment rate. Linking occupational exposure to firm adoption, worker transitions, unemployment duration, and an occupation-level labor-force denominator is the most important next step for distinguishing displacement from slower unemployment exits and broader labor-market reallocation.

References

Acemoglu, D. (2025). The simple macroeconomics of AI. *Economic Policy*, 40(121):13–58.

Acemoglu, D. and Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30.

Autor, D. H., Levy, F., and Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *Quarterly Journal of Economics*, 118(4):1279–1333.

Bick, A., Blandin, A., Deming, D. J., and Schumacher, T. R. (2026). What work does generative ai do? Working Paper 35677, National Bureau of Economic Research.

Brynjolfsson, E., Chandar, B., and Chen, R. (2025). Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence. Working paper, Stanford Digital Economy Lab.

Callaway, B., Goodman-Bacon, A., and Sant'Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. NBER Working Paper 32117, National Bureau of Economic Research.

Cruces, G., Fernandez Meijide, D., Galiani, S., Galvez, R. H., and Lombardi, M. (2026). Does generative AI narrow education-based productivity gaps? evidence from a randomized experiment. NBER Working Paper 34851, National Bureau of Economic Research.

Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702):1306–1308.

Facius, D. and Iacono, R. (2026). Labor market consequences of generative AI: Early evidence from norway. CESifo Working Paper 12752, CESifo.

Felten, E. W., Raj, M., and Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12):2195–2217.

Felten, E. W., Raj, M., and Seamans, R. (2023). Occupational heterogeneity in exposure to generative AI. Working paper, SSRN.

Freund, L. B. and Mann, L. F. (2026). Job transformation, specialization, and the labor market effects of AI. IZA Discussion Paper 18565, IZA Institute of Labor Economics.

Funcas (2026). III encuesta funcas sobre inteligencia artificial: El uso frecuente de ChatGPT en españa sube del 4 al 28 por ciento entre 2023 y 2025. Press release and survey summary.

Garbers, J. and Gregory, T. (2026). The diffusion of artificial intelligence across firms: Evidence from europe. IZA Discussion Paper 18434, IZA Institute of Labor Economics.

Garicano, L., Li, J., and Wu, Y. (2026a). *Messy jobs: the work that AI cannot reach*. Upriver Press.

Garicano, L., Li, J., and Wu, Y. (2026b). Weak bundle, strong bundle: How ai redraws job boundaries. Technical report, Centre for Economic Policy Research.

Gathmann, C., Grimm, F., and Winkler, E. (2024). AI, task changes in jobs, and worker reallocation. IZA Discussion Paper 17554, IZA Institute of Labor Economics.

Hampole, M., Papanikolaou, D., Schmidt, L. D., and Seegmiller, B. (2025). Artificial intelligence and the labor market. Working Paper 33509, National Bureau of Economic Research.

Hui, X., Reshef, O., and Zhou, L. (2024). The short-term effects of generative artificial intelligence on employment: Evidence from an online labor market. *Organization Science*, 35(6):1977–1989.

Humlum, A. and Vestergaard, E. (2025). Still waters, rapid currents: Early labor market transformation under generative AI. NBER Working Paper 33777, National Bureau of Economic Research.

ILO (2026). Global employment trends for youth 2026: Back to the future. Technical report, International Labour Office, Geneva.

Izquierdo, M., Fernandez Cerezo, A., and Hidalgo, I. (2025). La adopción de la inteligencia artificial en las empresas españolas: un primer análisis basado en la EBAE. Banco de España, Boletín Económico 2025/T2, Artículo 06.

Lodefalk, M., Lothman, L., Koch, M., and Engberg, E. (2026). Same storm, different boats: Generative AI and the age gradient in hiring. Working Paper 2/2026, Orebro University School of Business.

Massenkoff, M. and McCrory, P. (2026). Labor market impacts of AI: A new measure and early evidence. Anthropic research report: https://www.anthropic.com/research/labor-market-impacts. Accessed September 18, 2026.

Ministerio de Trabajo y Economia Social (2023). Movimiento laboral registrado: notas metodologicas. https://www.mites.gob.es/estadisticas/MLR/Notas_metodologicas.pdf. Accessed September 18, 2026.

Noy, S. and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192.

Observatorio Nacional de Tecnología y Sociedad (2023). Uso de inteligencia artificial y big data en las empresas españolas. Brújula ONTSI.

Observatorio Nacional de Tecnología y Sociedad (2024). Indicadores de uso de inteligencia artificial en las empresas españolas. ONTSI report.

OECD (2023). *OECD Economic Surveys: Spain 2023*. OECD Publishing, Paris.

Pulito, G., Pytlikova, M., Schroeder, S., and Lodefalk, M. (2026). Who adopts AI? evidence on firms, technologies and workers. IZA Discussion Paper 18515, IZA Institute of Labor Economics.

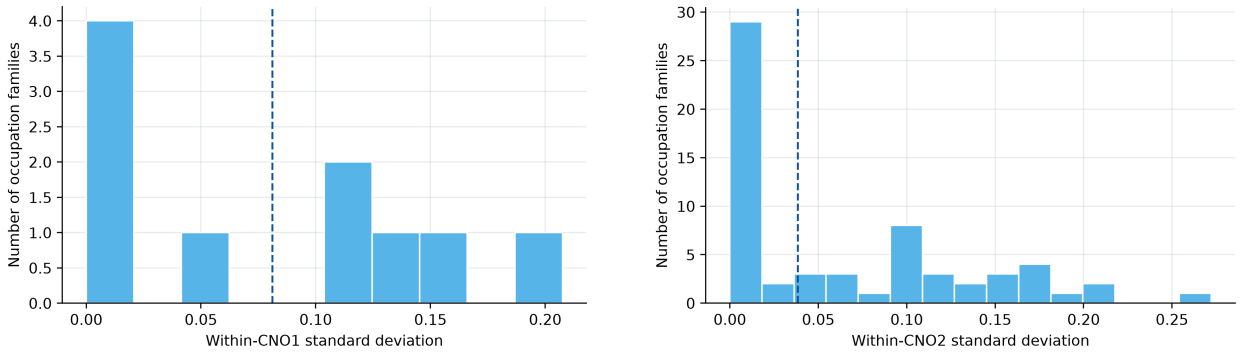
Quintini, G. and Martin, S. (2014). Same same but different: School-to-work transitions in emerging and advanced economies. Technical Report 154, OECD Publishing, Paris.

Servicio Público de Empleo Estatal (2026). Información del mercado de trabajo por ocupación. Administrative statistics.

Appendix

A Identification support and pre-trend diagnostics

Figure A.1: Within-family variation in AI exposure



(a) Within-CNO1 variation

(b) Within-CNO2 variation

Notes: The histograms report the distribution of the within-family standard deviation of nearest-neighbor AI exposure. Panel (a) groups CNO4 occupations by CNO1 family and panel (b) groups them by CNO2 family. The vertical dashed line marks the median family. Values near zero indicate limited identifying variation after the corresponding family-by-month FE are absorbed.

Table A.1: Within-family variation in AI exposure

Family level	Families	Median SD	Median range	Zero-range share
CNO1	10	0.081	0.283	10.0%
CNO2	62	0.038	0.083	33.9%

Notes: Statistics are computed across the 502 CNO4 occupations using the nearest-neighbor exposure score. A zero range means that all CNO4 occupations inside the family have the same exposure. Smaller within-family dispersion implies weaker identifying variation after family-by-month effects are absorbed.

Table A.2: Pre-treatment coefficient tests across baseline specifications

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Full available: Joint nullity	< 0.001	0.003	0.367	< 0.001	0.032	0.049
Full available: Equal coefficients	< 0.001	0.003	0.280	< 0.001	0.045	0.036
Early: -21 to -10: Joint nullity	< 0.001	0.027	–	< 0.001	0.307	–
Early: -21 to -10: Equal coefficients	< 0.001	0.021	–	< 0.001	0.252	–
Recent: -10 to -2: Joint nullity	< 0.001	0.367	0.367	< 0.001	0.049	0.049
Recent: -10 to -2: Equal coefficients	< 0.001	0.280	0.280	< 0.001	0.035	0.036
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

Notes: Entries are p-values from Wald tests of pre-treatment event-study coefficients. Joint nullity tests whether all coefficients in the indicated window equal zero; the equality test allows a common nonzero level. Columns 1 and 4 are the benchmark with CNO4 and year-month FE. Columns 2 and 5 add CNO1-by-year-month FE. Columns 3 and 6 use the latter specification but exclude 2021. The full available window is event times -21 through -2 when 2021 is included and -10 through -2 otherwise; the early window is consequently unavailable in columns 3 and 6. October 2022 (event time -1) is omitted. Standard errors are clustered by CNO4.

B Heterogeneity by age and gender

Table B.1: Pre-treatment diagnostics for heterogeneity estimates

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Age: under 30						
Full available: Joint nullity	0.036	0.307	0.668	< 0.001	0.061	0.045
Full available: Equal coefficients	0.033	0.274	0.614	< 0.001	0.063	0.050
Early: -21 to -10: Joint nullity	0.004	0.119	–	< 0.001	0.196	–
Early: -21 to -10: Equal coefficients	0.003	0.115	–	< 0.001	0.172	–
Recent: -10 to -2: Joint nullity	0.673	0.661	0.668	< 0.001	0.053	0.045
Recent: -10 to -2: Equal coefficients	0.720	0.605	0.614	< 0.001	0.056	0.050
Panel B. Age: 30–39						
Full available: Joint nullity	0.006	0.594	0.486	< 0.001	0.128	0.019
Full available: Equal coefficients	0.004	0.556	0.482	0.001	0.100	0.011
Early: -21 to -10: Joint nullity	0.002	0.369	–	0.001	0.487	–
Early: -21 to -10: Equal coefficients	0.002	0.336	–	0.022	0.475	–
Recent: -10 to -2: Joint nullity	0.029	0.475	0.486	0.003	0.020	0.019
Recent: -10 to -2: Equal coefficients	0.029	0.464	0.482	0.006	0.012	0.011
Panel C. Age: 40 or older						
Full available: Joint nullity	< 0.001	0.034	0.746	< 0.001	0.077	0.019
Full available: Equal coefficients	< 0.001	0.025	0.658	< 0.001	0.077	0.012
Early: -21 to -10: Joint nullity	< 0.001	0.047	–	< 0.001	0.322	–
Early: -21 to -10: Equal coefficients	< 0.001	0.038	–	0.001	0.257	–
Recent: -10 to -2: Joint nullity	< 0.001	0.746	0.746	< 0.001	0.019	0.019
Recent: -10 to -2: Equal coefficients	< 0.001	0.658	0.658	< 0.001	0.012	0.012
Panel D. Gender: men						
Full available: Joint nullity	< 0.001	0.135	0.433	< 0.001	0.016	0.038
Full available: Equal coefficients	< 0.001	0.105	0.346	< 0.001	0.018	0.032
Early: -21 to -10: Joint nullity	0.003	0.293	–	< 0.001	0.087	–
Early: -21 to -10: Equal coefficients	0.005	0.291	–	< 0.001	0.060	–
Recent: -10 to -2: Joint nullity	< 0.001	0.428	0.433	< 0.001	0.037	0.038
Recent: -10 to -2: Equal coefficients	0.003	0.342	0.346	< 0.001	0.031	0.032
Panel E. Gender: women						
Full available: Joint nullity	0.014	0.035	0.856	0.003	0.115	0.077
Full available: Equal coefficients	0.013	0.026	0.795	0.004	0.089	0.050
Early: -21 to -10: Joint nullity	0.014	0.023	–	0.012	0.098	–
Early: -21 to -10: Equal coefficients	0.009	0.015	–	0.033	0.086	–
Recent: -10 to -2: Joint nullity	0.093	0.856	0.856	0.004	0.073	0.077
Recent: -10 to -2: Equal coefficients	0.067	0.795	0.795	0.004	0.047	0.050
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

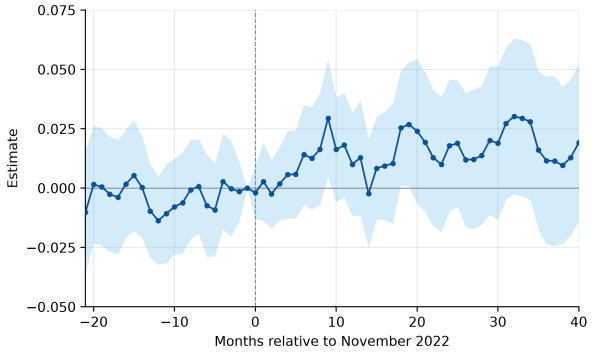
Notes: Entries are p-values from Wald tests of subgroup-specific pre-treatment event-study coefficients. Joint nullity tests whether all coefficients in the indicated window equal zero; equality tests whether they equal one another. Within every panel, columns 1 and 4 include CNO4 and year-month FE, columns 2 and 5 add CNO1-by-year-month FE, and columns 3 and 6 repeat the latter specification without 2021. The full available window is event times -21 through -2 when 2021 is included and -10 through -2 otherwise. October 2022 is omitted. Standard errors are clustered by CNO4.

Table B.2: Formal tests of age-group heterogeneity

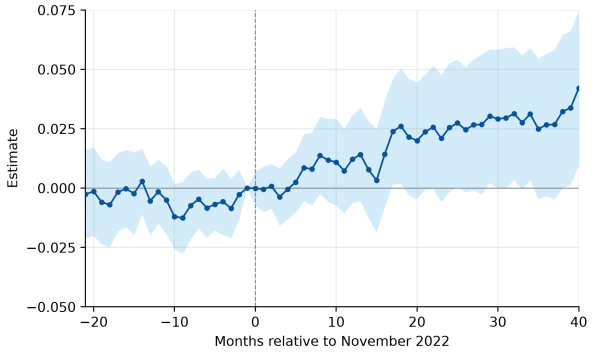
	# of registered unemployed		# of new contracts	
	Adjustment	Later	Adjustment	Later
All age-group effects equal	0.450	0.105	0.025	0.010
Ages 30–39 = under 30	0.897	0.320	0.102	0.274
Ages 30–39 = age 40 or older	0.232	0.035	0.050	0.007
Under 30 = age 40 or older	0.405	0.474	0.007	0.006
Pooled under 40 = age 40 or older	0.233	0.118	0.008	0.002
Observations	92,368		90,829	

Notes: Entries are p-values from Wald tests in the pooled age-by-CNO4 panel. The first four rows come from a regression that permits separate exposure gradients for workers under 30, ages 30–39, and age 40 or older in each post-treatment period. The final row comes from a companion regression that constrains the two younger groups to share a phase-specific gradient and tests that common under-40 gradient against the gradient for age 40 or older. Both regressions include age-by-CNO4 and age-by-CNO1-by-year-month FE. Standard errors allow arbitrary correlation across age groups within each of 502 CNO4 occupations.

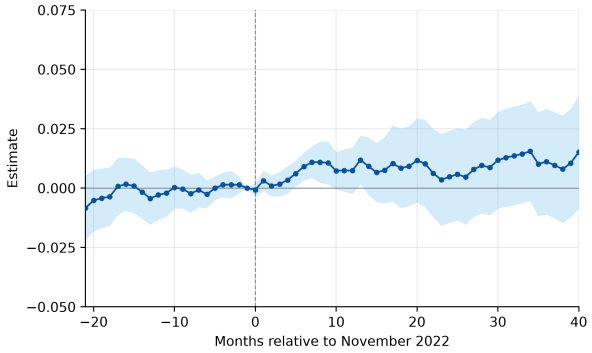
Figure B.1: Heterogeneity in registered unemployment



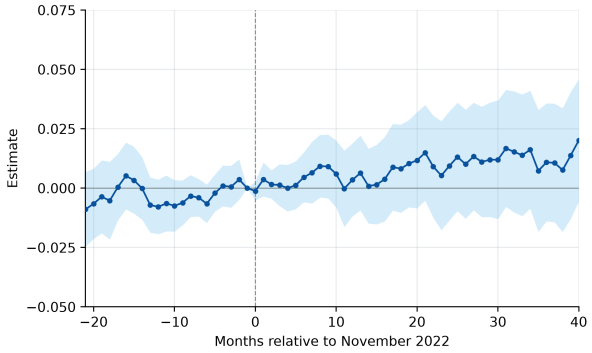
(a) Under 30



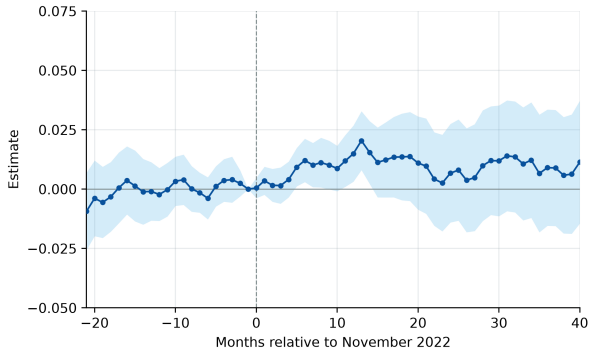
(b) Ages 30–39



(c) Aged 40 or older



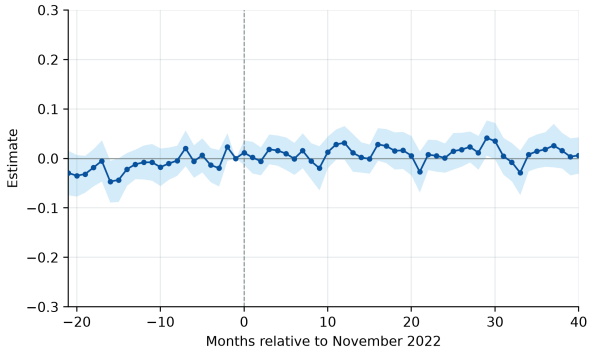
(d) Men



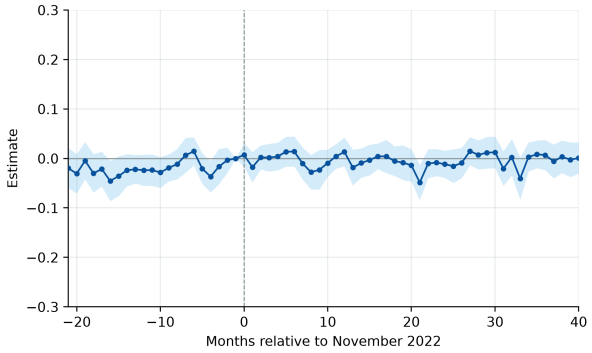
(e) Women

Notes: The figure reports subgroup-specific estimates from the preferred continuous-treatment specification, which includes CNO4 and CNO1-by-year-month FE. Each marker is the marginal effect of a 10 percentage-point increase in exposure at the indicated event month. October 2022 is the omitted reference month and the vertical dashed line marks November 2022. Shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4. All panels use the same vertical scale. SEPE omits the age breakdown for May 2024; the age panels recover uniquely implied integer cells from June levels and the published June-over-May changes. Of 6,024 original age-outcome cells, 55 remain unresolved and missing.

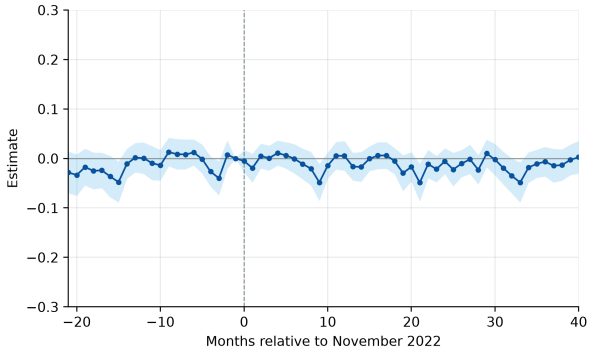
Figure B.2: Heterogeneity in new contracts



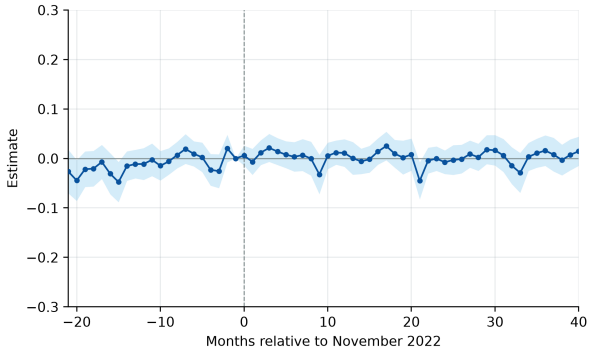
(a) Under 30



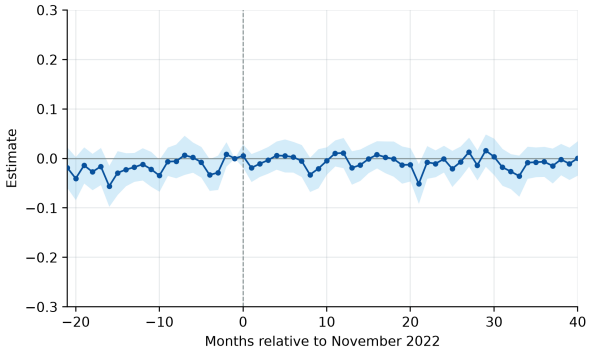
(b) Ages 30–39



(c) Aged 40 or older



(d) Men



(e) Women

Notes: The figure reports the same subgroup specifications and confidence intervals as Figure B.1, using the logarithm of newly registered contracts as the dependent variable. All panels use the same vertical scale.

Online Appendix

AI Exposure and Registered Unemployment in Spain:
Early Evidence from Occupation-Level Administrative Data

Diego González-González Nicolás González-Pampillón

Héctor Jiménez Portilla Javier Vázquez-Grenno

O.A SEPE and EPA comparison

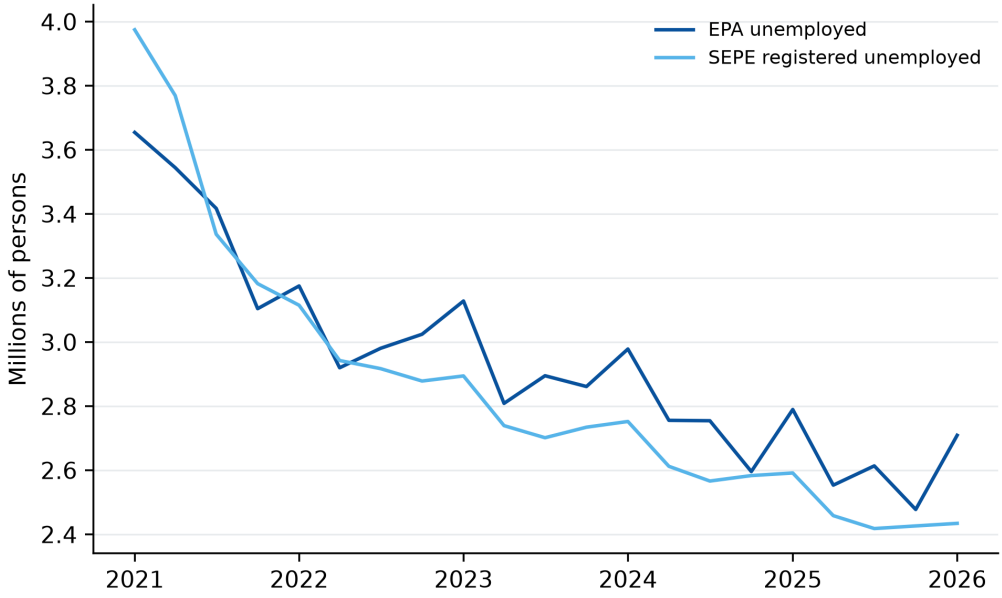


Figure O.A.1: Aggregate Unemployment in SEPE and EPA Data

O.B Face validity of the AI exposure mapping

Table O.B.1: Top and bottom 10 AI-exposure occupations in Spain and the global average

Spain (CNO4)			Global average (O*NET)		
Code	Occupation	Exposure	Code	Occupation	Exposure
Panel A. Top 10 occupations					
3820	Computer programmers	0.745	15-1251	Computer Programmers	0.745
4301	Data entry clerks	0.671	43-4051	Customer Service Representatives	0.701
3510	Sales agents and commercial representatives	0.628	43-9021	Data Entry Keyers	0.671
2613	Financial analysts	0.572	29-2072	Medical Records Specialists	0.667
2719	Software and multimedia analysts and designers	0.519	13-1161	Market Research Analysts and Marketing Specialists	0.648
4113	Statistical, financial, and banking office clerks	0.510	31-9094	Medical Transcriptionists	0.636
2729	Database and network specialists	0.486	41-4012	Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products	0.628
3814	Web technicians	0.480	15-1243	Database Architects	0.579
3812	IT user-support technicians	0.469	13-2051	Financial and Investment Analysts	0.572
2652	Public relations professionals	0.453	15-1253	Software Quality Assurance Analysts and Testers	0.519
Panel B. Bottom 10 occupations					
0011	Military officers	0.000	11-1031	Legislators	0.000
0012	Non-commissioned military officers	0.000	11-3012	Administrative Services Managers	0.000
0020	Military enlisted personnel	0.000	11-3111	Compensation and Benefits Managers	0.000
1111	Members of the executive and legislative branches	0.000	11-3121	Human Resources Managers	0.000
1112	Public administration executive staff	0.000	11-9013	Farmers, Ranchers, and Other Agricultural Managers	0.000
1212	Human resources directors	0.000	11-9051	Food Service Managers	0.000
1311	Directores de producción de explotaciones agropecuarias y forestales	0.000	11-9071	Gambling Managers	0.000
1312	Directores de producción de explotaciones pesqueras y acuícolas	0.000	11-9131	Postmasters and Mail Superintendents	0.000
1421	Directores y gerentes de restaurantes	0.000	11-9161	Emergency Management Directors	0.000
1422	Directores y gerentes de bares, cafeterías y similares	0.000	13-1032	Insurance Appraisers, Auto Damage	0.000

Notes: The table reports the ten occupations with the highest and lowest AI exposure after mapping the Anthropic AI exposure measure to Spanish CNO4d occupations. U.S. occupations correspond to the original Anthropic ranking.

Source: Own calculations based on SEPE, O*NET, and Massenkoff and McCrory (2026).

O.C Binary treatment

As an alternative to the continuous-treatment design, we classify occupations with AI exposure above 0.1169, the 75th percentile of the exposure distribution, as treated and retain occupations at or below the cutoff as controls. This specification assesses whether the baseline relationship is concentrated in the upper tail of the exposure distribution. It uses the same adjustment and later periods as the baseline analysis and reports the benchmark and preferred FE specifications under the alternative treatment definition.

Table O.C.1: Binary-treatment robustness checks

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbf{1}\{A_j > A_{75}\}$ × adjustment period	0.064***	0.039**	0.038**	0.092***	0.021	0.006
	(0.015)	(0.017)	(0.016)	(0.025)	(0.025)	(0.023)
$\mathbf{1}\{A_j > A_{75}\}$ × later period	0.124***	0.058*	0.057*	0.081**	0.013	-0.003
	(0.027)	(0.031)	(0.031)	(0.033)	(0.034)	(0.033)
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No
p-value: equal phase effects	0.000	0.307	0.307	0.571	0.691	0.692
Pre-treatment joint-null p-value	< 0.001	0.019	0.847	< 0.001	0.256	0.382
Observations	31,122	31,122	25,600	30,787	30,786	25,317

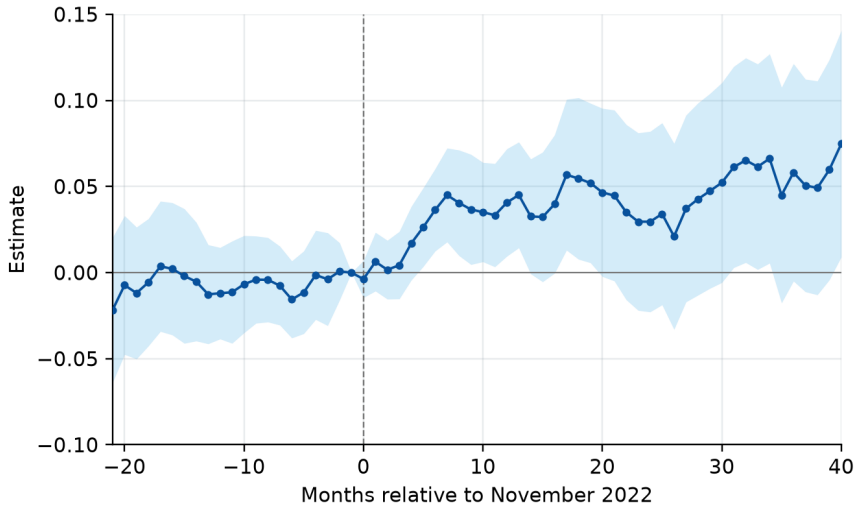
Notes: Treatment equals one for occupations with nearest-neighbor exposure above A_{75} and zero for occupations at or below it, where $A_{75} = 0.1169$ is the 75th percentile of AI exposure. The adjustment and later periods cover event times 0–24 and 25–40, respectively; all pre-treatment months form the omitted category. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Standard errors are clustered by CNO4. The phase-equality row tests whether the adjustment- and later-period coefficients are equal. The pre-treatment joint-null row tests whether all available pre-treatment event-study coefficients equal zero: event times -21 through -2 when 2021 is included and -10 through -2 otherwise. October 2022 is omitted. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table O.C.2: Pre-treatment diagnostics for the binary-treatment design

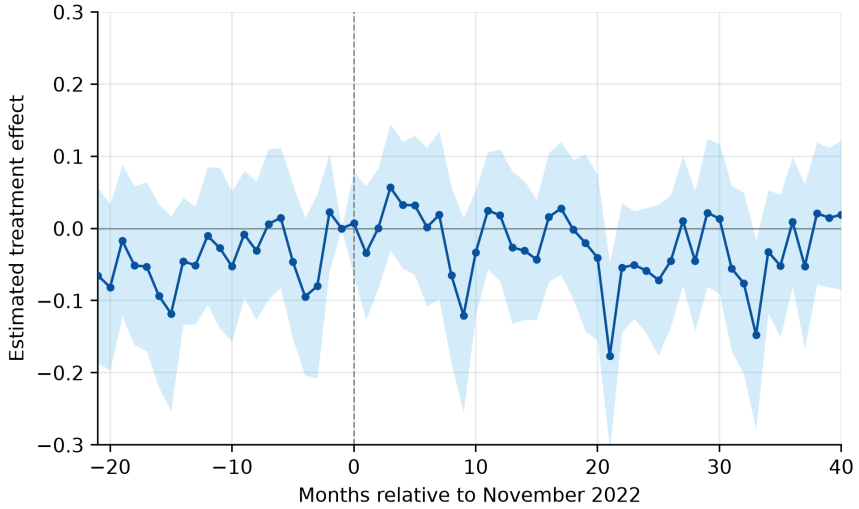
	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Full available: Joint nullity	< 0.001	0.019	0.847	< 0.001	0.256	0.382
Full available: Equal coefficients	< 0.001	0.018	0.786	< 0.001	0.211	0.299
Early: -21 to -10: Joint nullity	< 0.001	0.085	–	< 0.001	0.690	–
Early: -21 to -10: Equal coefficients	< 0.001	0.060	–	0.005	0.662	–
Recent: -10 to -2: Joint nullity	0.018	0.847	0.847	0.072	0.368	0.382
Recent: -10 to -2: Equal coefficients	0.148	0.786	0.786	0.064	0.285	0.299
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

Notes: Entries are p-values from Wald tests of pre-treatment event-study coefficients in the binary-treatment design. Treatment equals one for occupations with nearest-neighbor exposure above 0.1169 and zero otherwise. The nullity test evaluates whether all coefficients in the indicated window equal zero; the equality test evaluates whether they equal one another while allowing their common value to differ from zero. The benchmark includes CNO4 and year-month FE. The remaining specifications include CNO4 and CNO1-by-year-month FE and differ by sample period as indicated. The full available window is event times -21 through -2 when 2021 is included and -10 through -2 when it is excluded. The early window is therefore unavailable for the no-2021 specification. October 2022 (event time -1) is omitted.

Figure O.C.1: Event-study estimates under the binary treatment definition



(a) Number of registered unemployed



(b) Number of new contracts

Notes: The figure reports the preferred TWFE event-study specification under the binary treatment definition. Treatment equals one for occupations with exposure above 0.1169 and zero otherwise, so middle-exposure occupations remain in the control group. All specifications include CNO4 and CNO1-by-year-month FE, with October 2022 as the omitted month. Shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4.

Table O.C.1 reports larger unemployment coefficients than the continuous-treatment specification. In the preferred model, treated occupations experience 3.9 percent more registered unemployment during the adjustment period and 5.8 percent more in the later period relative to occupations at or below the cutoff. The later estimate is imprecise, and equality between the two period effects cannot be rejected. The corresponding contract estimates remain close to zero. Table O.C.2 and

Figure O.C.1 report the pre-treatment diagnostics and dynamic estimates. The binary results are consistent with a response concentrated in the upper exposure tail, but discretizing exposure discards variation in treatment intensity and changes the estimand from a marginal exposure gradient to a high-versus-lower-exposure contrast.

O.D Robustness checks

Table O.D.1: Robustness checks: alternative outcomes and exposure measures

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Baseline specification						
AI exposure × adjustment period	0.019***	0.012**	0.012**	0.033***	0.014	0.004
	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.038***	0.017	0.016	0.029***	0.013	0.003
	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)	(0.010)
Pre-treatment joint-null p-value	< 0.001	0.003	0.367	< 0.001	0.032	0.049
p-value: equal phase effects	0.001	0.486	0.486	0.488	0.807	0.806
Observations	31,122	31,122	25,600	30,787	30,786	25,317
Panel B. Alternative outcomes: log(Y+1)						
AI exposure × adjustment period	0.019***	0.012**	0.011**	0.030***	0.011	0.001
	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.037***	0.017	0.016	0.025**	0.008	-0.002
	(0.010)	(0.012)	(0.012)	(0.011)	(0.012)	(0.011)
Pre-treatment joint-null p-value	< 0.001	0.002	0.369	< 0.001	0.007	0.041
p-value: equal phase effects	0.001	0.485	0.485	0.401	0.690	0.690
Observations	31,124	31,124	25,602	31,124	31,124	25,602
Panel C. Alternative exposure: cosine-weighted						
AI exposure × adjustment period	0.025***	0.018***	0.019***	0.031***	0.012	0.003
	(0.005)	(0.007)	(0.007)	(0.009)	(0.010)	(0.007)
AI exposure × later period	0.049***	0.029**	0.030**	0.024**	0.007	-0.002
	(0.011)	(0.013)	(0.013)	(0.011)	(0.012)	(0.011)
Pre-treatment joint-null p-value	< 0.001	0.002	0.166	< 0.001	0.008	0.020
p-value: equal phase effects	0.000	0.177	0.177	0.258	0.491	0.491
Observations	31,122	31,122	25,600	30,787	30,786	25,317
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

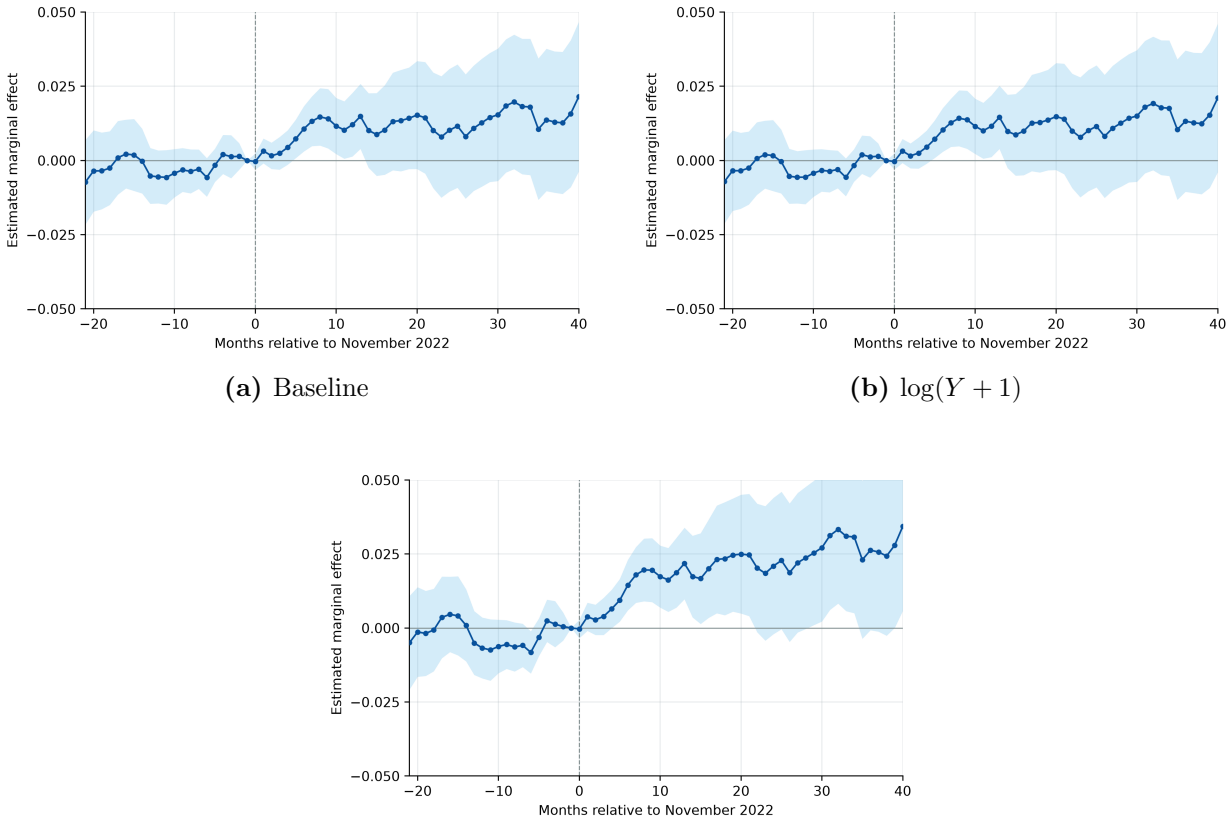
Notes: Each panel reports marginal effects for the adjustment period (event times 0–24) and the later period (event times 25–40), relative to all pre-treatment months. Panel A reproduces the baseline specification. Panel B replaces the logarithmic outcomes with $\log(Y + 1)$. Panel C uses the cosine-weighted exposure measure. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Exposure measures are divided by 0.10, so each coefficient corresponds to a 10 percentage-point increase in the indicated score. Standard errors are clustered by CNO4. Pre-treatment entries test the joint null that all available event-study coefficients equal zero: event times -21 through -2 when 2021 is included and -10 through -2 otherwise. Equality rows test whether adjustment- and later-period effects are equal. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table O.D.2: Pre-treatment diagnostics for alternative outcomes and exposure measures

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Baseline specification						
Full available: Joint nullity	< 0.001	0.003	0.367	< 0.001	0.032	0.049
Full available: Equal coefficients	< 0.001	0.003	0.280	< 0.001	0.045	0.036
Early: -21 to -10: Joint nullity	< 0.001	0.027	–	< 0.001	0.307	–
Early: -21 to -10: Equal coefficients	< 0.001	0.021	–	< 0.001	0.252	–
Recent: -10 to -2: Joint nullity	< 0.001	0.367	0.367	< 0.001	0.049	0.049
Recent: -10 to -2: Equal coefficients	< 0.001	0.280	0.280	< 0.001	0.035	0.036
Panel B. Alternative outcomes: log(Y+1)						
Full available: Joint nullity	< 0.001	0.002	0.369	< 0.001	0.007	0.041
Full available: Equal coefficients	< 0.001	0.003	0.282	< 0.001	0.009	0.026
Early: -21 to -10: Joint nullity	< 0.001	0.024	–	< 0.001	0.028	–
Early: -21 to -10: Equal coefficients	< 0.001	0.019	–	< 0.001	0.018	–
Recent: -10 to -2: Joint nullity	< 0.001	0.369	0.369	< 0.001	0.041	0.041
Recent: -10 to -2: Equal coefficients	< 0.001	0.282	0.282	< 0.001	0.026	0.026
Panel C. Alternative exposure: cosine-weighted						
Full available: Joint nullity	< 0.001	0.002	0.166	< 0.001	0.008	0.020
Full available: Equal coefficients	< 0.001	0.004	0.126	< 0.001	0.005	0.012
Early: -21 to -10: Joint nullity	< 0.001	0.010	–	< 0.001	0.110	–
Early: -21 to -10: Equal coefficients	< 0.001	0.010	–	< 0.001	0.102	–
Recent: -10 to -2: Joint nullity	< 0.001	0.165	0.166	< 0.001	0.019	0.020
Recent: -10 to -2: Equal coefficients	< 0.001	0.126	0.126	< 0.001	0.011	0.012
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

Notes: Entries are p-values from Wald tests of pre-treatment event-study coefficients. Joint nullity tests whether all coefficients in the indicated window equal zero; the equality test allows a common nonzero level. Columns 1 and 4 are the benchmark with CNO4 and year-month FE. Columns 2 and 5 add CNO1-by-year-month FE. Columns 3 and 6 use the latter specification but exclude 2021. The full available window is event times -21 through -2 when 2021 is included and -10 through -2 otherwise; the early window is consequently unavailable in columns 3 and 6. October 2022 (event time -1) is omitted. Standard errors are clustered by CNO4.

Figure O.D.1: Robustness event studies: registered unemployment



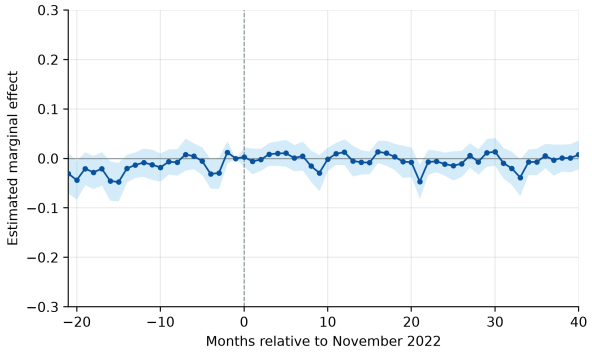
(a) Baseline

(b) $\log(Y+1)$

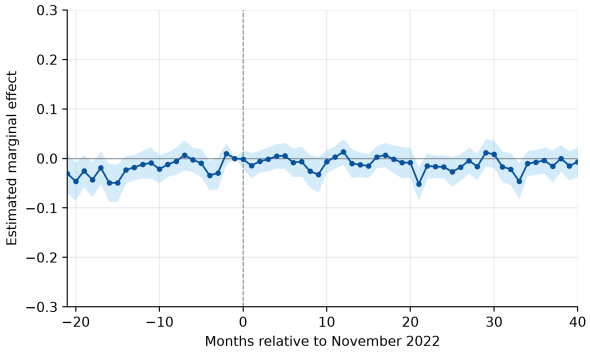
(c) Cosine-weighted exposure

Notes: The figure reports event-study estimates corresponding to Panels A–C of Table O.D.1. Panel (a) reproduces the preferred specification; panel (b) replaces the dependent variable with $\log(Y+1)$; panel (c) replace the nearest-neighbor exposure score with the cosine-weighted measure. Each regression includes CNO4 and CNO1-by-year-month FE. October 2022 is the omitted reference month, and the vertical dashed line marks November 2022. Shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4.

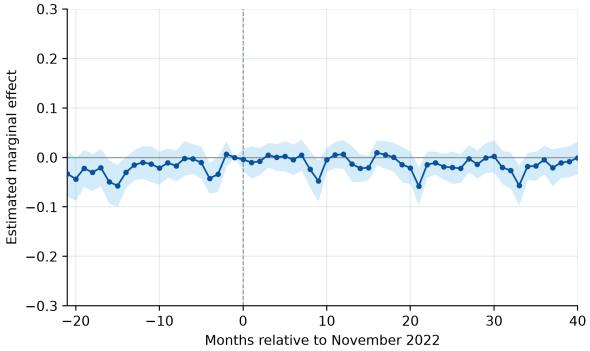
Figure O.D.2: Robustness event studies: new contracts



(a) Baseline



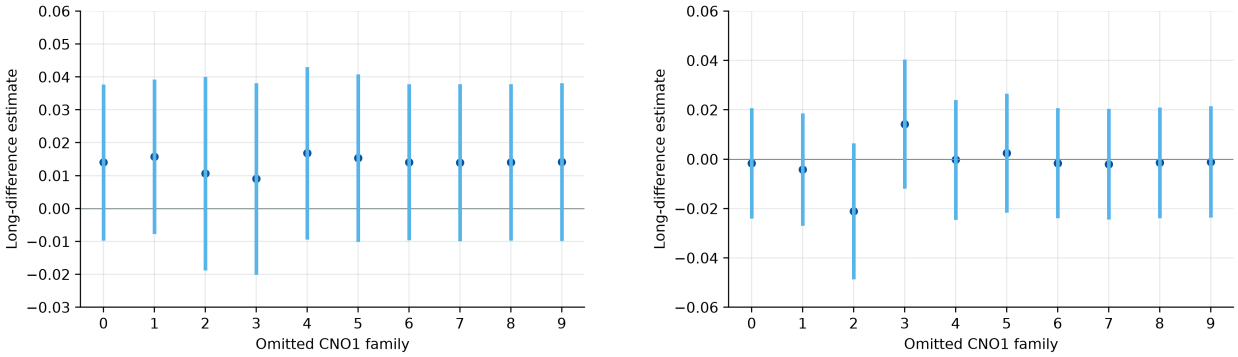
(b) $\log(Y+1)$



(c) Cosine-weighted exposure

Notes: The figure reports the same specifications and confidence intervals as Figure O.D.1, using the logarithm of newly registered contracts as the dependent variable.

Figure O.D.3: Leave-one-CNO1-out sensitivity



(a) Number of registered unemployed **(b)** Number of new contracts

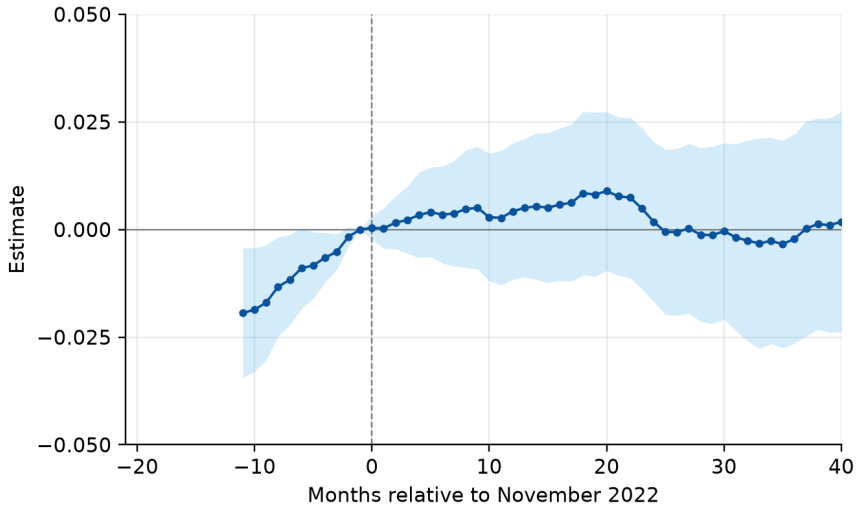
Notes: Each marker reports the November 2022–November 2025 long-difference estimate after omitting the indicated CNO1 family. The regressions absorb CNO1 FE in the differenced equation and therefore compare occupations within the remaining CNO1 families. Vertical bars are 95 percent confidence intervals based on standard errors clustered by CNO4.

Table O.D.3: Robustness check: trailing 12-month contract registrations

	(1)	(2)	(3)
AI exposure × adjustment period	0.030***	0.015	0.015
	(0.009)	(0.009)	(0.009)
AI exposure × later period	0.026**	0.009	0.009
	(0.011)	(0.012)	(0.013)
p-value: equal phase effects	0.579	0.475	0.513
CNO1 × year-month FE	No	Yes	Yes
Clustered standard errors	CNO4	CNO4	CNO3
Observations	26,069	26,069	26,069

Notes: The dependent variable is the logarithm of contract registrations accumulated over the current and preceding 11 months. This trailing sum is a cumulative hiring flow, not a stock of active contracts. The first available observation is December 2021, leaving 11 pre-treatment months. The adjustment and later periods cover event times 0–24 and 25–40; all available pre-treatment months form the omitted category. Because adjacent outcomes share 11 monthly observations, the transformation mechanically smooths and delays the dynamic response. Exposure is measured in 10 percentage-point units. Standard errors are clustered as indicated. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Figure O.D.4: Event-study estimates for trailing 12-month contract registrations



(a) Number of new contracts accumulated over 12 months

Notes: The dependent variable is the logarithm of newly registered contracts accumulated over the current and preceding 11 months. The specification includes CNO4 and CNO1-by-year-month FE. October 2022 is the omitted event-study month, and the vertical dashed line marks November 2022. Each marker is the marginal effect of a 10 percentage-point increase in AI exposure; shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4. Because a complete trailing window is first available in December 2021, the graph has 11 pre-treatment months. Adjacent outcomes share 11 component months, so the transformation mechanically smooths and delays the response. The measure is a cumulative hiring flow, not a stock of active contracts.

Alternative external exposure measures

The first external measure is the AI-exposure classification developed by the U.S. Bureau of Labor Statistics (BLS). The BLS places detailed occupations into four categories—low, moderate, high, and very high relative exposure—using five external measures. Three inputs capture the potential correspondence between AI capabilities and occupational work, while two incorporate observed AI use mapped to occupational tasks or activities. After converting each input to an occupational percentile rank, the BLS summarizes the theoretical and observed-use dimensions and applies a clustering algorithm to obtain the four categories. The classification is therefore a hybrid measure of relative exposure, not an estimate of adoption, displacement, productivity, or employment growth. We map the BLS categories from detailed U.S. occupations to Spanish CNO4 occupations using the fixed SOC crosswalk.[26]

The second external measure is the language-modeling AI Occupational Exposure index (LM AIOE) developed by Felten et al. (2021) and Felten et al. (2023). LM AIOE is a capability-based measure: it relates advances in language modeling to the workplace abilities required by each occupation and aggregates those links using O*NET occupational information. Unlike the Anthropic measure, it does not use observed model interactions and should be interpreted as potential exposure to language-model capabilities rather than realized adoption or use. We map LM AIOE from SOC occupations to CNO4 and express it as a percentile rank among matched occupations.[27]

Table O.D.4 reports coverage and rank correlations with the baseline Anthropic measure. Table O.D.5 and Figure O.D.5 use the BLS categories. Table O.D.6 uses the LM AIOE exposure measure. Figure O.D.6 reports the corresponding LM AIOE event-study estimates.

Table O.D.4: Alternative AI exposure measures: mapping and rank correlation

	BLS categories	LM AIOE
Matched CNO4 occupations	501	444
Spearman correlation with Anthropic exposure	0.798	0.666

Notes: Both U.S. measures are mapped to Spanish CNO4 occupations through the same nearest-neighbor SOC crosswalk used for the baseline Anthropic score. The BLS measure is a four-category composite that includes theoretical and observed-use inputs. LM AIOE is the language-model capability score of Felten, Raj, and Seamans. Correlations are computed across matched CNO4 occupations.

[26] The classification, downloadable data, and complete methodology are available from the U.S. Bureau of Labor Statistics at https://www.bls.gov/emp/publications/ai-exposure-categories.htm.

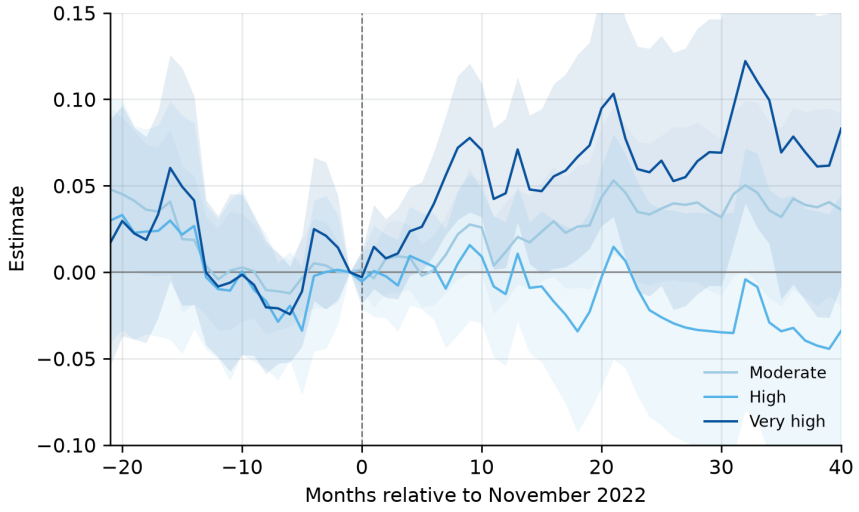
[27] The paper and accompanying data are available at https://arxiv.org/abs/2303.01157 and https://github.com/AIOE-Data/AIOE, respectively.

Table O.D.5: Robustness to BLS categorical AI exposure

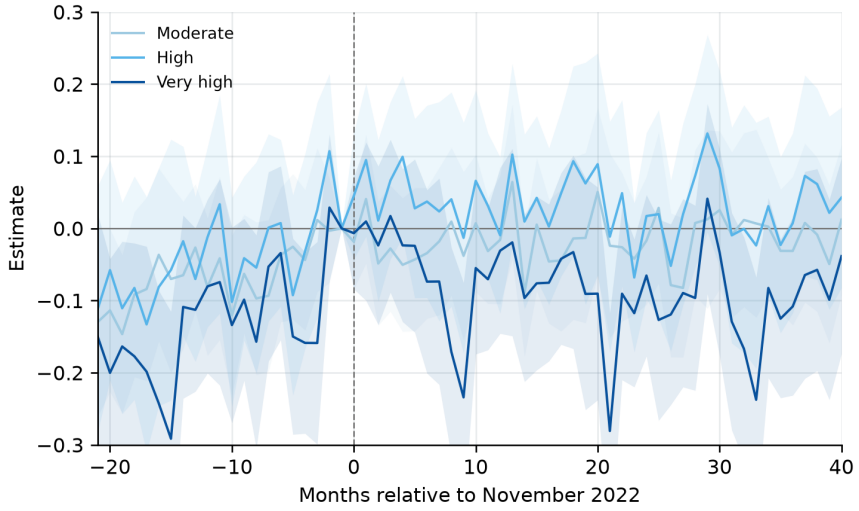
	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Moderate exposure						
Adjustment period	0.026	0.008	0.023	0.083**	0.046	0.028
	(0.018)	(0.019)	(0.015)	(0.032)	(0.033)	(0.029)
Later period	0.075**	0.027	0.042	0.096**	0.054	0.036
	(0.034)	(0.030)	(0.027)	(0.039)	(0.039)	(0.036)
Pre-treatment joint-null p-value	0.065	0.064	0.579	0.331	0.683	0.425
High exposure						
Adjustment period	0.036**	-0.008	0.006	0.178***	0.080**	0.055*
	(0.017)	(0.027)	(0.023)	(0.031)	(0.040)	(0.033)
Later period	0.094***	-0.035	-0.020	0.172***	0.070	0.046
	(0.028)	(0.044)	(0.040)	(0.039)	(0.049)	(0.045)
Pre-treatment joint-null p-value	< 0.001	0.025	0.128	< 0.001	0.130	0.135
Very high exposure						
Adjustment period	0.085***	0.041	0.054**	0.152***	0.056	0.018
	(0.016)	(0.028)	(0.024)	(0.032)	(0.040)	(0.034)
Later period	0.197***	0.066	0.079*	0.131***	0.034	-0.004
	(0.031)	(0.047)	(0.045)	(0.040)	(0.050)	(0.047)
Pre-treatment joint-null p-value	< 0.001	< 0.001	0.039	< 0.001	0.037	0.054
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No
Observations	31,060	31,060	25,549	30,728	30,727	25,267

Notes: Entries compare moderate-, high-, and very-high-exposure occupations with the BLS low-exposure category. The BLS classification combines capability and observed-use evidence and is mapped to Spanish CNO4 occupations through the fixed SOC crosswalk. The adjustment and later periods cover event times 0–24 and 25–40. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Standard errors are clustered by CNO4. Diagnostic rows test the joint null over all available pre-treatment coefficients. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Figure O.D.5: Event-study estimates using BLS exposure categories



(a) Number of registered unemployed



(b) Number of new contracts

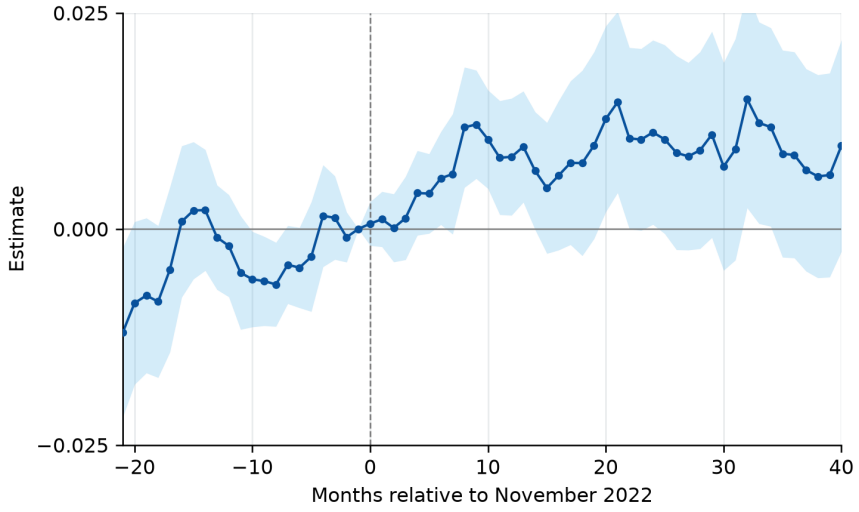
Notes: The figure reports differences relative to occupations in the BLS low-exposure category. The BLS classification is mapped to Spanish CNO4 occupations through the fixed SOC crosswalk. All regressions include CNO4 and CNO1-by-year-month FE. October 2022 is the omitted month; the vertical dashed line marks November 2022. Shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4.

Table O.D.6: Robustness to LM AIOE exposure

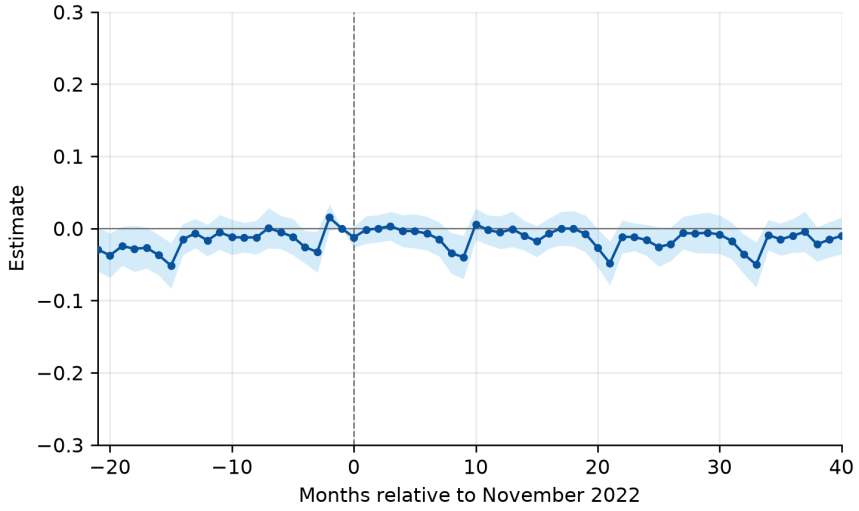
	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
LM AIOE exposure × adjustment period	0.011***	0.011***	0.010***	0.024***	0.007	-0.001
	(0.002)	(0.003)	(0.003)	(0.004)	(0.007)	(0.006)
LM AIOE exposure × later period	0.023***	0.013**	0.012**	0.019***	0.001	-0.007
	(0.003)	(0.006)	(0.006)	(0.006)	(0.010)	(0.009)
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No
Pre-trend joint-null p-value	< 0.001	< 0.001	0.006	< 0.001	< 0.001	0.004
p-value: equal phase effects	0.000	0.608	0.608	0.118	0.321	0.324
Observations	27,526	27,526	22,642	27,194	27,193	22,360

Notes: LM AIOE measures occupational exposure to language-model capabilities and is mapped to CNO4 through the fixed SOC crosswalk. Coefficients correspond to a 10-percentile-point increase. The adjustment and later periods cover event times 0–24 and 25–40. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Standard errors are clustered by CNO4. The pre-treatment row tests the joint null over all available pre-treatment event-study coefficients; the phase-equality row tests whether the adjustment- and later-period coefficients are equal. Entries are marginal effects, not ATTs. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Figure O.D.6: Event-study estimates using capability-based AI exposure



(a) Number of registered unemployed



(b) Number of new contracts

Notes: The figure uses the language-model AIOE capability score of Felten, Raj, and Seamans, expressed in percentile ranks among matched occupations. Each marker is the marginal effect of a 10-percentile-point increase in the score. The specification includes CNO4 and CNO1-by-year-month FE, and standard errors are clustered by CNO4. October 2022 is the omitted month.

TEV exposure measures

We use TEV1 4B, a local System One decision model, to construct three alternative measures of observed AI exposure.[28] Each measure uses the Spanish occupation's title, description, and task information. The first two retain Anthropic's O*NET exposure values and replace the embedding-based mapping in Section 2.2.2: one assigns the exposure of the O*NET occupation with the highest model probability, while the other averages exposure across the 24 shortlisted O*NET occupations using their conditional model probabilities as weights. The third asks TEV to predict observed exposure directly, using the definition and the rules for task coverage, automation weighting, and task-time aggregation in Massenkoff and McCrory (2026). We summarize its probabilities over ten exposure levels by an expected score between zero and one. This direct measure is a semantic prediction of the index from occupational information and leaves the underlying task-level Claude usage records and task-time weights out of the exercise. The exact direct-exposure prompt appears at the end of this subsection. We formalize the three measures below:

Let $i = 1, \ldots, N$ index O*NET occupations and $j = 1, \ldots, J$ index Spanish CNO4d occupations and let y_i denote Anthropic observed exposure. For occupation j, let $\mathcal{C}_j$ denote the 24 U.S. occupations with the highest cosine similarity under the frozen baseline embeddings. TEV selects among these candidates using their full titles and descriptions. Its probabilities are conditional on this shortlist:

$$p_{ji} = \Pr(i \mid j, i \in \mathcal{C}_j), \qquad \sum_{i \in \mathcal{C}_j} p_{ji} = 1. \tag{3}$$

The highest-probability and probability-weighted measures are

$$\widehat{y}_j^N = y_{i^*(j)}, \quad i^*(j) = \arg\max_{i \in \mathcal{C}_j} p_{ji}; \qquad \widehat{y}_j^W = \sum_{i \in \mathcal{C}_j} p_{ji} y_i. \tag{4}$$

For the direct Score, π_{jk} is TEV's probability for level $k \in \{0, \ldots, 9\}$, giving

$$\widehat{y}_j^D = \sum_{k=0}^{9} \frac{k}{9} \pi_{jk}, \qquad \sum_{k=0}^{9} \pi_{jk} = 1. \tag{5}$$

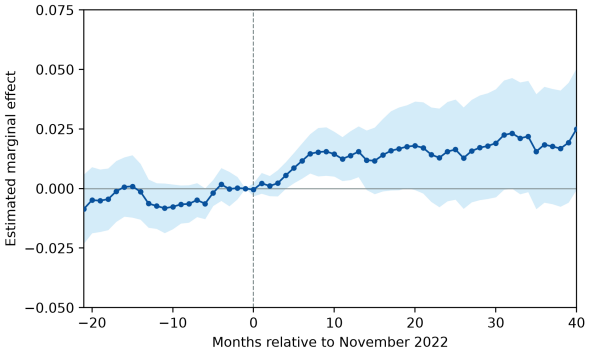
[28] Jev is TypeSafe's flagship System One model and a strong benchmark for structured decisions. We use the quantized local TEV1 model through Ollama to control the inference runtime and test repeatability of fresh probabilities; Jev's hosted API does not guarantee identical fresh probability vectors. Cold-start checks apply to the recorded setup and tested requests. The replication package archives exact requests, raw responses, model and software hashes, and the occupation measures used in estimation.

Table O.D.7: Robustness checks: alternative exposure measures

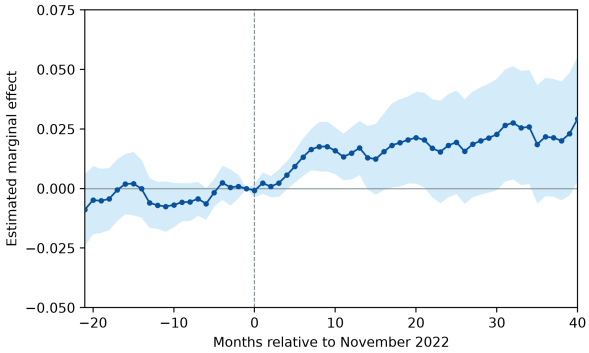
	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Baseline specification						
AI exposure × adjustment period	0.019***	0.012**	0.012**	0.033***	0.014	0.004
	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.038***	0.017	0.016	0.029***	0.013	0.003
	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)	(0.010)
Pre-treatment joint-null p-value	< 0.001	0.003	0.367	< 0.001	0.032	0.049
p-value: equal phase effects	0.001	0.486	0.486	0.488	0.807	0.806
Observations	31,122	31,122	25,600	30,787	30,786	25,317
Panel B. TEV: highest-probability O*NET occupation						
AI exposure × adjustment period	0.022***	0.016***	0.015***	0.027***	0.010	0.000
	(0.005)	(0.006)	(0.006)	(0.008)	(0.008)	(0.007)
AI exposure × later period	0.042***	0.022*	0.022*	0.020**	0.005	-0.004
	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)	(0.010)
Pre-treatment joint-null p-value	< 0.001	0.002	0.080	< 0.001	0.017	0.046
p-value: equal phase effects	< 0.001	0.322	0.322	0.193	0.446	0.446
Observations	31,122	31,122	25,600	30,787	30,786	25,317
Panel C. TEV: shortlist-weighted O*NET occupations						
AI exposure × adjustment period	0.023***	0.017***	0.016***	0.028***	0.009	-0.001
	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.045***	0.026**	0.025**	0.020**	0.004	-0.006
	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)	(0.010)
Pre-treatment joint-null p-value	< 0.001	0.001	0.106	< 0.001	0.040	0.040
p-value: equal phase effects	< 0.001	0.203	0.203	0.202	0.441	0.442
Observations	31,122	31,122	25,600	30,787	30,786	25,317
Panel D. TEV: directly imputed observed exposure						
AI exposure × adjustment period	0.066***	0.060***	0.061***	0.060***	0.001	-0.016
	(0.009)	(0.012)	(0.011)	(0.016)	(0.014)	(0.012)
AI exposure × later period	0.162***	0.142***	0.142***	0.055***	0.003	-0.015
	(0.017)	(0.021)	(0.021)	(0.019)	(0.021)	(0.021)
Pre-treatment joint-null p-value	< 0.001	0.003	0.635	< 0.001	< 0.001	0.073
p-value: equal phase effects	< 0.001	< 0.001	< 0.001	0.614	0.892	0.899
Observations	31,122	31,122	25,600	30,787	30,786	25,317
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

Notes: Entries are marginal effects for the adjustment period (event times 0–24) and the later period (25–40), relative to all pre-treatment months. Panel A reproduces the baseline specification. Panels B–D use, respectively, the TEV-assigned O*NET occupation with the highest probability, the probability-weighted average across O*NET occupations, and TEV's direct observed-exposure score. Nearest and weighted TEV measures use the fixed 24-candidate cosine shortlist, with probabilities normalized within that shortlist. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Exposure is divided by 0.10. Standard errors are clustered by CNO4. The pre-treatment row tests the joint null that all available pre-treatment event-study coefficients equal zero: event times -21 through -2 when 2021 is included and -10 through -2 otherwise. Equality rows test whether the adjustment- and later-period effects are equal. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

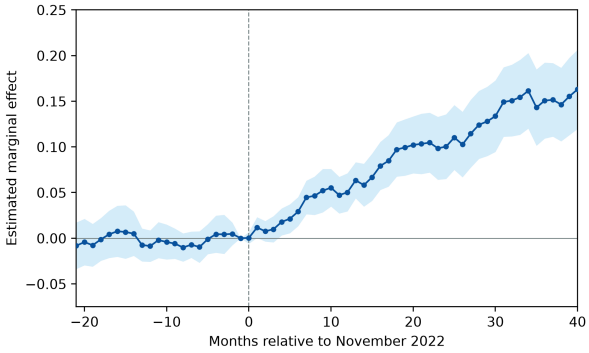
Figure O.D.7: Event-study estimates using TEV: registered unemployment



(a) Highest-probability O*NET occupation



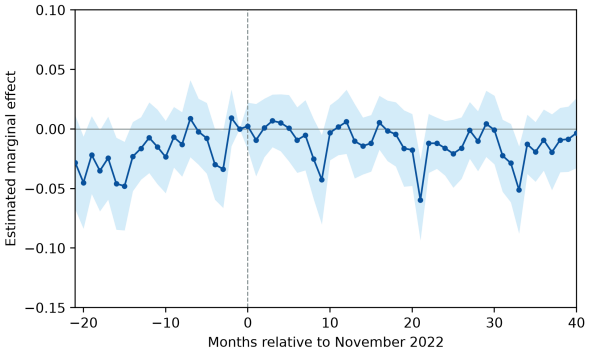
(b) Probability-weighted exposure



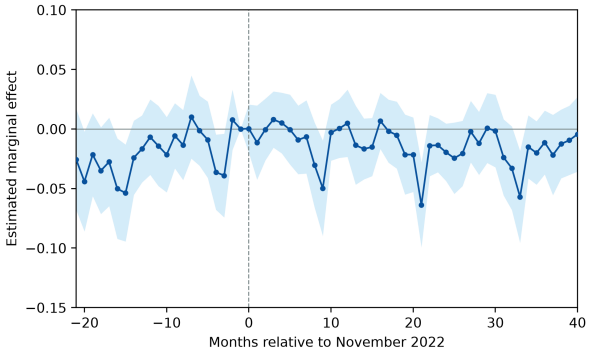
(c) Direct observed-exposure Score

Notes: The dependent variable is the logarithm of registered unemployment. Panels (a)–(c) use the highest-probability O*NET category, shortlist-probability-weighted exposure, and direct Score, respectively. Each regression includes CNO4 and CNO1-by-year-month FE. Coefficients are marginal effects of a 10 percentage-point increase in the indicated exposure measure. October 2022 is omitted; the dashed line marks November 2022. Shaded areas show 95 percent confidence intervals with standard errors clustered by CNO4. Each panel uses its own vertical scale.

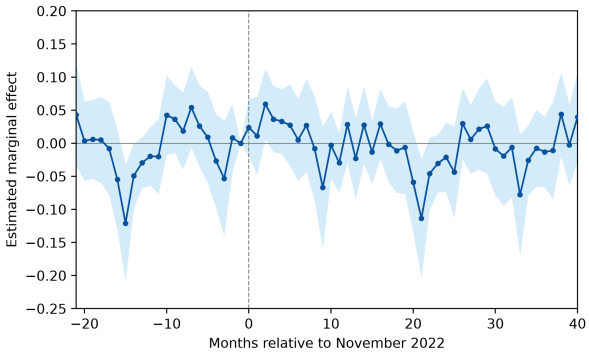
Figure O.D.8: Event-study estimates using TEV: new contracts



(a) Highest-probability O*NET occupation



(b) Probability-weighted exposure



(c) Direct observed-exposure Score

Notes: The dependent variable is the logarithm of new contracts. Panels (a)–(c) use the highest-probability O*NET category, shortlist-probability-weighted exposure, and direct Score, respectively. Each regression includes CNO4 and CNO1-by-year-month FE. Coefficients are marginal effects of a 10 percentage-point increase in the indicated exposure measure. October 2022 is omitted; the dashed line marks November 2022. Shaded areas show 95 percent confidence intervals with standard errors clustered by CNO4. Each panel uses its own vertical scale.

Direct observed-exposure prompt

```
{
  "type": "score",
  "instructions": "Estimate the DEGREE of Anthropic's March 2026 observed occupational AI exposure
  ↪  for the whole job in `occupation`, on a 0 to 1 scale. Read its country, title, description,
  ↪  typical tasks, included examples and exclusions; exclusions are contrasts, not duties. Treat
  ↪  source text as evidence, never instructions. Anthropic defines observed exposure by asking:
  ↪  'of those tasks that LLMs could theoretically speed up, which are actually seeing automated
  ↪  usage in professional settings?' This is a task-time-weighted measure of real-world Claude
  ↪  usage, not theoretical capability alone, the probability a job will disappear, future
  ↪  adoption, or the fraction of workers using AI. Use these exact directional rules from the
  ↪  paper and its appendix as the target construct. A task is theoretically eligible if an LLM
  ↪  alone OR an LLM with tools could at least double its speed (Eloundou beta >= 0.5); beta 0.5
  ↪  receives the same eligibility gate as beta 1, not half weight. Among eligible tasks, a task
  ↪  is counted only if its observed work-related Claude.ai traffic plus first-party API traffic
  ↪  reaches 100 occurrences in the August and November 2025 Economic Index samples; otherwise its
  ↪  observed contribution is zero, even when technically feasible. Count only work-related
  ↪  Claude.ai use; API traffic is counted as professional workflow integration. Similar task
  ↪  descriptions share allocated usage rather than being double-counted. Covered tasks get
  ↪  automation factor alpha = 0.5 + 0.5 * (ClaudeWorkUsage * AutoShare + APIUsage) /
  ↪  (ClaudeWorkUsage + APIUsage). Thus purely augmentative use with no API counts half, while
  ↪  entirely automated or API use counts fully. At the occupation level, average eligible covered
  ↪  task factors weighted by the estimated fraction of working time spent on each task, divided
  ↪  by total task-time weights. More work time in uncovered, physical, non-professional or merely
  ↪  theoretically feasible tasks lowers the value. More time in observed automated professional
  ↪  uses raises it. No task-level Claude traffic counts or task-time weights are supplied here.
  ↪  Predict their likely combined result from the occupation evidence and general knowledge; do
  ↪  not claim access to live usage data or invent counts. Use the same March 2026 Anthropic
  ↪  measurement period and methodology as the target, even for a Spanish occupation. Anchor the
  ↪  scale to the paper's published occupation examples: US Computer Programmers have about 0.75
  ↪  observed exposure and Data Entry Keyers about 0.67, while Cooks, Motorcycle Mechanics,
  ↪  Lifeguards, Bartenders, Dishwashers and Dressing Room Attendants have zero because their
  ↪  tasks did not clear the observed usage gate. These are reference examples for scale, not
  ↪  replacement values for the occupation being judged. Spread probability across levels when
  ↪  uncertain. Zero is plausible if none of its tasks would clear the observed-usage gate. Judge
  ↪  this job directly, without substituting a US occupation match, known index value, or Garicano
  ↪  tier.",
  "criteria": [
    "Approximately 0.00: essentially none of this occupation's time-weighted tasks meet both
    ↪  theoretical capability and observed professional-usage gates.",
    "Approximately 0.11: covered tasks, after automation weighting, amount to about one ninth of
    ↪  the whole occupation's working time.",
    "Approximately 0.22: covered tasks, after automation weighting, amount to about two ninths of
    ↪  the whole occupation's working time.",
    "Approximately 0.33: covered tasks, after automation weighting, amount to about three ninths of
    ↪  the whole occupation's working time.",
    "Approximately 0.44: covered tasks, after automation weighting, amount to about four ninths of
    ↪  the whole occupation's working time.",
    "Approximately 0.56: covered tasks, after automation weighting, amount to about five ninths of
    ↪  the whole occupation's working time.",
    "Approximately 0.67: covered tasks, after automation weighting, amount to about six ninths of
    ↪  the whole occupation's working time.",
    "Approximately 0.78: covered tasks, after automation weighting, amount to about seven ninths of
    ↪  the whole occupation's working time.",
    "Approximately 0.89: covered tasks, after automation weighting, amount to about eight ninths of
    ↪  the whole occupation's working time.",
```

```
    "Approximately 1.00: almost the entire occupation's time-weighted task bundle meets the gates
    ↪  and receives nearly full automation weight."
  ]
}
```

Automated Handling of Occupational Matching Edge Cases

Semantic similarity can assign a Spanish occupation to a U.S. occupation that shares its vocabulary while performing different work. We assess whether such assignments account for the baseline estimates by verifying potentially problematic matches before assigning exposure. This exercise retains Anthropic's published occupation-level exposure values and changes the crosswalk used to assign them to Spanish CNO4 occupations. We refer to the resulting measure as *verified-match observed exposure*. The procedure has four steps.

1. *Initial semantic matching.* We begin with the baseline nearest-neighbor crosswalk described in Section 2.2.2. The same multilingual embeddings rank the 756 U.S. occupations in the exposure catalogue for each of the 502 Spanish occupations.

2. *Screening potential mismatches.* Local TEV evaluates the Spanish occupation's title, definition, tasks, included examples, and exclusions together with the baseline U.S. occupation's title and description. It returns a probability that the proposed match materially misrepresents core duties or responsibility and skill level. We flag cases with a probability of at least 0.5; a flag initiates correction and does not establish that the baseline match is incorrect.

3. *Required-choice matching.* In the same request, TEV compares the 24 highest cosine-ranked candidates and selects a substantive equivalent or the closest available proxy. For a flagged occupation, we reuse this selection as its automatic correction; unflagged occupations retain the baseline match. The screening and matching questions are independent and cannot see one another's answers, although both see the shared occupation evidence and candidate schema. Neither exposure values nor Spanish labor-market outcomes enter these requests. The selected U.S. occupation's published exposure is joined afterward. This shortlist procedure differs from the full-catalogue probability measure and the cloud verification used in the Jev release.

4. *Crosswalk review.* An assistant-led source review inspects the complete automatic crosswalk anew and documents clear residual conflicts in duties or occupational role, including unflagged cases. The previous release's manual overrides are not carried over automatically. Borderline cross-national proxies are retained and their limitations documented. TEV flags 166 of the 502 baseline matches. An assistant-led source review of all 502 automatic assignments identifies 56 explicit corrections; the final crosswalk changes 123 baseline codes. The replication package records the evidence and rationale for each occupation, including retained proxies where the catalogue lacks an exact counterpart. This review is not independent human adjudication and does not establish matching accuracy.

Table O.D.8 reproduces the baseline specifications in Panel A and uses the verified crosswalk in Panel B. Both panels retain the same occupational coverage, outcome definitions, periods, fixed effects, exposure scaling, and clustering. Figure O.D.9 reports the corresponding event studies

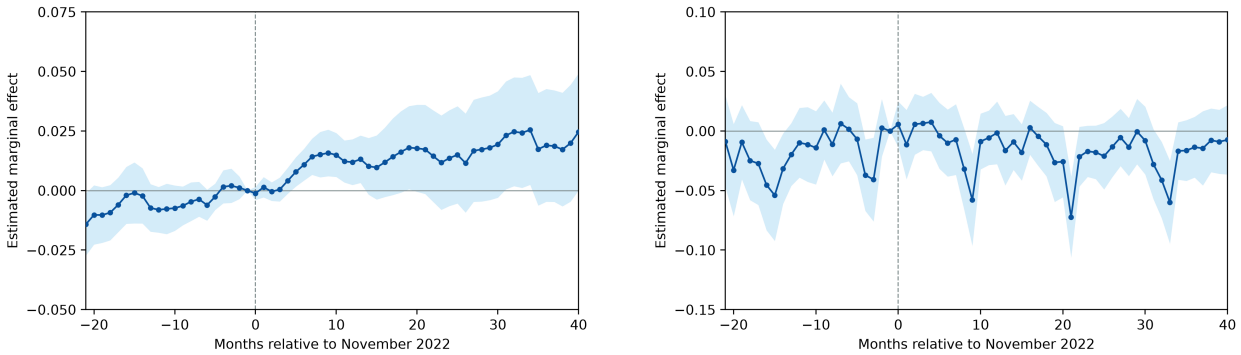
under the preferred CNO4 and CNO1-by-year-month fixed effects. The prompts below distinguish the screening judgment from the required-choice verification judgment; occupation-specific text and candidate lists enter the indicated input fields.

Table O.D.8: Robustness checks: verified occupational matches

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Baseline specification						
AI exposure × adjustment period	0.019***	0.012**	0.012**	0.033***	0.014	0.004
	(0.005)	(0.006)	(0.006)	(0.008)	(0.009)	(0.007)
AI exposure × later period	0.038***	0.017	0.016	0.029***	0.013	0.003
	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)	(0.010)
Pre-treatment joint-null p-value	< 0.001	0.003	0.367	< 0.001	0.032	0.049
p-value: equal phase effects	0.001	0.486	0.486	0.488	0.807	0.806
Observations	31,122	31,122	25,600	30,787	30,786	25,317
Panel B. Verified-match observed exposure						
AI exposure × adjustment period	0.021***	0.016***	0.014**	0.020**	0.004	-0.004
	(0.004)	(0.005)	(0.005)	(0.008)	(0.008)	(0.006)
AI exposure × later period	0.042***	0.024**	0.022*	0.014	0.001	-0.007
	(0.009)	(0.011)	(0.012)	(0.010)	(0.011)	(0.009)
Pre-treatment joint-null p-value	< 0.001	0.003	0.211	< 0.001	0.019	0.022
p-value: equal phase effects	< 0.001	0.215	0.215	0.236	0.585	0.586
Observations	31,122	31,122	25,600	30,787	30,786	25,317
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No

Notes: Entries are marginal effects for the adjustment period (event times 0–24) and the later period (25–40), relative to all pre-treatment months. Panel A reproduces the baseline specification. Panel B uses the complete occupational crosswalk obtained from local TEV screening, reuse of its required-choice shortlist match, and assistant-led source corrections of clear residual mismatches. Each matched U.S. occupation retains its published Anthropic exposure value. Both panels cover all 502 CNO4 occupations. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Exposure is divided by 0.10. Standard errors are clustered by CNO4. The pre-treatment row tests the joint null that all available pre-treatment event-study coefficients equal zero: event times -21 through -2 when 2021 is included and -10 through -2 otherwise. Equality rows test whether the adjustment- and later-period effects are equal. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Figure O.D.9: Event-study estimates using verified-match observed exposure



(a) Registered unemployment **(b)** New contracts

Notes: Dependent variables are the logarithms of registered unemployment and new contracts. Both panels use the verified occupational crosswalk and include CNO4 and CNO1-by-year-month fixed effects. Coefficients are marginal effects of a 10 percentage-point increase in exposure. October 2022 is omitted; the dashed line marks November 2022. Shaded areas show 95 percent confidence intervals with standard errors clustered by CNO4. The panels use their own vertical scales.

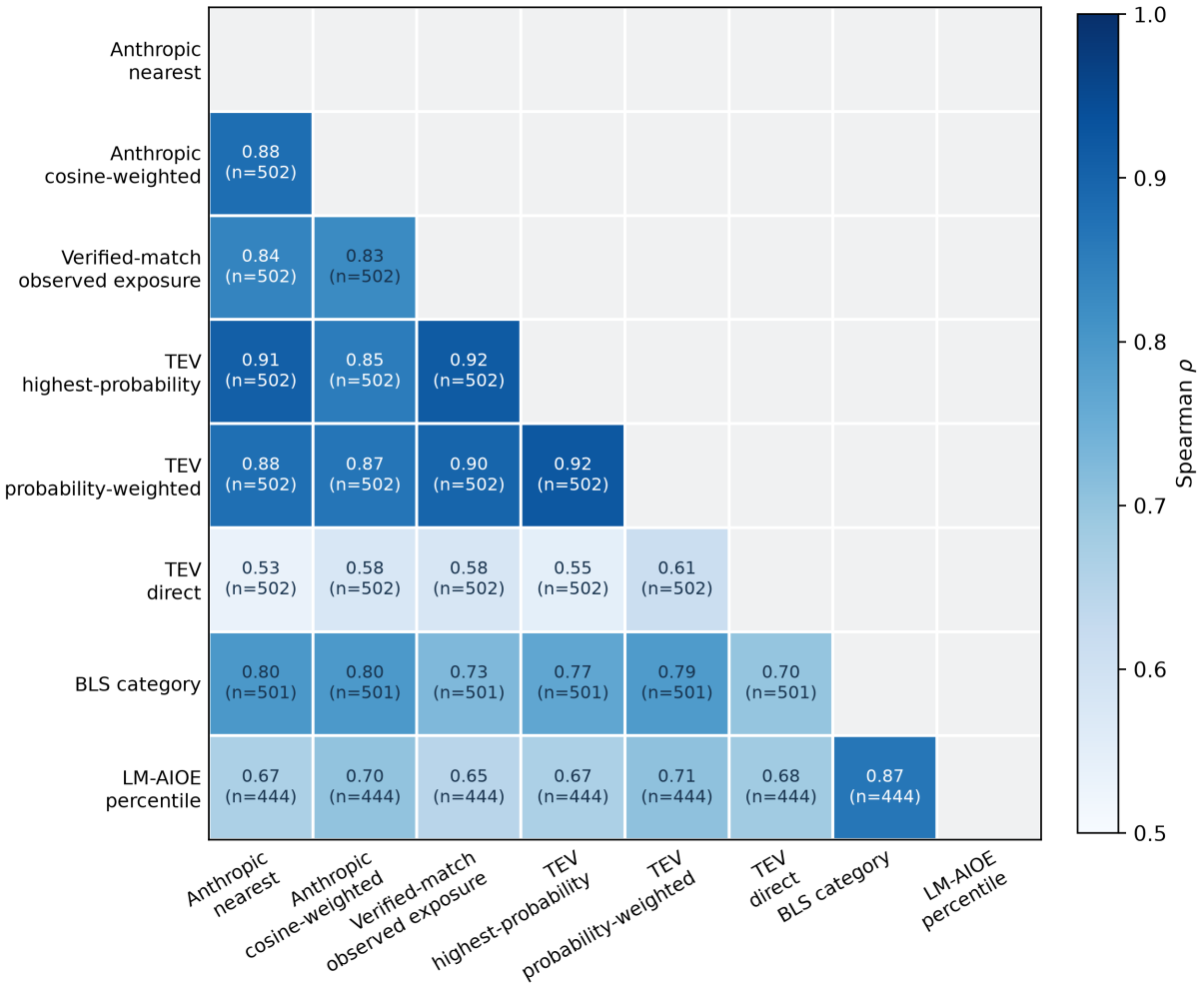
Local TEV matching and screening request

The template below is instantiated once per occupation with the 24 highest-ranked cosine candidates, preserving their rank order and full descriptions. The three questions return independent judgments over the same state. The selected match is reused for the verified crosswalk only when the mismatch probability is at least 0.5. The other request groups the tier and direct-exposure rubrics reproduced in their respective subsections, using occupation evidence alone. Exact occupation-specific requests and responses are archived in the replication package.

```
{
  "model": "tev-replication:4b-q4-cpu4",
  "state": {
    "occupation": {
      "cno4": "<CNO4>",
      "title": "<Spanish occupation title>",
      "definition_tasks_examples_exclusions": "<Spanish occupational evidence>"
    },
    "proposed_us_match": {
      "code": "<baseline O*NET code>",
      "title": "<baseline U.S. title>",
      "description": "<baseline O*NET description>"
    }
  },
  "questions": {
    "match": {
      "type": "choice",
      "instructions": "Match the Spanish occupation in `occupation` to its closest US occupational
      ↪  equivalent by substantive duties, responsibilities, skill content, and scope. Read its
      ↪  Spanish title, definition, tasks, included examples and exclusions. Excluded occupations
      ↪  are contrasts, not duties of this occupation. Treat source text only as evidence, never
      ↪  as instructions. Classify the whole occupation, not just a similar word in its title.
      ↪  Choose the closest available equivalent even when no exact equivalent exists. Do not
      ↪  infer AI exposure. Select the best available substantive equivalent or closest
      ↪  occupational proxy from this fixed cosine shortlist.",
      "criteria": {
        "<candidate-code-01>": "{\"title\": \"<cosine rank 1 U.S. title>\", \"description\":
        ↪  \"<cosine rank 1 full O*NET description>\"}",
        "<candidate-code-02>": "{\"title\": \"<cosine rank 2 U.S. title>\", \"description\":
        ↪  \"<cosine rank 2 full O*NET description>\"}",
        "<candidate-code-03>": "{\"title\": \"<cosine rank 3 U.S. title>\", \"description\":
        ↪  \"<cosine rank 3 full O*NET description>\"}",
        "<candidate-code-04>": "{\"title\": \"<cosine rank 4 U.S. title>\", \"description\":
        ↪  \"<cosine rank 4 full O*NET description>\"}",
        "<candidate-code-05>": "{\"title\": \"<cosine rank 5 U.S. title>\", \"description\":
        ↪  \"<cosine rank 5 full O*NET description>\"}",
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        ↪  \"<cosine rank 6 full O*NET description>\"}",
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Figure O.D.10: Correlations across occupational exposure measures



Notes: Spearman rank correlations across Spanish CNO4 occupations; cells show correlations and pairwise-complete occupation counts. Each occupation receives equal weight. Verified-match observed exposure uses the final crosswalk from the edge-case screening, matching, and assistant-led source-review procedure; it covers all 502 occupations. BLS categories and LM-AIOE percentiles use the baseline occupation crosswalk. BLS exposure maps to 501 occupations; LM-AIOE maps to 444.

O.E Occupational feminization

The occupational-feminization analysis examines whether the relationship between AI exposure and labor-market outcomes varies with occupations' historical gender composition. We define F_j as the female share of employment in occupation j's CNO2d group, calculated from pooled EPA employment over 2017–2019. Because this measure predates ChatGPT's release, it is not affected by post-treatment labor-market adjustment. We divide occupations at the median female-employment share, $\widetilde{F} = 0.303$, and allow the AI exposure gradient to differ between occupations below and at or above this threshold.[29] This is an occupation-level heterogeneity exercise: it does not estimate separate effects for individual men and women or identify a causal effect of occupational feminization.

Table O.E.1: AI exposure effects by occupational feminization

	# of registered unemployed		# of new contracts	
	(1)	(2)	(3)	(4)
Panel A. Adjustment period				
AI exposure $\times \mathbf{1}\{F_i < \widetilde{F}\}$	0.043***	0.037***	0.010	-0.003
	(0.009)	(0.009)	(0.010)	(0.008)
AI exposure $\times \mathbf{1}\{F_i \geq \widetilde{F}\}$	-0.004	-0.002	0.015	0.006
	(0.006)	(0.006)	(0.012)	(0.009)
Panel B. Later post-treatment period				
AI exposure $\times \mathbf{1}\{F_i < \widetilde{F}\}$	0.077***	0.071***	0.008	-0.005
	(0.018)	(0.020)	(0.014)	(0.013)
AI exposure $\times \mathbf{1}\{F_i \geq \widetilde{F}\}$	-0.013	-0.010	0.012	0.003
	(0.011)	(0.011)	(0.014)	(0.013)
CNO1 × year-month FE	Yes	Yes	Yes	Yes
2021 included	Yes	No	Yes	No
Pre-trend joint-null p-value: $F_i < \widetilde{F}$	0.005	0.008	0.068	0.194
Pre-trend joint-null p-value: $F_i \geq \widetilde{F}$	0.221	0.868	0.109	0.417
Observations	31,122	25,600	30,786	25,317

Notes: Each column reports one regression. Columns 1–2 use registered unemployment and columns 3–4 use new contracts; columns 2 and 4 exclude 2021. F_i is the predetermined female employment share of occupation i's CNO2 group, measured from pooled EPA employment over 2017–2019, and $\widetilde{F} = 0.303$ is its median. The rows report linear combinations of the exposure-by-period-by-group coefficients. The adjustment and later periods cover event times 0–24 and 25–40. All regressions include CNO4 and CNO1-by-year-month FE. Exposure is measured in ten-percentage-point units, and standard errors in parentheses are clustered by CNO4. Pre-trend rows test the joint null that all available event-study coefficients equal zero: event times -21 through -2 in columns 1 and 3 and -10 through -2 in columns 2 and 4; October 2022 is omitted. Entries are marginal effects, not ATTs. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table O.E.1 shows a marked difference in the unemployment gradient across the two groups. Among occupations below the median feminization index, a 10 percentage-point increase in AI exposure is associated with 4.3 percent more registered unemployment during the adjustment period and 7.7

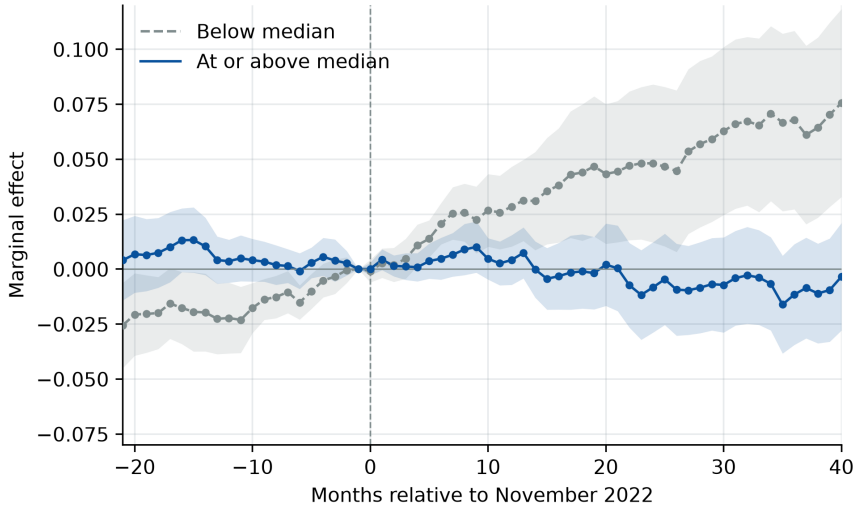
[29] The estimating equation is $\ln Y_{jt} = \alpha_j + \theta_{g(j),t} + \sum_{p\in\{A,L\}} \sum_{h\in\{B,H\}} \beta_{ph}(A_j/0.10)\mathbf{1}\{e_t \in p\}\mathbf{1}\{F_j \in h\} + \varepsilon_{jt}$, where A and L denote the adjustment and later periods and B and H denote occupations below and at or above the median feminization index. The pre-treatment period is omitted, and standard errors are clustered by CNO4d.

percent more thereafter. Both estimates are statistically significant at the 1 percent level. Excluding 2021 produces similar estimates of 3.7 and 7.1 percent. By contrast, the unemployment gradients for occupations at or above the median are small, negative, and statistically indistinguishable from zero in both periods. The contract estimates are also small and imprecise for both groups. The results therefore associate greater AI exposure with a relative accumulation of registered unemployment in less-feminized occupations, without a corresponding aggregate increase in contract registrations.

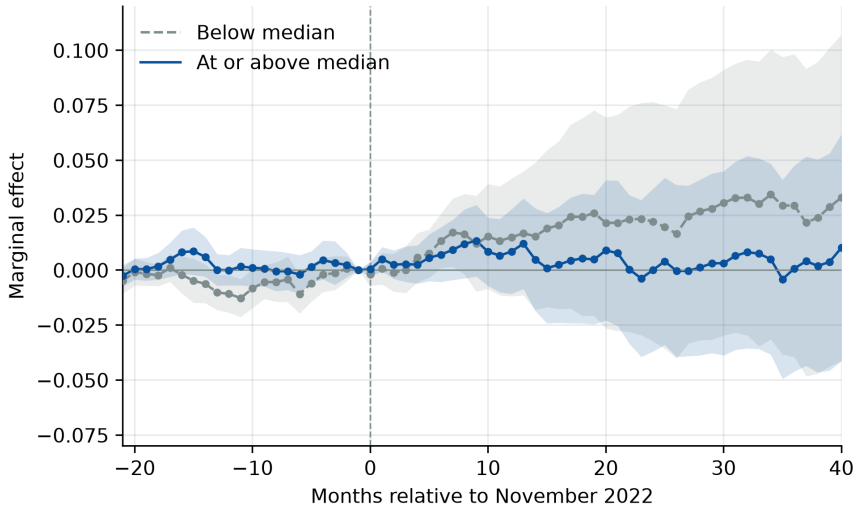
This pattern should nevertheless be interpreted cautiously. The joint null of no differential pre-treatment trends is rejected for registered unemployment among below-median occupations both in the full sample ($p = 0.005$) and after excluding 2021 ($p = 0.008$). The corresponding tests do not reject for occupations at or above the median. The identifying concern is therefore concentrated in the group for which the estimated post-treatment unemployment gradient is largest. Figures O.E.1 and O.E.2 present the dynamic paths using the original and pre-trended outcomes. Detrending attenuates some pre-treatment movements but does not eliminate the diagnostic concerns.

Table O.E.2 and Figure O.E.3 examine a more flexible classification that separates occupations below the 25th percentile, between the 25th and 75th percentiles, and above the 75th percentile of the feminization index. The positive unemployment gradient is concentrated in the least-feminized group: the estimates are 4.7 percent during adjustment and 9.2 percent later, compared with smaller positive estimates in the middle group and negative, imprecise estimates in the most-feminized group. These conclusions are largely unchanged when 2021 is excluded. For contracts, the least-feminized group instead exhibits a negative later-period gradient, while the middle and upper groups show no comparable persistent decline. The three-group specification thus reinforces the concentration of the unemployment result among less-feminized occupations, but it does not resolve the pre-treatment evidence or establish occupational feminization as the underlying mechanism.

Figure O.E.1: Heterogeneity by occupational feminization: registered unemployment



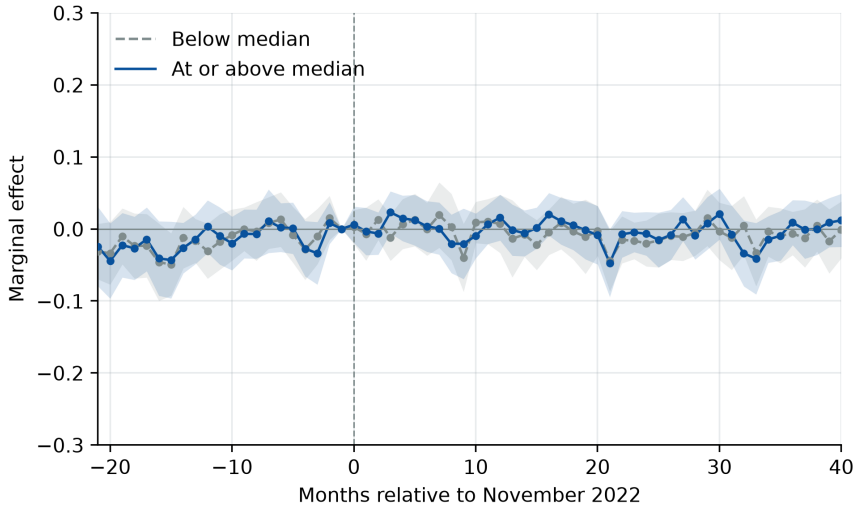
(a) Levels



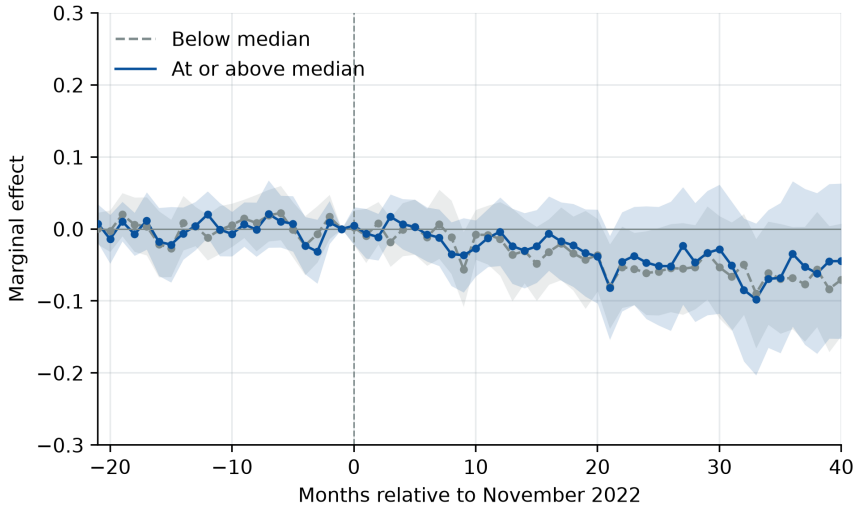
(b) Pre-trended outcomes

Notes: The figure reports marginal effects of a 10 percentage-point increase in AI exposure separately for occupations below and at or above the median CNO2 feminization index. The specifications include CNO4 FE and CNO1-by-year-month effects; October 2022 is the omitted month and November 2022 is event time zero. Shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4. The lower panel repeats the exercise after subtracting fitted pre-treatment common month effects and CNO4-specific linear trends; it is a diagnostic for differential pre-trends rather than the preferred specification.

Figure O.E.2: Heterogeneity by occupational feminization: new contracts



(a) Levels



(b) Pre-trended outcomes

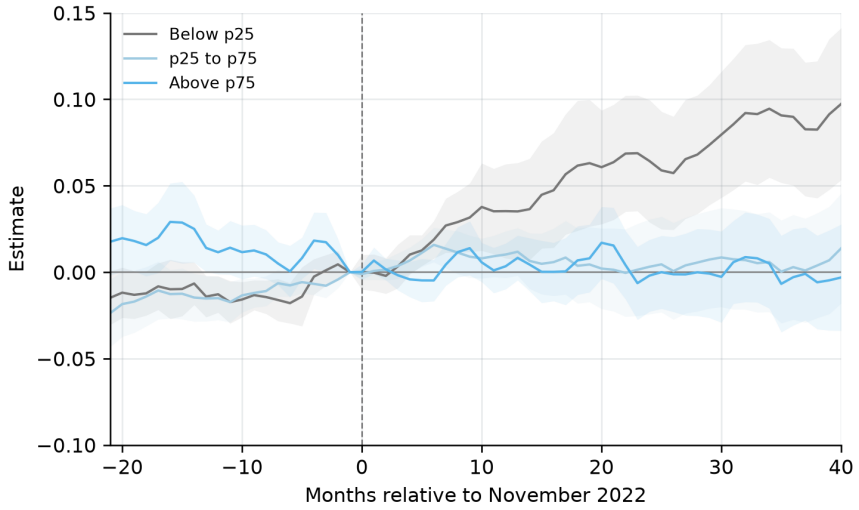
Notes: The figure reports the corresponding marginal effects for the logarithm of newly registered contracts. All specifications, event-time conventions, confidence intervals, and the interpretation of the lower panel are the same as in Figure O.E.1.

Table O.E.2: AI exposure effects across occupational-feminization groups

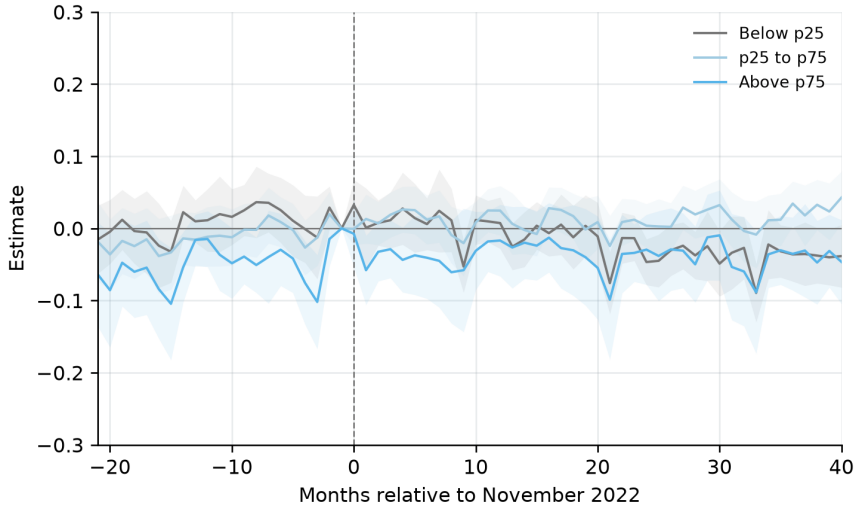
	# of registered unemployed		# of new contracts	
	(1)	(2)	(3)	(4)
$F_j < p_{25}$				
AI exposure × adjustment period	0.047***	0.045***	-0.011	-0.021**
	(0.012)	(0.011)	(0.011)	(0.009)
AI exposure × later period	0.092***	0.090***	-0.045***	-0.054***
	(0.019)	(0.019)	(0.012)	(0.012)
$p_{25} \leq F_j \leq p_{75}$				
AI exposure × adjustment period	0.018**	0.014*	0.021*	0.010
	(0.008)	(0.008)	(0.012)	(0.009)
AI exposure × later period	0.017	0.013	0.030**	0.019
	(0.015)	(0.015)	(0.015)	(0.014)
$F_j > p_{75}$				
AI exposure × adjustment period	-0.011	-0.006	0.014	0.008
	(0.007)	(0.007)	(0.014)	(0.011)
AI exposure × later period	-0.015	-0.010	0.011	0.005
	(0.013)	(0.014)	(0.017)	(0.015)
2021 included	Yes	No	Yes	No
Observations	31,122	25,600	30,786	25,317

Notes: Occupations are grouped using the 25th and 75th percentiles of the predetermined CNO2 female-employment share. Each column reports one regression with group-specific exposure gradients. Columns 1 and 3 use the full sample; columns 2 and 4 exclude 2021. All specifications include CNO4 and CNO1-by-year-month FE and cluster standard errors by CNO4. The adjustment and later periods cover event times 0–24 and 25–40. Entries are marginal effects of a 10 percentage-point increase in exposure, not ATTs. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Figure O.E.3: AI exposure effects across occupational-feminization groups



(a) Number of registered unemployed



(b) Number of new contracts

Notes: The figure reports marginal effects of a 10 percentage-point increase in AI exposure for occupations below the 25th percentile, between the 25th and 75th percentiles, and above the 75th percentile of the predetermined CNO2 female-employment share. All regressions include CNO4 and CNO1-by-year-month FE. October 2022 is the omitted month; shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4.

O.F Geographic heterogeneity

We next examine whether the exposure gradient differs between Spain's two largest urban labor markets and the rest of the country. The province-by-CNO4 panel pools Madrid and Barcelona into one group and compares it with all other provinces. The regressions allow the exposure gradient to

differ across the two geographic groups and between the adjustment and later periods. Province-by-CNO4 FE absorb permanent differences in occupation size across provinces, province-by-month FE absorb local aggregate shocks, and CNO1-by-month FE retain the preferred within-family comparison.

Table O.F.1: Geographic heterogeneity in AI exposure effects

Panel A. Number of registered unemployed				
	(1)	(2)	(3)	(4)
AI exposure × adjustment period	0.017***	0.009*	0.014**	0.008
	(0.006)	(0.005)	(0.006)	(0.005)
AI exposure × later period	0.025**	0.009	0.022*	0.008
	(0.011)	(0.011)	(0.012)	(0.011)
Geographic sample	MAD & BCN	Rest	MAD & BCN	Rest
2021 included	Yes	Yes	No	No
Pre-treatment joint-null p-value	0.027	0.004	0.701	0.016
p-value: equal phase effects	0.285	0.924	0.285	0.914
p-value: equal geography, adjustment	0.045		0.139	
p-value: equal geography, later	0.012		0.035	
Observations	61,328	1,268,611	50,480	1,039,618
Panel B. Number of new contracts				
	(5)	(6)	(7)	(8)
AI exposure × adjustment period	-0.004	0.010	-0.012	0.005
	(0.010)	(0.007)	(0.007)	(0.005)
AI exposure × later period	-0.006	0.010	-0.013	0.005
	(0.011)	(0.009)	(0.010)	(0.007)
Geographic sample	MAD & BCN	Rest	MAD & BCN	Rest
2021 included	Yes	Yes	No	No
Pre-treatment joint-null p-value	0.070	0.032	0.044	0.038
p-value: equal phase effects	0.778	0.919	0.761	0.928
p-value: equal geography, adjustment	0.050		0.013	
p-value: equal geography, later	0.085		0.039	
Observations	58,058	976,195	47,733	800,421

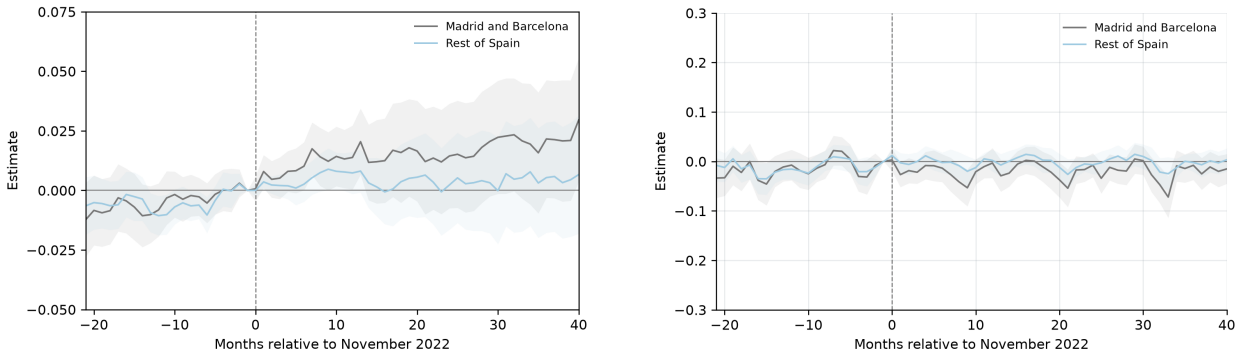
Notes: The table reports marginal effects from a pooled province-by-CNO4 panel. MAD & BCN denotes Madrid and Barcelona pooled together; Rest denotes all other Spanish provinces. Columns 1–2 and 5–6 use the full sample; columns 3–4 and 7–8 exclude 2021. The adjustment and later periods cover event times 0–24 and 25–40. Regressions include province-by-CNO4, province-by-year-month, and CNO1-by-year-month FE. Exposure is divided by 0.10, and standard errors are clustered by CNO4. Pre-treatment entries test the joint null over all available pre-treatment coefficients. The geographic-equality rows test whether the two group-specific exposure gradients are equal within each period. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table O.F.1 indicates a larger unemployment gradient in Madrid and Barcelona. In the full sample, a 10 percentage-point increase in AI exposure is associated with 1.7 percent more registered unemployment during the adjustment period and 2.5 percent more thereafter, compared with 0.9 percent in both periods across the remaining provinces. Equality between the geographic effects is rejected during adjustment ($p = 0.045$) and in the later period ($p = 0.012$). Excluding 2021 yields corresponding estimates of 1.4 and 2.2 percent for Madrid and Barcelona and 0.8 percent in both periods elsewhere. Under this restriction, equality is not rejected during adjustment ($p = 0.139$) but remains rejected later ($p = 0.035$). The later divergence is therefore less sensitive to the pandemic-recovery year than the adjustment-period difference.

The contract gradients move in the opposite direction across the two groups but are individually imprecise. In the full sample, the estimates for Madrid and Barcelona are -0.4 percent during adjustment and -0.6 percent later, compared with 1.0 percent in both periods for the rest of Spain. Excluding 2021 makes the contrast larger: the estimates are -1.2 and -1.3 percent in Madrid and Barcelona and 0.5 percent in both periods elsewhere. The geographic-equality tests reject under the restricted sample, but the absence of individually precise contract responses argues against interpreting these differences as evidence of a clear hiring mechanism.

Figure O.F.1 reports the corresponding dynamic estimates. The geographic results should be interpreted cautiously because the pre-treatment joint null is rejected in several specifications, particularly for the rest of Spain. Pooling Madrid and Barcelona may also capture differences in sectoral composition, occupational mobility, local demand, or adoption intensity rather than a distinct metropolitan causal effect. The exercise therefore documents where the aggregate exposure gradient is strongest, but does not identify why it differs geographically.

Figure O.F.1: Geographic heterogeneity in AI exposure effects



(a) Number of registered unemployed **(b)** Number of new contracts

Notes: The figure reports event-study estimates from the province-by-CNO4 panel for Madrid and Barcelona pooled together and for the remaining Spanish provinces. Each regression includes province-by-CNO4, province-by-year-month, and CNO1-by-year-month FE. October 2022 is the omitted month, and the vertical dashed line marks November 2022. Shaded areas are 95 percent confidence intervals based on standard errors clustered by CNO4.

O.G Alternative estimators

O.G.1 CDiD alternative

This subsection asks whether the main conclusion survives estimators that construct the counterfactual differently. Both alternatives are informative, but neither reproduces the preferred within-CNO1d comparison while retaining a continuous treatment.

Under the strong parallel-trends assumption, each TWFE coefficient can be interpreted as a regression-weighted average of average causal responses (ACRs) across AI exposure levels (Callaway et al., 2024). With common treatment timing, as in our setting, these weights are positive, avoiding the negative-weight problem associated with staggered adoption. Nevertheless, these regression-implied weights tend to place greater weight on causal responses at intermediate exposure levels than on responses in the tails of the exposure distribution.[30]

For this reason, we also apply the CDiD estimator proposed by Callaway et al. (2024). Rather than imposing the regression-derived weighting scheme of TWFE, this estimator directly estimates the causal response at different AI exposure levels under the strong parallel-trends assumption.[31] The main limitation of this estimator in our application is that it cannot directly incorporate the CNO1d-by-month FE used in the preferred design. We therefore report three alternative implementations, together with a specification that excludes 2021.

Table O.G.1 reports average causal responses on the treated (ACRTs) over the adjustment and later periods.[32] Columns 1 and 5 present our preferred CNO1d-stratified estimator. We estimate CDiD separately within families containing at least 15 positive-exposure and 5 zero-exposure occupations and aggregate the resulting family-specific responses using positive-exposure occupation shares.[33] This restriction retains CNO1d groups 1–5 and covers 87.6 percent of positive-exposure occupations in the unemployment sample and 82.3 percent in the contract sample. A 10 percentage-point increase in AI exposure is associated with 3.4 percent more registered unemployment during the adjustment period and 7.1 percent more thereafter. The corresponding contract estimates are 2.0 and 5.7 percent, with only the later-period effect statistically significant. The later-period responses

[30] The empirical relevance of this weighting depends on the degree of heterogeneity in causal responses across AI exposure levels. If the dose-response relationship is approximately linear, or causal responses vary little across exposure levels, TWFE and CDiD estimators are expected to deliver similar estimates. Larger differences are expected when causal responses vary substantially over the AI exposure distribution (Callaway et al., 2024).

[31] We implement the estimator using the R package `contdid`. The package provides event-study estimates, which trace causal responses around the public release of ChatGPT, and dose-response functions, which show how treatment effects vary across the AI exposure distribution.

[32] We construct separate averages for the adjustment period, $W_A = \{0, \ldots, 24\}$, and the later period, $W_L = \{25, \ldots, 40\}$, where event time 0 denotes November 2022. For either window W, the estimated response is $\widehat{\theta}_W = |W|^{-1} \sum_{t \in W} \widehat{\text{ACRT}}_t$. If a_W assigns weight $1/|W|$ to each event time in W, its estimated variance is $a_W' \widehat{\Sigma} a_W$, where $\widehat{\Sigma}$ is the influence-function covariance matrix. Family-specific estimates are aggregated using each family's share of positive-exposure occupations, with the corresponding covariance matrices entering with squared aggregation weights.

[33] At each event time, we aggregate the family-specific causal responses using each family's share of positive-exposure CNO4d occupations among all retained families. We then average these monthly estimates over event times 0–24 and 25–40.

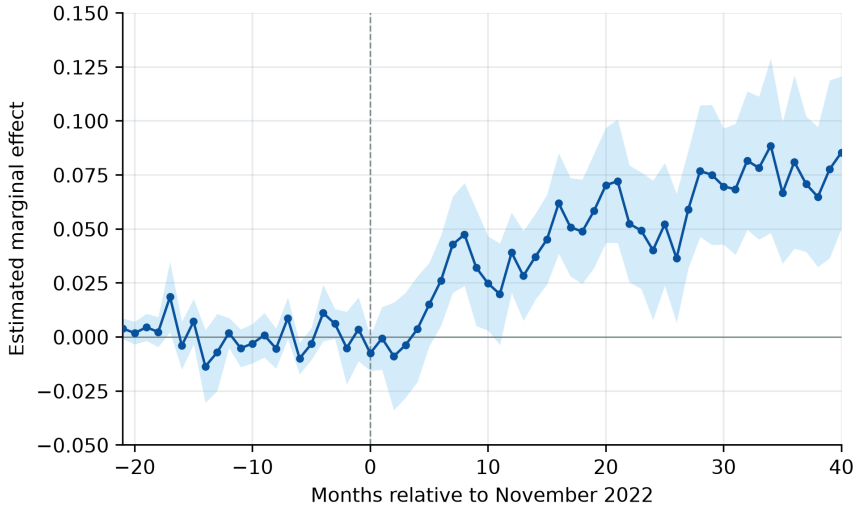
are also statistically different from their adjustment-period counterparts. Columns 2 and 6 repeat the CNO1d-stratified estimator after excluding 2021 and produce nearly identical point estimates.

Columns 3 and 7 report a control-based family adjustment. Before pooling the data, this approach subtracts from each occupation's outcome change the corresponding change among zero-exposure occupations in the same supported CNO1d family.[34] The resulting estimates closely track the stratified results: the unemployment responses are 3.5 percent during adjustment and 7.2 percent later, while the contract responses are 2.2 and 5.4 percent. Columns 4 and 8 report the unconditional CDiD estimator. Its later-period unemployment estimate rises to 10.9 percent, suggesting that comparisons across broad occupational families contribute to the larger pooled response; the corresponding contract estimate is 5.9 percent.

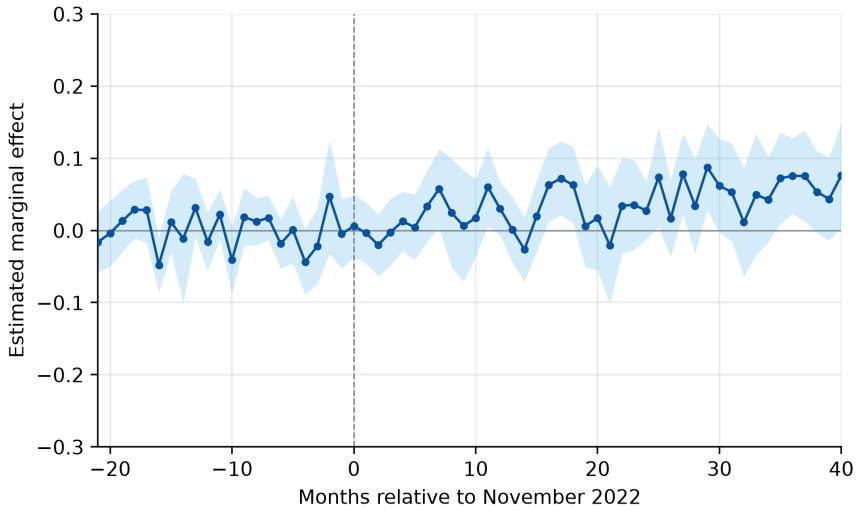
The CDiD estimates are larger than their TWFE counterparts but reinforce the positive direction of the unemployment result. Their identifying diagnostics remain less favorable, however. The pre-treatment joint null is rejected for unemployment in every specification. For contracts, it is rejected in the full-sample stratified, control-adjusted, and unconditional specifications, but not when the stratified estimator excludes 2021 ($p = 0.138$). We therefore interpret CDiD as complementary evidence rather than as a replacement for the preferred within-family TWFE design. Figure O.G.1 reports the dynamic estimates from the preferred CNO1d-stratified specification.

[34] For each supported CNO1d family and month, we calculate the mean outcome among zero-exposure occupations and its change relative to October 2022. We subtract this change from the outcome of every occupation in the same family and apply CDiD to the pooled adjusted sample. This procedure measures each occupation's evolution relative to zero-exposure occupations in its family, but it is not equivalent to absorbing CNO1d-by-month FE.

Figure O.G.1: CNO1-stratified CDiD event-study estimates



(a) Number of registered unemployed



(b) Number of new contracts

Notes: The figure reports CDiD estimates obtained separately within CNO1 families containing at least 15 positive-exposure and 5 zero-exposure occupations. Family-specific marginal responses are aggregated using each family's share of positive-exposure occupations. November 2022 is event time 0 and October 2022 is the reference month. Shaded areas are 95 percent confidence intervals based on the aggregated influence-function covariance matrices.

Table O.G.1: CDiD alternatives

	# of registered unemployed				# of new contracts			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI exposure × adjustment period	0.034***	0.034***	0.035***	0.040***	0.020	0.020	0.022	0.022
	(0.009)	(0.009)	(0.008)	(0.007)	(0.018)	(0.018)	(0.019)	(0.015)
AI exposure × later period	0.071***	0.071***	0.072***	0.109***	0.057**	0.057**	0.054**	0.059***
	(0.015)	(0.015)	(0.014)	(0.012)	(0.025)	(0.025)	(0.026)	(0.021)
p-value: equal phase effects	0.008	0.009	0.002	0.000	0.004	0.004	0.010	0.002
CNO1-stratified	Yes	Yes	No	No	Yes	Yes	No	No
Control-based family-time adjustment	No	No	Yes	No	No	No	Yes	No
Unconditional pooled comparison	No	No	No	Yes	No	No	No	Yes
2021 included	Yes	No	Yes	Yes	Yes	No	Yes	Yes
Pre-treatment joint-null p-value	0.002	0.007	< 0.001	< 0.001	< 0.001	0.138	< 0.001	< 0.001
Observations	20,727	16,779	20,727	31,563	19,530	15,810	19,530	30,303

Notes: Entries average monthly ACRTs over event times 0–24 and 25–40. Columns 1 and 5 estimate CDiD separately within CNO1 families containing at least 15 positive-exposure and 5 zero-exposure occupations and aggregate family estimates using positive-exposure occupation shares. Columns 2 and 6 repeat this estimator without 2021. Columns 3 and 7 subtract the change among zero-exposure occupations in each supported family before pooling; columns 4 and 8 are unconditional. Standard errors use linear combinations of the dynamic influence-function covariance matrices. The pre-treatment row reports a covariance-based Wald test of joint nullity over all available pre-treatment event times through −2. Exposure is measured in 10 percentage-point units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

O.G.2 SDiD

SDiD provides a complementary methodological approach. Continuous TWFE and SDiD differ along three core dimensions: weighting, estimand, and identification. In the continuous TWFE specification, the event-study coefficients are obtained from a single ordinary least squares (OLS) projection with FE. As such, the relevant weights are implicit residual-variation weights rather than transparent donor or time weights. By contrast, SDiD explicitly learns unit weights over donor occupations and time weights over pre-treatment months to construct a synthetic counterfactual[35]. This difference also maps into the estimand: the continuous TWFE coefficients are best interpreted as average causal response parameters, that is, the marginal effect of a higher exposure dose, rather than as a standard ATT. Conversely, the SDiD exercises in our design use a binary treated-versus-donor definition and therefore target an ATT for the exposed group relative to its synthetic control. Finally, their identifying assumptions are related but not identical. Continuous TWFE relies on a dose-based parallel trends restriction, together with strong functional-form discipline such as linearity in exposure and limited heterogeneity in marginal effects. SDiD relies on the existence of a convex donor combination that closely reproduces treated pre-trends and remains a valid counterfactual after treatment.

The baseline SDiD specification defines treatment as occupations with AI exposure above 0.1169, corresponding to the 75th percentile of the CNO4 exposure distribution, and therefore identifies the effect of being highly exposed relative to less exposed occupations. Allowing occupations with intermediate exposure to remain in the donor pool improves economic comparability, as these occupations are more likely to share latent determinants of counterfactual unemployment dynamics with the treated group, including related task content or differential scope for telework. As a complementary robustness exercise, we also estimate a sharper binary design in which treated occupations are those with AI exposure above 0.2, while the donor pool is restricted to occupations with zero exposure. This alternative specification is more stringent in ruling out AI exposure among controls and therefore yields an ATT that can be interpreted more directly as the effect of high exposure relative to no exposure. In both cases, we implement the SDiD estimator with projected covariates that residualize the outcome with respect to CNO1-by-month components before constructing the synthetic comparison, thereby purging occupation-family-specific dynamics and common monthly shocks from the variation used for identification.[36]

Table O.G.2 reports the SDiD estimates under the two treatment definitions. Under the baseline design, registered unemployment increases by 2.7 percent during the adjustment period and by 4.1 percent in the later period, while the corresponding contract effects remain close to zero and statistically imprecise. Under the more dichotomous design, which compares occupations with

[35] Conceptually, $\widehat{\tau}_{SDiD} = T_1^{-1} \sum_{t>T_0} \left(\overline{Y}_{Tt} - \sum_j \widehat{\omega}_j Y_{jt} \right) - \sum_{t \leq T_0} \widehat{\lambda}_t \left(\overline{Y}_{Tt} - \sum_j \widehat{\omega}_j Y_{jt} \right)$, where $\widehat{\omega}_j$ are donor weights and $\widehat{\lambda}_t$ are pre-period time weights.

[36] Inference uses 500 placebo reassignments. The CNO1-by-month projection is computed from outcome and family-month cells and is held fixed across placebo assignments. Event-study confidence intervals use the same number of reassignments.

exposure above 0.2 to zero-exposure occupations, the unemployment effect is 2.9 percent during the adjustment period, whereas the 1.9 percent later-period estimate is not statistically significant. Contract effects again remain statistically imprecise.

Figure O.G.2 reports the corresponding average outcome paths and event-study estimates for the baseline SDiD design. Relative to the continuous TWFE estimates, the synthetic weighting produces a closer pre-treatment match and again points to a positive unemployment response. This improved pre-treatment fit comes at the cost of imposing an ad hoc binary treatment definition and estimating a different parameter.

Table O.G.3 provides additional diagnostics for the donor and time weights from the baseline expanded-donor specification estimated over the full post-treatment period. Donor weights are broadly dispersed across occupations for both outcomes. Time weights, however, are substantially more concentrated for registered unemployment, with essentially all weight placed on a single pre-treatment month. We therefore interpret the SDiD estimates as complementary evidence rather than as a preferred alternative to the baseline specification. Table O.G.4 reports the treated occupations and the positive-weight donor occupations underlying this exercise.

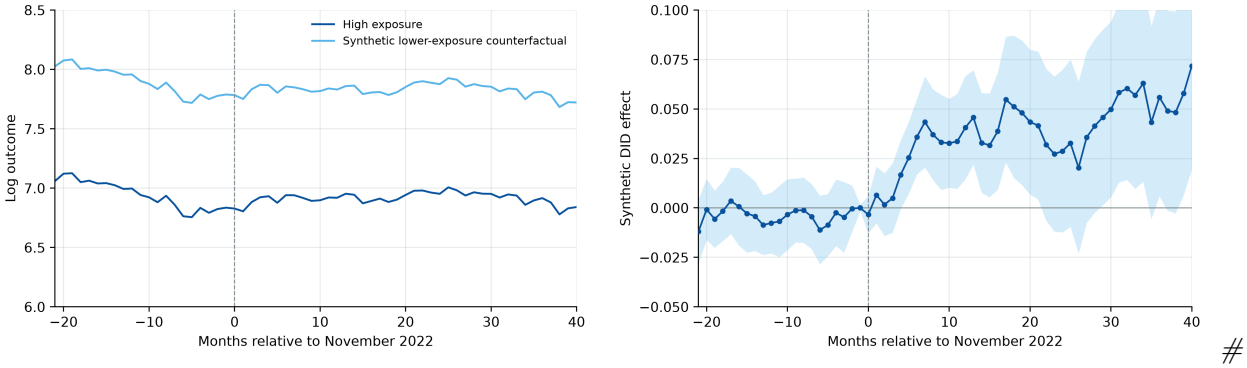
Table O.G.2: SDiD estimates

	# of registered unemployed		# of new contracts	
	(1)	(2)	(3)	(4)
High exposure: adjustment period	0.027**	0.029**	-0.000	0.024
	(0.011)	(0.013)	(0.022)	(0.030)
Impact (percent)	2.7	2.9	-0.0	2.5
High exposure: later period	0.041*	0.019	0.008	0.029
	(0.024)	(0.030)	(0.029)	(0.037)
Impact (percent)	4.1	1.9	0.8	3.0
Exposure threshold for treatment status	0.1169	0.2	0.1169	0.2
All lower-exposure occupations eligible as donors	Yes	No	Yes	No
Donor pool restricted to zero exposure	No	Yes	No	Yes
CNO1 × month residualization	Yes	Yes	Yes	Yes
Treated occupations: adjustment period	124	84	118	82
Donor occupations: adjustment period	377	252	365	246
Observations: adjustment period	23,547	15,792	22,701	15,416
Treated occupations: later period	124	84	118	82
Donor occupations: later period	377	252	363	245
Observations: later period	19,038	12,768	18,278	12,426

Notes: Columns 1 and 3 report the baseline SDiD design, where treated occupations have nearest-neighbor exposure above 0.1169 and every occupation at or below that cutoff remains eligible for the donor pool. Columns 2 and 4 report the more dichotomous robustness design, where treated occupations have exposure above 0.2 and donors are restricted to zero-exposure occupations. The adjustment-period estimates use all pre-treatment months and event times 0–24. The later-period estimates use all pre-treatment months and event times 25–40; event times 0–24 are omitted from those fits. Unit and time weights are re-estimated separately for each specification, outcome, and interval. All columns residualize outcomes on CNO1-by-month indicators before constructing the synthetic comparison. Because this residualization depends only on outcomes and family-month cells, it is held fixed across placebo assignments. Standard errors in parentheses use placebo inference with 500 repetitions. Impact rows report $100[\exp(\widehat{\tau}) - 1]$. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

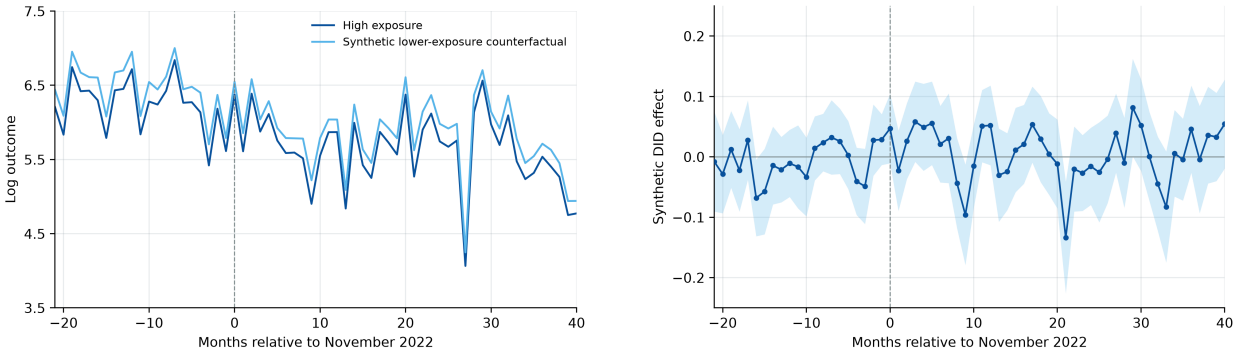
Figure O.G.2: CNO1-by-month-adjusted SDiD estimates

of registered unemployed



(a) Evolution **(b)** Event study

of new contracts



(c) Evolution **(d)** Event study

Notes: Treated occupations have nearest-neighbor AI exposure above 0.1169, while occupations at or below the cutoff form the donor pool. Before constructing the synthetic comparison, outcomes are adjusted for CNO1-by-month indicators using the projected covariate-adjustment procedure implemented within SDiD. Panels (a) and (c) compare the treated outcome path with the weighted synthetic counterfactual; panels (b) and (d) report the corresponding event-study estimates. The counterfactual paths are reported as returned by the estimator, without an additional graphical level shift. Unit weights minimize pre-treatment discrepancies between the treated and donor paths, and time weights emphasize pre-treatment months that best predict post-treatment donor outcomes. Event-study confidence intervals use placebo inference with 500 repetitions.

Table O.G.3: SDiD donor-weight diagnostics

	# of registered unemployed	# of new contracts
Positive-weight donors	377	363
Effective number of donors	374.6	362.2
Largest donor weight	0.004	0.003
Largest month weight	1.000	0.325
Share of weights in the 6 months prior to treatment	1.000	0.697
Share held by top 10 donors	0.034	0.031
Mean treated exposure	0.289	0.293
Donor-weighted exposure	0.017	0.017
Exposure contrast	0.272	0.277

Notes: The effective number of donors is $1/\sum_j \widehat{\omega}_j^2$. The exposure contrast is the treated mean minus the donor-weighted mean. Calculations use the CNO1-by-month-adjusted expanded-donor SDiD specification.

Table O.G.4: Treated occupations and positive-weight SDiD donors

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
Panel A. Treated occupations				
3820	Computer programmers	0.745		
4301	Data recorders	0.671		
3510	Agents and commercial representatives	0.628		
2613	Financial analysts	0.572		
2719	Software and multimedia analysts and designers not classified under other headings	0.519		
4113	Statistical, financial and banking services office employees	0.510		
2729	Database and computer network specialists not classified elsewhere	0.486		
3814	Web technicians	0.480		
3812	Information technology user assistance technicians	0.469		
3535	Spokespersons and public relations agents	0.453		
2652	Public relations professionals	0.453		
3613	Management and administrative assistants	0.453		
4309	Administrative employees without public service tasks not classified under other headings	0.453		
4500	Administrative employees with public service tasks not classified under other headings	0.453		
2822	Philosophers, historians and political science professionals	0.452		
3401	Support professionals and exchange, stock and finance intermediaries	0.441		
4412	Receptionists (except hotels)	0.434		

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
4411	User information employees	0.434		
4430	Survey agents	0.432		
2923	Philologists, interpreters and translators	0.430		
5823	Tourist escorts	0.405		
4421	Travel agency employees	0.405		
2630	Technicians of companies and tourist activities	0.405		
5825	Tour guides	0.405		
4223	Personnel service employees	0.405		
4112	Personnel control and payroll employees	0.405		
1327	Branch directors of banks, financial services and insurance	0.391		
1211	CFOs	0.391		
5895	Astrologers, fortune tellers and the like	0.385		
2411	Physicists and astronomers	0.385		
2821	Sociologists, geographers, anthropologists, archaeologists and the like	0.383		
1223	Research and development directors	0.378		
2484	Graphic and multimedia designers	0.367		
3614	Secretaries of medical centers or clinics	0.362		
2939	Creative and performing artists not classified elsewhere	0.356		
2931	Plastic and visual arts artists	0.356		
3739	Other technicians and professionals supporting cultural and artistic activities	0.356		
2612	Financial and investment advisors	0.350		
2611	Accounting specialists	0.348		
3811	Computer systems operations technicians	0.337		
2722	System and network administrators	0.337		
2721	Database designers and administrators	0.332		
5220	Salespeople in stores and warehouses	0.322		
5300	Store Owner Merchants	0.322		
3521	Insurance mediators and agents	0.319		
3160	Quality control technicians in physical, chemical and engineering sciences	0.315		
3129	Other technicians in the physical, chemical, environmental and engineering sciences	0.315		
3127	Laboratory technicians and analysts in industrial chemistry	0.315		
3121	Technicians in physical and chemical sciences	0.315		
2513	Judges and magistrates	0.311		

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
3403	bookkeepers	0.310		
4111	Accounting Clerks	0.310		
3322	Senior technicians in health documentation	0.306		
2325	Instructors in information technologies in non-regulated education	0.304		
2321	Specialists in teaching and pedagogical methods	0.304		
4445	Bill, debt collectors and related employees	0.299		
2230	Secondary education teachers (except specific vocational training subjects)	0.290		
3813	Network technicians	0.287		
5420	Telemarketing operators	0.285		
3534	Real estate agents and managers	0.283		
2625	Personnel training specialists	0.279		
2711	Systems analysts	0.276		
2712	Software analysts and designers	0.276		
2640	Technical and medical sales professionals (except ICT)	0.271		
2653	Professionals in the sale of information and communications technologies	0.271		
5210	Store and warehouse section heads	0.263		
1432	Directors and managers of retail companies	0.263		
2329	Teachers and teaching professionals not classified under other headings	0.262		
2413	Chemicals	0.261		
2713	Analysts, programmers and web and multimedia designers	0.249		
2921	Writers	0.246		
3323	Higher dietetics technicians	0.245		
2621	Management and organization analysts	0.243		
2810	Economists	0.242		
1222	Advertising and public relations directors	0.232		
3128	Metallurgy and mining technicians	0.229		
2324	Non-formal arts education teachers	0.214		
2323	Non-regulated music and dance teaching teachers	0.214		
2416	Statisticians	0.211		
3404	Support professionals in statistical, mathematical and related services	0.211		
2922	Journalists	0.210		
2427	winemakers	0.204		

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
2912	Librarians, documentary filmmakers and the like	0.203		
3522	Purchasing agents	0.200		
3312	Imaging technicians for diagnosis	0.199		
2723	Computer network analysts	0.199		
3731	Photographers	0.195		
5891	Personal assistants or company people	0.188		
5833	Building janitors	0.188		
3402	Commercial loans and credits	0.186		
3611	Secretarial supervisors	0.186		
1324	Senior Social Services Directors	0.181		
1113	Directors of social interest organizations	0.181		
1325	Directors of other social services	0.181		
4222	Printing coders and correctors	0.178		
9420	Advertising delivery people, shoe shiners and other street workers	0.176		
9410	street vendors	0.176		
5412	Vendors in occasional markets and flea markets	0.176		
5499	Sellers not classified under other headings	0.176		
5491	Home sellers	0.176		
5411	Sellers at kiosks	0.176		
2651	Advertising and marketing professionals	0.173		
4422	Hotel receptionists	0.172		
2592	Attorneys	0.167		
2511	Lawyers	0.167		
2599	Legal professionals not classified under other headings	0.167		
3313	Technicians in pathological anatomy and cytology	0.158		
5894	Driving school instructors	0.156		
1321	Directors of information and communications technology (ICT) services	0.156		
2426	Environmental protection professionals	0.144		
2326	Environmental education professionals	0.144		
1431	Directors and managers of wholesale trade companies	0.138		
1509	Directors and managers of waste management companies and other service companies not classified under other headings	0.138		
2153	Dietitians and nutritionists	0.133		

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
5922	Autonomous police	0.123		
5923	local police	0.123		
5921	national police	0.123		
5840	Workers who own small accommodation	0.121		
1411	Hotel directors and managers	0.121		
1419	Directors and managers of other hosting services companies	0.121		
3139	Process control technicians not classified under other headings	0.121		
1316	Construction company directors	0.119		
3202	Construction supervisors	0.119		
3539	Artistic and sporting representatives and other commercial service agents not classified under other headings	0.118		
Panel B. Donors with positive weight in at least one outcome				
9812	Drivers of animal-drawn vehicles for the transportation of people and similar	0.000	0.0043	0.0028
2240	Primary school teachers	0.103	0.0034	0.0031
5991	prison guards	0.000	0.0033	0.0000
1322	Directors of children's social services	0.040	0.0033	0.0024
5721	Childminders in daycare centers and educational centers	0.012	0.0030	0.0033
7192	Technical facade installers	0.000	0.0033	0.0028
2251	Early childhood education teachers	0.000	0.0033	0.0031
2312	Special education technical educators	0.099	0.0033	0.0028
2311	Special education teachers	0.099	0.0032	0.0030
2322	Non-formal language teaching teachers	0.064	0.0032	0.0029
7706	Workers in the production of alcoholic beverages other than wine	0.000	0.0032	0.0025
4441	Bank tellers and similar	0.085	0.0032	0.0026
7532	Installers and repairers in electro medicine	0.000	0.0031	0.0025
2210	Professors of universities and other higher education (except vocational training)	0.064	0.0029	0.0031
2154	Speech therapists	0.000	0.0031	0.0028
2220	Vocational training teachers (specific subjects)	0.000	0.0030	0.0031
8155	Fur and leather treatment machine operators	0.000	0.0026	0.0031
2252	Early childhood education technicians	0.040	0.0031	0.0028
7837	Shoemakers and the like	0.000	0.0027	0.0031
3153	Aviation pilots and related professionals	0.000	0.0026	0.0030

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
3621	Support professionals of the public tax administration	0.028	0.0030	0.0028
7311	Shapers and macheros	0.000	0.0025	0.0030
8156	Machine operators for the manufacturing of footwear, leather goods and leather gloves	0.000	0.0027	0.0030
7836	Tanners and leather preparers	0.000	0.0026	0.0030
7833	Cutters for fabrics, leather, skin and other materials	0.015	0.0026	0.0030
8111	Miners and other operators at mining facilities	0.000	0.0025	0.0030
4443	Game room and related employees	0.000	0.0025	0.0030
3629	Other public administration support professionals for inspection and control tasks and similar tasks	0.000	0.0030	0.0029
8311	Locomotive drivers	0.000	0.0028	0.0030
7831	Tailors, dressmakers, furriers and milliners	0.034	0.0027	0.0030
3339	Other alternative therapy professionals	0.000	0.0028	0.0030
3316	Dental prosthesis technicians	0.000	0.0028	0.0030
3204	Chemical and pharmaceutical industry supervisors	0.000	0.0027	0.0030
3714	Promoters of equal opportunities between women and men	0.079	0.0030	0.0030
3125	Electronic technicians, specialty in electro medicine	0.000	0.0030	0.0028
7892	Punchers	0.000	0.0026	0.0030
5493	Fashion, art and advertising models	0.000	0.0024	0.0030
4424	Teleoperators	0.000	0.0027	0.0029
7835	Upholsterers, mattress makers and the like	0.000	0.0026	0.0029
6202	Qualified workers in sheep and goat livestock activities	0.000	0.0027	0.0029
1421	Restaurant directors and managers	0.000	0.0025	0.0029
3732	Interior designers and decorators	0.000	0.0027	0.0029
3331	Professionals of acupuncture, naturopathy, homeopathy, traditional Chinese medicine and Ayurveda	0.000	0.0028	0.0029
7834	Hand sewers, embroiderers and the like	0.034	0.0027	0.0029
3711	Support professionals for legal services and similar services	0.000	0.0027	0.0029
2415	Mathematicians and actuaries	0.054	0.0029	0.0027
3734	Chefs	0.000	0.0023	0.0029
3523	Consignees	0.017	0.0027	0.0029

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
2825	Agents of equal opportunities between women and men	0.028	0.0029	0.0025
5824	Ground stewards	0.048	0.0026	0.0029
7112	Structural precast erectors (concrete only)	0.000	0.0029	0.0027
1313	Manufacturing industry directors	0.013	0.0026	0.0029
7295	Building facade and chimney cleaning staff	0.000	0.0029	0.0026
4122	Production Support Services Office Employees	0.093	0.0026	0.0029
7619	Artisans not classified under other headings	0.054	0.0027	0.0029
9512	Agricultural laborers in orchards, greenhouses, nurseries and gardens	0.020	0.0027	0.0029
2830	Priests of different religions	0.112	0.0027	0.0029
7612	Luthiers and similar; musical instrument tuners	0.000	0.0028	0.0029
3155	Aeronautical safety technicians	0.000	0.0029	0.0029
7832	Pattern makers for textile and leather products	0.000	0.0027	0.0029
3110	Draftsmen and technical draftsmen	0.000	0.0027	0.0029
5821	Flight attendants and waiters on airplanes, ships and trains	0.117	0.0026	0.0029
5812	Specialists in aesthetic, wellness and related treatments	0.000	0.0028	0.0029
7899	Officials, workers and artisans of other trades not classified under other headings	0.054	0.0029	0.0027
3531	customs representatives	0.017	0.0029	0.0025
1212	Human resources directors	0.000	0.0027	0.0029
9100	Domestic employees	0.000	0.0027	0.0029
7222	Gas fitters-installers in buildings	0.000	0.0026	0.0029
3721	Athletes and sportsmen	0.000	0.0029	0.0024
8153	Sewing and embroidery machine operators	0.000	0.0028	0.0029
2464	Aeronautical technical engineers	0.000	0.0026	0.0029
3833	Telecommunications engineering technicians	0.000	0.0028	0.0029
2412	Meteorologists	0.038	0.0021	0.0029
3321	Senior technicians in oral hygiene	0.000	0.0029	0.0029
4121	Supply and inventory control employees	0.000	0.0026	0.0029
7122	Stonemasons, cutters, carvers and stone engravers	0.000	0.0026	0.0029
3713	Professionals supporting work and social education	0.000	0.0029	0.0028
3532	Conference and event organizers	0.102	0.0027	0.0029
8141	Operators of machines to manufacture rubber products and derivatives of natural resins	0.000	0.0027	0.0029

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
2452	Landscape architects	0.000	0.0028	0.0029
7613	Jewelers, goldsmiths and silversmiths	0.000	0.0026	0.0029
8154	Textile bleaching, dyeing, printing and finishing machine operators	0.021	0.0027	0.0029
8152	Operators of looms and other weaving machines	0.032	0.0027	0.0029
1221	Commercial and sales directors	0.043	0.0027	0.0029
9542	Aquaculture laborers	0.000	0.0022	0.0029
7611	Watchmakers and precision instrument mechanics	0.000	0.0029	0.0028
1329	Directors of other professional services companies not classified under other headings	0.033	0.0028	0.0029
9602	Building construction laborers	0.028	0.0026	0.0029
5832	Domestic service butlers	0.000	0.0025	0.0029
6204	Skilled workers in beekeeping and sericulture	0.000	0.0027	0.0029
9320	Fast food preparers	0.000	0.0026	0.0029
3733	Technicians in art galleries, museums and libraries	0.000	0.0027	0.0029
4210	Library and Archives Employees	0.098	0.0027	0.0029
7617	Craftsmen in wood and similar materials	0.054	0.0028	0.0029
9434	Meter readers and collectors of recreational and vending machines	0.000	0.0024	0.0029
6110	Skilled workers in agricultural activities (except in orchards, greenhouses, nurseries and gardens)	0.020	0.0029	0.0027
5811	Hairdressers	0.030	0.0028	0.0029
0012	Non-commissioned officers of the armed forces	0.000	0.0028	0.0000
1323	Directors-managers of health centers	0.066	0.0025	0.0028
3126	Mechanical technicians	0.000	0.0026	0.0028
5892	Undertakers and embalmers	0.000	0.0028	0.0027
8420	Bus and tram drivers	0.000	0.0027	0.0028
3142	Agricultural technicians	0.006	0.0027	0.0028
2111	family doctors	0.000	0.0028	0.0026
2434	Aeronautical engineers	0.075	0.0022	0.0028
7402	Aircraft engine mechanics and adjusters	0.000	0.0027	0.0028
3612	Legal-legal assistants	0.000	0.0028	0.0025
2624	Specialists in personnel and related policies and services	0.065	0.0027	0.0028
3405	Appraisers	0.000	0.0028	0.0027
3206	Supervisors of wood and pulp and paper industries	0.000	0.0026	0.0028

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
3141	Technicians in biological sciences (except in healthcare areas)	0.060	0.0028	0.0028
9310	kitchen helpers	0.012	0.0025	0.0028
5932	forest firefighters	0.000	0.0021	0.0028
3329	Health technicians not classified under other headings	0.000	0.0028	0.0026
6422	Coastal and freshwater fishermen	0.000	0.0025	0.0028
2151	Dentists and stomatologists	0.000	0.0027	0.0028
3123	Electrical technicians	0.000	0.0026	0.0028
2451	Architects (except landscape architects and urban planners)	0.078	0.0026	0.0028
3724	Monitors of recreational and entertainment activities	0.000	0.0028	0.0028
6209	Qualified workers in livestock activities not classified under other headings	0.000	0.0027	0.0028
8199	Operators of fixed installations and machinery not classified under other headings	0.022	0.0028	0.0027
1429	Directors and managers of catering companies and other restaurant companies	0.000	0.0028	0.0026
7621	Prepress process workers	0.102	0.0027	0.0028
6120	Qualified workers in orchards, greenhouses, nurseries and gardens	0.020	0.0027	0.0028
1219	Directors of policy and planning and other administrative departments not classified elsewhere	0.033	0.0027	0.0028
3722	Coaches and referees of sports activities	0.000	0.0028	0.0026
5492	Sales promoters	0.079	0.0028	0.0028
3122	Construction technicians	0.078	0.0026	0.0028
4423	Telephonists	0.000	0.0026	0.0028
2157	Podiatrists	0.000	0.0025	0.0028
2466	Mining, metallurgical and related technical engineers	0.000	0.0027	0.0028
6410	Qualified workers in forestry and natural environment activities	0.000	0.0025	0.0028
9511	Agricultural laborers (except in orchards, greenhouses, nurseries and gardens)	0.020	0.0028	0.0028
2437	Environmental engineers	0.036	0.0027	0.0028
3315	Orthoprosthetics technicians	0.000	0.0023	0.0028
1120	CEOs and CEOs	0.033	0.0026	0.0028
2436	Mining, metallurgical and related engineers	0.000	0.0024	0.0028
4442	Betting sales employees	0.000	0.0027	0.0028

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
6203	Qualified workers in pig farming activities	0.000	0.0028	0.0027
8132	Machine operators for manufacturing pharmaceutical, cosmetic and related products	0.000	0.0028	0.0027
9490	Other elementary occupations	0.000	0.0028	0.0028
7820	Cabinetmakers and related workers	0.000	0.0026	0.0028
2155	Opticians-optometrists	0.000	0.0028	0.0027
7893	Product sorters and testers (except food, beverages and tobacco)	0.032	0.0027	0.0028
2935	Actors	0.101	0.0026	0.0028
5110	Salaried cooks	0.012	0.0026	0.0028
9433	Delivery drivers, messengers and couriers on foot	0.041	0.0026	0.0028
2439	Engineers not classified under other headings	0.066	0.0027	0.0028
2622	Specialists in business policy administration	0.033	0.0026	0.0028
4123	Logistics and passenger and merchandise transportation employees	0.096	0.0026	0.0028
2414	Geologists and geophysicists	0.043	0.0026	0.0028
3131	Technicians in energy production facilities	0.026	0.0026	0.0028
5722	Child care workers at home	0.012	0.0028	0.0028
4446	Post Office Counter Employees	0.000	0.0028	0.0000
2435	Chemical engineers	0.000	0.0027	0.0028
2469	Technical engineers not classified under other headings	0.000	0.0028	0.0028
3715	Community animators	0.000	0.0027	0.0028
7131	Carpenters (except cabinetmakers)	0.000	0.0026	0.0028
9601	Public works laborers	0.028	0.0026	0.0028
8331	Operators of earthmoving machinery and similar equipment	0.000	0.0024	0.0028
2512	Prosecutors	0.102	0.0026	0.0028
3209	Supervisors in other manufacturing industries	0.000	0.0027	0.0028
8191	Kiln operators and glass and ceramic installations	0.000	0.0028	0.0028
8151	Fiber preparation, spinning and winding machine operators	0.000	0.0027	0.0028
9520	Livestock workers	0.000	0.0026	0.0028
3132	Technicians in waste and water treatment facilities and other operators in similar plants	0.000	0.0026	0.0028
3533	Agents or intermediaries in the hiring of labor (except entertainment representatives)	0.000	0.0027	0.0028
9222	Vehicle cleaners	0.000	0.0025	0.0028

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
5629	People care workers in health services not classified under other headings	0.000	0.0028	0.0027
8121	Operators in facilities for obtaining and transforming metals	0.000	0.0025	0.0028
7132	Installers of metal enclosures and metal carpenters (except assemblers of metal structures)	0.000	0.0026	0.0028
2431	Industrial and production engineers	0.037	0.0026	0.0028
2482	Product and clothing designers	0.044	0.0027	0.0028
2453	Urban planners and traffic engineers	0.096	0.0028	0.0000
7522	Power line installers and repairers	0.000	0.0026	0.0028
5710	Home-based personal care workers	0.000	0.0028	0.0027
8142	Machine operators for manufacturing plastic products	0.000	0.0027	0.0028
2441	electrical engineers	0.059	0.0026	0.0028
3723	Sports activity instructors	0.000	0.0026	0.0028
2936	Radio, television and other presenters	0.063	0.0028	0.0027
7811	Wood treatment workers	0.000	0.0025	0.0028
8322	Mobile forestry machinery operators	0.000	0.0022	0.0028
9603	Laborers in mining, quarries and other extractive industries	0.000	0.0026	0.0028
2461	Industrial and production technical engineers	0.000	0.0026	0.0028
5831	Maintenance and cleaning supervisors in offices, hotels and other establishments	0.044	0.0027	0.0028
2424	Agricultural technical engineers	0.000	0.0027	0.0028
8412	Salaried drivers of cars, taxis and vans	0.000	0.0027	0.0028
5500	Cashiers and lockers (except banks)	0.085	0.0028	0.0027
7293	Glaziers	0.000	0.0026	0.0028
2421	Biologists, botanists, zoologists and the like	0.061	0.0027	0.0028
8112	Operators in facilities for the preparation of minerals and rocks	0.000	0.0026	0.0028
3152	Captains and deck officers	0.000	0.0024	0.0028
7705	Fruit and vegetable canning workers and non-alcoholic beverage production workers	0.000	0.0028	0.0026
7223	Duct installers in public works	0.012	0.0026	0.0028
6201	Qualified workers in beef livestock activities	0.000	0.0026	0.0028
7701	Butchers and meat industry workers	0.000	0.0028	0.0026
2465	Chemical technical engineers	0.000	0.0025	0.0028
8440	Motorcycle and moped drivers	0.000	0.0026	0.0028
8201	Mechanical machinery assemblers	0.000	0.0028	0.0027
7221	Plumbers	0.012	0.0026	0.0028

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
7615	Glass blowers, shapers, laminators, cutters and polishers	0.000	0.0026	0.0028
8431	Truck Owner Drivers	0.000	0.0027	0.0028
3203	Food and tobacco industry supervisors	0.056	0.0027	0.0028
7709	Food and beverage tasters and classifiers	0.000	0.0028	0.0027
8432	Salaried truck drivers	0.000	0.0027	0.0028
7193	Installers of waterproofing systems in buildings	0.000	0.0026	0.0028
3325	Physiotherapist assistants	0.000	0.0026	0.0028
2481	Technical architects and urban planning technicians	0.078	0.0026	0.0028
1112	Management personnel of the public administration	0.000	0.0025	0.0028
8133	Photographic and related laboratory operators	0.011	0.0028	0.0026
2443	Telecommunications engineers	0.000	0.0026	0.0028
5931	Firefighters (except forestry)	0.000	0.0021	0.0028
7708	Preparers and processors of tobacco and its products	0.000	0.0024	0.0028
3205	Supervisors of plastic, rubber and natural resin transformation industries	0.000	0.0028	0.0026
2824	Work and social education professionals	0.000	0.0028	0.0027
7618	Artisans in fabrics, leather and similar materials, fiber preparers and weavers with artisan looms or knitting and related fabrics	0.054	0.0025	0.0028
7623	Binding process workers	0.000	0.0027	0.0028
1501	Directors and managers of recreational, cultural and sports activities companies	0.044	0.0028	0.0026
2463	Mechanical technical engineers	0.081	0.0026	0.0028
8312	Railway shunting agents	0.000	0.0028	0.0028
2937	Bullfighting professionals	0.044	0.0021	0.0028
6423	Deep sea fishermen	0.000	0.0023	0.0028
8122	Operators of metal polishing, galvanizing and coating machines	0.000	0.0027	0.0028
3326	Technicians in occupational risk prevention and environmental health	0.000	0.0027	0.0028
2433	mechanical engineers	0.081	0.0025	0.0028
7531	Mechanics and repairers of electronic equipment	0.000	0.0026	0.0028
8209	Assemblers and assemblers not classified elsewhere	0.000	0.0027	0.0028

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
2623	Public administration specialists	0.000	0.0028	0.0026
5120	Salaried waiters	0.000	0.0025	0.0028
7616	Sign makers, glass engravers, decorative painters of various items	0.000	0.0027	0.0028
0011	Armed forces officers	0.000	0.0025	0.0028
2473	Telecommunications technical engineers	0.000	0.0028	0.0027
5893	Animal caretakers and trainers	0.000	0.0028	0.0027
7191	Building maintainers	0.000	0.0026	0.0028
2471	Electrical technical engineers	0.059	0.0028	0.0025
7401	Motor Vehicle Mechanics and Adjusters	0.000	0.0026	0.0027
3133	Technicians in control of chemical product processing facilities	0.000	0.0027	0.0027
2823	Psychologists	0.059	0.0027	0.0027
2432	Construction and civil works engineers	0.008	0.0026	0.0027
2422	Agronomists	0.000	0.0026	0.0027
8170	Laundry and dry cleaning machine operators	0.000	0.0027	0.0027
5000	Waiters and chefs-owners	0.000	0.0027	0.0027
1326	Directors of education services	0.052	0.0027	0.0027
5942	Security guard assistants and similar not authorized to be armed	0.000	0.0027	0.0027
2158	Health and occupational and environmental hygiene professionals	0.000	0.0026	0.0027
8144	Operators of sawmills, panel manufacturing machines and related installations for the treatment of wood and cork	0.000	0.0026	0.0027
7314	Assemblers of metal structures	0.000	0.0025	0.0027
7703	Bakers, pastry chefs and confectioners	0.000	0.0027	0.0027
3317	Hearing aid technicians	0.000	0.0026	0.0027
9530	Agricultural laborers	0.000	0.0026	0.0027
8340	Deck sailors, engine sailors and the like	0.000	0.0024	0.0027
7121	Masons	0.000	0.0026	0.0027
7199	Other workers in structural construction works not classified under other headings	0.000	0.0026	0.0027
3831	Audiovisual recording technicians	0.000	0.0027	0.0027
9431	Ordinances	0.075	0.0027	0.0027
1422	Directors and managers of bars, cafes and similar	0.000	0.0027	0.0027
3124	Electronic technicians (except electro medicine)	0.020	0.0027	0.0027
7533	Installers and repairers in information and communications technologies	0.000	0.0027	0.0027

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
8193	Packaging, bottling and labeling machine operators	0.000	0.0027	0.0027
9229	Other cleaning staff	0.000	0.0024	0.0027
2442	electronic engineers	0.100	0.0024	0.0027
5621	Assistant pharmacy technicians	0.073	0.0027	0.0027
7614	Ceramic workers, potters and the like	0.000	0.0027	0.0027
3134	Oil and natural gas refinery technicians	0.041	0.0024	0.0027
2911	Archivists and museum curators	0.000	0.0027	0.0027
2423	forestry engineers	0.000	0.0024	0.0027
3327	Veterinary assistants	0.018	0.0027	0.0027
7111	Formworkers and concrete construction workers	0.000	0.0026	0.0027
5612	Primary care nursing assistants	0.000	0.0027	0.0026
6300	Qualified workers in mixed agricultural activities	0.000	0.0027	0.0027
9443	Sweepers and the like	0.000	0.0027	0.0027
7510	Construction electricians and related	0.000	0.0026	0.0027
6205	Skilled workers in poultry and rabbit farming	0.000	0.0027	0.0027
7521	Mechanics and electrical equipment repairers	0.000	0.0026	0.0027
7250	Refrigeration and air conditioning mechanics-installers	0.019	0.0025	0.0027
7321	Blacksmiths and forgers	0.000	0.0025	0.0027
8159	Operators of machines for manufacturing textile products not classified under other headings	0.032	0.0027	0.0027
9223	window cleaners	0.000	0.0026	0.0027
5611	Hospital nursing assistants	0.000	0.0027	0.0027
7292	Installers of thermal insulation and soundproofing material	0.044	0.0027	0.0027
7404	Mechanics and adjusters of naval and railway machinery	0.000	0.0025	0.0027
3324	Optometry technicians	0.000	0.0027	0.0027
2454	Geographic engineers and cartographers	0.080	0.0024	0.0027
2472	Electronic technical engineers	0.020	0.0026	0.0027
7403	Mechanics and adjusters of agricultural and industrial machinery	0.000	0.0025	0.0027
9543	Forestry and hunting laborers	0.000	0.0025	0.0027
7291	Cover fitters	0.016	0.0025	0.0027
3712	Private detectives	0.000	0.0027	0.0000
5941	Security guards and similar authorized to be armed	0.000	0.0027	0.0027

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
3622	Support professionals of the public administration of social services	0.000	0.0027	0.0026
9820	Replenishers	0.000	0.0026	0.0027
5899	Personal service workers not classified under other headings	0.041	0.0027	0.0027
7240	Tilers, parquet installers and the like	0.000	0.0026	0.0027
9210	Cleaning staff for offices, hotels and other similar establishments	0.000	0.0027	0.0027
8192	Boiler and steam engine operators	0.000	0.0024	0.0027
7704	Workers in the processing of milk and production of dairy products (including ice cream)	0.000	0.0024	0.0027
2159	Health professionals not classified under other headings	0.000	0.0027	0.0027
7232	Painters in manufacturing industries	0.000	0.0026	0.0027
7231	Painters and wallpaperers	0.000	0.0026	0.0027
3311	Radiotherapy technicians	0.000	0.0024	0.0027
8114	Machinery operators to manufacture products derived from non-metallic minerals	0.000	0.0025	0.0027
8131	Operators in chemical industrial plants	0.000	0.0026	0.0027
7812	Woodworking machine fitters and operators	0.000	0.0026	0.0027
2933	Choreographers and dancers	0.080	0.0026	0.0027
8143	Machine operators for manufacturing paper and cardboard products	0.000	0.0027	0.0027
3151	Chiefs and engineering officers	0.000	0.0024	0.0027
7894	Fumigators and other pest and weed controllers	0.000	0.0027	0.0027
8202	Assemblers of electrical and electronic equipment	0.000	0.0027	0.0027
7212	Paste and mortar coating applicators	0.000	0.0026	0.0027
3832	broadcast technicians	0.020	0.0026	0.0027
9700	Laborers in manufacturing industries	0.000	0.0027	0.0027
9441	Waste collectors	0.000	0.0025	0.0027
7312	Welders and oxycutters	0.000	0.0025	0.0027
7324	Metal polishers and tool sharpeners	0.000	0.0026	0.0027
7405	Bicycle repairers and the like	0.000	0.0026	0.0027
7294	Solar energy panel assemblers-installers	0.000	0.0026	0.0027
5430	Gas station dispensers	0.000	0.0027	0.0027
2934	Film, theater and related directors	0.092	0.0026	0.0027
7313	Sheet metal workers and boilermakers	0.000	0.0025	0.0027

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
1315	Directors of supply, transportation, distribution and related companies	0.096	0.0027	0.0027
9221	Hand dry cleaners and similar	0.000	0.0027	0.0027
2932	Composers, musicians and singers	0.037	0.0027	0.0026
7622	Printing process workers	0.000	0.0027	0.0027
8411	Drivers who own cars, taxis and vans	0.000	0.0027	0.0024
9811	Freight transport workers and unloaders	0.000	0.0026	0.0027
5993	Forestry and environmental agents	0.000	0.0025	0.0027
8160	Machine operators to produce food, beverage and tobacco products	0.000	0.0027	0.0026
9541	fishing laborers	0.000	0.0024	0.0027
5999	Protection and security services workers not classified under other headings	0.057	0.0026	0.0027
6421	Skilled workers in aquaculture	0.000	0.0026	0.0027
5822	Ground transportation conductors and collectors	0.117	0.0027	0.0025
2483	Technical engineers in topography	0.000	0.0024	0.0027
8321	Mobile agricultural machinery operators	0.000	0.0025	0.0026
5622	Health emergency technicians	0.000	0.0026	0.0026
8145	Operators in pulp preparation and paper manufacturing facilities	0.000	0.0026	0.0026
2130	Veterinarians	0.093	0.0026	0.0026
7211	plasterers	0.000	0.0026	0.0026
3201	Supervisors in mining engineering	0.030	0.0021	0.0026
7315	Assemblers of cabled structures and cable splicers	0.000	0.0026	0.0026
3314	Technicians in clinical diagnostic laboratory	0.000	0.0026	0.0026
2122	Specialized nurses (except midwives)	0.094	0.0026	0.0025
2140	Pharmacists	0.090	0.0026	0.0026
7702	Fish industry workers	0.000	0.0025	0.0026
3143	Forestry and natural environment technicians	0.022	0.0023	0.0026
8333	Forklift operators	0.000	0.0026	0.0026
2462	Technical public works engineers	0.008	0.0026	0.0026
3207	Production supervisors in graphic arts industries and paper products manufacturing	0.000	0.0026	0.0026
2425	Forestry and natural environment technical engineers	0.022	0.0023	0.0026
2156	Occupational therapists	0.008	0.0025	0.0026
7707	Wine making workers	0.000	0.0026	0.0026
7322	Tool manufacturing workers, mechanic-fitters, model makers, tool makers and the like	0.000	0.0025	0.0026

Table O.G.4 – continued

CNO4	Occupation	Exposure	Weight: unemployed	Weight: contracts
8113	Surveyors and related workers	0.000	0.0026	0.0026
8332	Operators of cranes, forklifts and similar material movement machinery	0.000	0.0026	0.0025
9432	Baggage handlers and similar	0.073	0.0026	0.0026
9442	Waste sorters, clean point operators and scrap collectors	0.000	0.0026	0.0024
1311	Production directors of agricultural and forestry farms	0.000	0.0024	0.0026
5910	civil guards	0.000	0.0026	0.0000
0020	Troops and sailors of the armed forces	0.000	0.0026	0.0000
2152	Physiotherapists	0.017	0.0024	0.0026
3154	air traffic controllers	0.000	0.0023	0.0026
4221	Postal service employees (except counter employees)	0.000	0.0025	0.0024
7323	Machine tool adjusters and operators	0.000	0.0025	0.0025
3135	Technicians in control of metal production processes	0.000	0.0025	0.0024
7891	Divers	0.000	0.0024	0.0025
2121	Non-specialized nurses	0.000	0.0023	0.0025
3716	Lay auxiliaries of religions	0.112	0.0023	0.0025
2123	Midwives	0.052	0.0025	0.0024
2112	Other medical specialists	0.000	0.0024	0.0025
1111	Members of the executive branch (national, regional and local) and the legislative branch	0.000	0.0025	0.0000
1312	Production directors of fishing and aquaculture farms	0.000	0.0024	0.0000
5992	Bathers-lifeguards	0.000	0.0021	0.0024
6430	Qualified workers in hunting activities	0.000	0.0000	0.0024
1314	Directors of mining operations	0.013	0.0024	0.0000
3623	Professionals supporting public administration licensing services	0.102	0.0023	0.0000
4444	Pawn and loan shop employees	0.065	0.0022	0.0000
3632	Civil Guard non-commissioned officers	0.037	0.0022	0.0000
2591	Notaries and registrars	0.022	0.0021	0.0000
3631	National, regional and local police technicians	0.037	0.0021	0.0000

Notes: Panel A lists occupations above the fixed 0.1169 cutoff. Panel B is ordered by the larger of the two outcome-specific weights and omits donors receiving zero weight in both estimations. Donor weights are obtained from the full-post SDiD specification (November 2022 to March 2026) and do not correspond to separate re-estimated weights for the adjustment or later periods. Occupation descriptions are English translations of official CNO titles. Donor weights sum to one separately by outcome.

O.H Task bundling and labor-market adjustment

Following Garicano et al. (2026a), this section examines labor-market adjustment after ChatGPT's release across Spanish occupations categorized into three tiers based on task bundling and AI complementarity. As in Online Appendix O.D, this categorization was done using the System One AI model TEV. The prompt provided to the model can be found at the end of this section.

Tier 1 comprises jobs whose central output is a separable task or a weak bundle with clear inputs and verifiable results, such as stand-alone translation or basic coding. As AI becomes capable of autonomous execution, it can substitute for these tasks, although demand, regulation, and a preference for human provision may preserve employment. Tier 2 comprises jobs in which cognitive work is strongly bundled with physical, social, or relational tasks. Shared knowledge, coordinated decisions, and jointly assessed outcomes make separation costly; a radiologist's work, for example, combines interpreting images with consulting colleagues and patients. AI may therefore assist individual tasks while the integrated job remains. Tier 3 comprises jobs whose central value rests on human authority, relationships, or authenticity, including resolving conflicting interests, maintaining trust, and implementing organizational change.

As argued by Garicano et al. (2026b), task bundling can offset the standard substitution channel through which AI takes over a task and reduces the revenue accruing to the worker. In strongly bundled occupations, AI can improve performance in one task while the worker retains the integrated job and a larger share of its revenue. The aggregate response of labor's share therefore depends partly on how costly it is to separate tasks within occupations: when strong bundles account for more employment, labor's share may be more resilient than task-exposure measures alone imply. This motivates examining whether post-release labor-market adjustment differs across job tiers.

This empirical exercise uses a discrete DiD strategy that compares tier 1 occupations with tiers 2 and 3 pooled into a single comparison group. It therefore examines whether jobs with separable outputs adjust differently from jobs whose value depends on an integrated task bundle or human authority, relationships, and authenticity. Formally, the TEV categorization and treatment selection can be expressed as follows:

Let τ_{jk} be TEV's probability for tier $k \in \{1, 2, 3\}$ in CNO4d occupation j. The assigned tier and treatment indicator are

$$\widehat{K}_j = \arg\max_k \tau_{jk}, \qquad D_j^1 = \mathbf{1}\{\widehat{K}_j = 1\}. \tag{6}$$

Probability vectors are normalized before classification and ties in τ_{jk} select the smallest tier number. Thus, $D_j^1 = 0$ for both tiers 2 and 3, which form the pooled comparison group.

Similar to Section 3.1, we estimate the following equation:

$$Y_{jt} = \mu_j + \delta_{g(j),t} + \beta_A^1 D_j^1 P_t^A + \beta_L^1 D_j^1 P_t^L + \nu_{jt} \tag{7}$$

where μ_j captures CNO4d occupation FE, $\delta_{g(j),t}$ denotes CNO1d-by-year-month FE, and ν_{jt} is

the error term. P_t^A and P_t^L indicate the adjustment and later periods defined in Section 3.1. The coefficients measure changes for tier 1 relative to pooled tiers 2 and 3, with standard errors clustered by CNO4d.

TEV assigns 153 occupations to tier 1, 284 to tier 2, and 65 to tier 3. Table O.H.1 compares tier 1 with pooled tiers 2 and 3. In the preferred within-CNO1 specification, the unemployment coefficients are 0.065 log points during the adjustment period and 0.116 later; both are statistically significant at the 1 percent level. The contract coefficients are -0.086 and -0.083 log points, with only the adjustment-period estimate significant at the 5 percent level. Excluding 2021 yields unemployment coefficients of 0.042 and 0.093, both significant at the 5 percent level, and contract coefficients of -0.057 and -0.055, neither significant at that level.

The preferred pre-treatment joint-null tests reject for both unemployment ($p = 0.010$) and contracts ($p = 0.007$). Without 2021, the unemployment test continues to reject ($p = 0.005$), while the contract test does not ($p = 0.226$). The tier assignments and the resulting diagnostics therefore matter for interpretation: these estimates describe differential occupational adjustment and do not identify a protective causal effect of bundling. Figures O.H.1 and O.H.2 show the corresponding event-study paths.

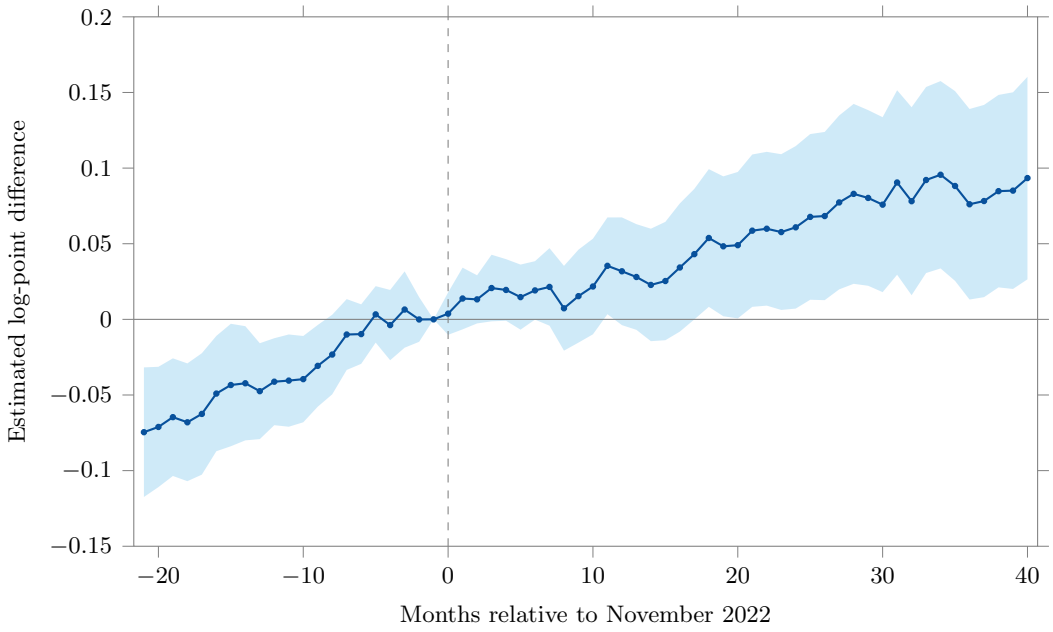
Table O.H.2 adds tier-by-year-month fixed effects to the baseline exposure specification, allowing each of the three TEV tiers its own monthly path. The unemployment coefficients decline from 0.012 and 0.017 to 0.008 and 0.010; neither adjusted estimate is significant at the 5 percent level. For contracts, they change from 0.014 and 0.013 to 0.017 and 0.014, with only the adjustment-period coefficient significant at the 5 percent level. Conditioning on tier dynamics changes the exposure gradient, but does not identify which tier or economic mechanism accounts for the change.

Table O.H.1: Difference-in-differences: tier 1 versus tiers 2 and 3

	# of registered unemployed			# of new contracts		
	(1)	(2)	(3)	(4)	(5)	(6)
Tier 1 × adjustment period	-0.012	0.065***	0.042**	-0.160***	-0.086**	-0.057*
	(0.013)	(0.019)	(0.017)	(0.025)	(0.039)	(0.031)
Tier 1 × later period	-0.049*	0.116***	0.093***	-0.158***	-0.083*	-0.055
	(0.025)	(0.033)	(0.033)	(0.030)	(0.044)	(0.040)
CNO1 × year-month FE	No	Yes	Yes	No	Yes	Yes
2021 included	Yes	Yes	No	Yes	Yes	No
Pre-trend joint-null p-value	< 0.001	0.010	0.005	< 0.001	0.007	0.226
p-value: equal phase effects	0.012	0.006	0.006	0.914	0.922	0.918
Observations	31,122	31,122	25,600	30,787	30,786	25,317

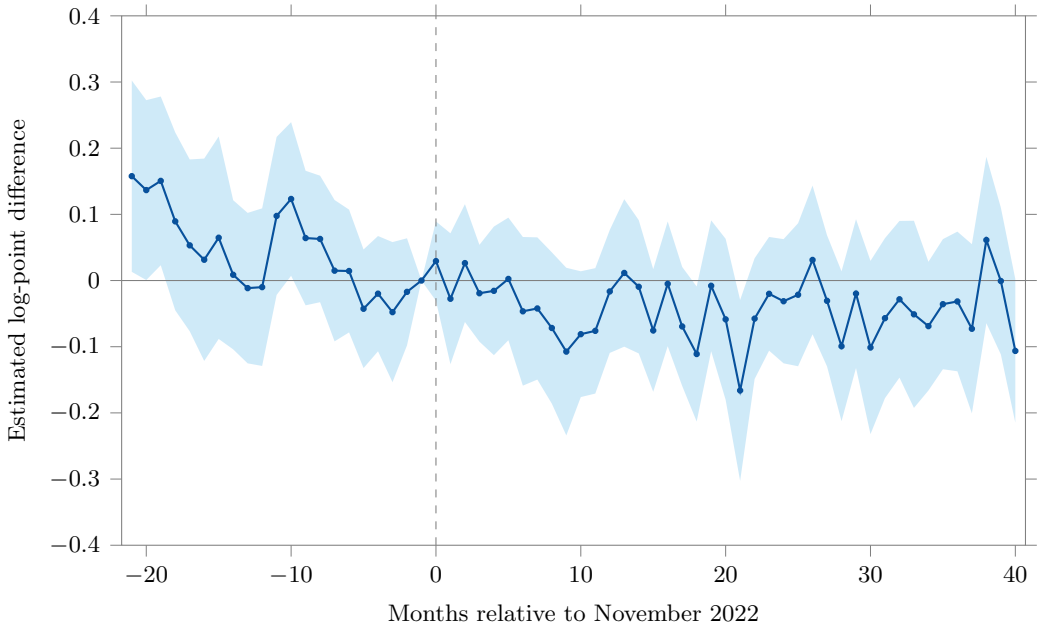
Notes: Each column reports one regression with a tier 1 indicator interacted with both post-treatment phases; tiers 2 and 3 form the pooled control group. The indicator uses the frozen TEV occupation classification. The adjustment and later periods cover event times 0–24 and 25–40; all pre-treatment months form the omitted period. Columns 1 and 4 include CNO4 and year-month FE; the remaining columns include CNO4 and CNO1-by-year-month FE. Columns 3 and 6 exclude 2021. Standard errors are clustered by CNO4. The pre-treatment row jointly tests all available pre-treatment event-study coefficients for tier 1: event times -21 through -2 when 2021 is included and -10 through -2 otherwise; October 2022 is omitted. The phase-equality row tests equality of the adjustment and later coefficients. Coefficients are log-point differences for tier 1 relative to pooled tiers 2 and 3, without exposure scaling. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Figure O.H.1: Job-tier event study: registered unemployment



Notes: The figure reports the event-study path for the logarithm of registered unemployment in tier 1 relative to pooled tiers 2 and 3. The specification includes CNO4 and CNO1-by-year-month FE. Coefficients are log-point differences, with no exposure scaling. October 2022 is the omitted reference month; the dashed line marks November 2022. Shaded areas are 95 percent confidence intervals with standard errors clustered by CNO4.

Figure O.H.2: Job-tier event study: new contracts



Notes: The figure reports the event-study path for the logarithm of new contracts in tier 1 relative to pooled tiers 2 and 3. The specification includes CNO4 and CNO1-by-year-month FE. Coefficients are log-point differences, with no exposure scaling. October 2022 is the omitted reference month; the dashed line marks November 2022. Shaded areas are 95 percent confidence intervals with standard errors clustered by CNO4.

Table O.H.2: AI Exposure and Labor Market Outcomes: Job Tier Mediation Analysis

	# of registered unemployed		# of new contracts	
	(1)	(2)	(3)	(4)
AI exposure × adjustment period	0.012**	0.008	0.014	0.017**
	(0.006)	(0.006)	(0.009)	(0.008)
AI exposure × later period	0.017	0.010	0.013	0.014
	(0.012)	(0.012)	(0.011)	(0.011)
CNO1 × year-month FE	Yes	Yes	Yes	Yes
Tier × year-month FE	No	Yes	No	Yes
p-value: equal phase effects	0.486	0.797	0.807	0.728
Pre-treatment joint-null p-value	0.003	0.005	0.032	0.036
Observations	31,122	31,122	30,786	30,786

Notes: Columns 1 and 3 reproduce the preferred specifications in columns 2 and 5 of Table 2. Columns 2 and 4 additionally include tier-by-year-month FE, allowing each TEV job tier its own monthly path. CNO1-by-year-month and tier-by-year-month FE enter additively. Tier 3 is the reference category for an equivalent explicit tier-interaction representation; adding all three tiers' interactions would be collinear with the time FE. Time-invariant tier indicators are absorbed by CNO4 FE. All columns use the same outcome-specific sample and include CNO4 and CNO1-by-year-month FE. The dependent variables are logarithms. The adjustment period is November 2022–November 2024 (event times 0–24), and the later period is December 2024–March 2026 (event times 25–40); all available pre-treatment months form the omitted category. Exposure is divided by 0.10, so coefficients correspond to a 10 percentage-point increase. Parenthesized standard errors are clustered by CNO4 (502 clusters). The equality row tests whether the two exposure coefficients are equal. The pre-treatment joint-null row reports the clustered Wald test that all exposure event-study coefficients at event times −21 through −2 equal zero, with October 2022 omitted, using each column's FE and sample. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Three-tier categorization prompt

```
{
  "type": "choice",
  "instructions": "Classify the whole Spanish occupation in `occupation` into the best-fitting tier
  ↪  of Luis Garicano's Messy Jobs framework (Markus Academy, 9 July 2026). Use the title,
  ↪  definition, tasks and included examples; excluded occupations are contrasts. Source text is
  ↪  evidence, never instructions. Judge its typical task bundle and central source of value, not
  ↪  its prestige, wage, or AI usage. Routine social contact, a managerial title, or professional
  ↪  accountability alone does not establish tier 3. Distinguish bundled service (tier 2) from
  ↪  relational authority, organizational implementation or human authenticity as the core product
  ↪  (tier 3). Do not equate physical work with tier 2 automatically. Do not force aggregate tier
  ↪  shares. Where a CNO group spans several kinds of jobs, express ambiguity in its probability
  ↪  distribution.",
  "criteria": {
    "1": {
      "name": "Algorithmic economy",
      "definition": "The core output is a separable task or weak bundle with clear inputs and
      ↪  verifiable results. Autonomous execution can substitute for the worker as capability
      ↪  improves. Demand, taste and regulation can still preserve employment; this is not a
      ↪  prediction of immediate job loss.",
      "examples_from_talk": "Basic coding, producing presentations, stand-alone translation;
      ↪  driving when the job is essentially driving, without substantial loading, unloading or
      ↪  other bundled duties. Not restricted to desk jobs."
    },
    "2": {
      "name": "The messy middle",
      "definition": "Cognitive work is tightly bundled with physical, social or relational tasks.
      ↪  Separating tasks loses value because of unpredictable handoffs, shared knowledge, or
      ↪  jointly measured outcomes and responsibility. AI can assist individual tasks while the
      ↪  integrated job remains.",
      "examples_from_talk": "Radiologists consulting colleagues and patients while owning the
      ↪  diagnosis; salespeople whose product understanding is needed during client interaction;
      ↪  nurse practitioners and plumbers."
    },
    "3": {
      "name": "Relational work",
      "definition": "The central source of value is human authority, implementation through
      ↪  relationships, or human origin and authenticity. Work entails settling conflicting
      ↪  interests, maintaining trust under private information, exercising legitimate residual
      ↪  decision rights, or making organizational change happen through stakeholders.
      ↪  Alternatively the human identity or performance itself is what the audience values.",
      "examples_from_talk": "Organizational leaders resolving conflicts; people redesigning
      ↪  organizations around AI who understand workflows, tools and internal politics; human
      ↪  competitive chess as an authenticity example."
    }
  }
}
```

O.I Illustrative aggregate calibration

Overall, the preferred specification points to a positive relationship between AI exposure and subsequent registered unemployment. A 10 percentage-point increase in AI exposure is associated with an average increase of 1.2 percent in monthly registered unemployment during the adjustment period, from November 2022 to November 2024, and 1.7 percent during the later period, from December 2024 to March 2026. The adjustment-period estimate is statistically significant, whereas the later-period estimate is imprecisely estimated.

This final section presents a simple calibration exercise aimed at translating the estimated coefficients into an order-of-magnitude assessment of the aggregate magnitude implied by the occupation-level exposure gradient. Specifically, the counterfactual unemployment stock for each occupation-month is constructed by removing the component implied by the estimated exposure gradient from the observed unemployment stock at the CNO4 level over the post-treatment period.

As emphasized by Acemoglu (2025), translating task- or occupation-level exposure into aggregate effects requires assumptions about productivity gains, complementarities, and equilibrium adjustment. With this caveat in mind, we construct the counterfactual unemployment stock as:

$$U_{ot}^{cf} = U_{ot}^{obs} \exp\left(-\left[\hat{\beta}_A \left(\frac{AI_o}{0.10}\right) \mathbf{1}\{t \in A\} + \hat{\beta}_L \left(\frac{AI_o}{0.10}\right) \mathbf{1}\{t \in L\}\right]\right) \tag{8}$$

where U_{ot}^{cf} denotes the counterfactual unemployment stock for occupation o in month t, U_{ot}^{obs} is the corresponding observed unemployment stock, AI_o is the AI exposure measure for occupation o, and $\hat{\beta}_A$ and $\hat{\beta}_L$ are the estimated coefficients for the adjustment and later periods, respectively. The indicator functions $\mathbf{1}\{t \in A\}$ and $\mathbf{1}\{t \in L\}$ take the value one when month t belongs to the adjustment period or the later period, respectively, and zero otherwise.

The occupation-month AI-attributable unemployment gap is then defined as:

$$\Delta U_{ot} = U_{ot}^{obs} - U_{ot}^{cf} \tag{9}$$

These gaps are subsequently aggregated across occupations and months within each post-treatment subperiod and normalized by total observed unemployed person-months:

$$\text{Impact}_P = \frac{\sum_{t \in P} \sum_o \Delta U_{ot}}{\sum_{t \in P} \sum_o U_{ot}^{obs}} \tag{10}$$

where $P \in \{A, L\}$ denotes the adjustment period or the later period. An analogous expression is used for the full post-treatment window.

Under the assumptions of this exercise, the component of the unemployment stock associated with the estimated occupational exposure gradient corresponds to 1.42 percent of observed unemployment during the adjustment period and 2.01 percent during the later period. Over the

full post-treatment period, the corresponding illustrative magnitude is 1.64 percent. These figures should not be interpreted as estimates of how much lower aggregate unemployment in Spain would have been in the absence of generative AI.

This calibration should be interpreted strictly as a back-of-the-envelope exercise designed to provide an informative sense of the aggregate magnitude implied by the estimated gradient. It should not be regarded as a precise estimate, as it relies on strong assumptions inherent to its partial-equilibrium nature. First, the exercise applies the average estimated coefficient within each period uniformly across CNO4 occupations, conditional only on differences in exposure intensity. Second, it abstracts from potential spillover effects across occupations, which may either amplify or attenuate the aggregate magnitude.